

Orthotropic Plasticity for Aragonite Crystals: Complete Implementation Guide for MOOSE

Material Repository Documentation

July 6, 2026

Abstract

This document provides complete documentation for the orthotropic plasticity material model for aragonite crystals in the MOOSE finite element framework. The model implements a quadric yield surface (following Schwiedrzik et al. [1]) with full orthotropic elasticity and captures highly anisotropic mechanical behaviour with direction-dependent yield stresses ranging from 4100–5340 MPa in tension and 4500–5080 MPa in shear. Post-yield behaviour includes perfect plasticity, exponential/linear hardening, and exponential/piecewise softening with configurable delayed onset. The implementation uses a robust two-stage return mapping algorithm (Newton-Raphson with Primal CPP fallback) and optional Perzyna viscoplastic regularization for challenging convergence scenarios.

Key Feature: The implementation supports reading per-element Euler angles from exodus files, enabling realistic polycrystalline Representative Volume Element (RVE) simulations with spatially-varying crystal orientations. This allows direct simulation of EBSD-characterized microstructures with hundreds or thousands of grains, each with unique crystallographic orientation.

This document covers installation (conda-based, 5–10 minutes), mathematical formulation, complete parameter reference, exodus file reading workflow, validation procedures, and troubleshooting.

Implementation: C++ material classes `OrthotropicPlasticityStressUpdate` and `ComputeElasticityTensorCoupled` for MOOSE solid mechanics module.

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Chapter 1

Introduction

1.1 Overview

Coral aragonite exhibits extreme mechanical anisotropy at the single-crystal level, with:

- Young's moduli varying by factor of 2 ($E_1 = 140$ GPa vs $E_3 = 63$ GPa)
- Yield stresses varying by 30% across loading directions
- Direction-dependent post-yield softening behaviour
- Complex grain boundary interactions in polycrystals

This implementation provides a continuum-scale finite element model that captures these features, bridging molecular dynamics simulations to engineering-scale analysis.

1.2 Model Capabilities

Elastic behaviour:

- Full orthotropic elasticity (9 independent constants)
- **Material models:**
 - `ComputeElasticityTensorCoupled`: Explicit orthotropic constants with optional density scaling
 - `ComputeFabricElasticityTensor`: Fabric-tensor-based model for trabecular bone (Zysset and Curnier [2])
 - Both support Euler angle rotations for polycrystalline microstructures
- Porosity/density scaling (ρ^k power law)
- Per-element elastic constants
- Proper transformation between global and material frames

Yield behaviour:

- Smooth quadric yield surface (Schwiedrzik formulation [1])
- Six independent yield strengths (3 normal + 3 shear) for explicit orthotropic formulation
- **Two modes for porous materials:**

- Explicit yield strengths with density scaling ($\sigma = \sigma_0 \rho^p$)
- Fabric-based scaling for trabecular bone ($\sigma_i = \sigma_0 \rho^p m_i^{2q}$)
- Optional tension-compression asymmetry via linear term
- Associated flow rule (normality condition)
- Cortical correction for high-density bone ($\rho > 0.5$)

Post-yield behaviour:

- Perfect plasticity (no hardening/softening)
- Exponential hardening or softening
- Linear hardening or piecewise softening
- Configurable delayed onset (prevents initial instabilities)

Numerical robustness:

- Two-stage return mapping (Newton-Raphson + Primal CPP)
- Armejo line search for difficult cases
- Optional Perzyna viscoplastic regularization (6 modes)
- Adaptive convergence strategies

Polycrystalline microstructure support:

- Read per-element Euler angles from exodus files
- Direct import of EBSD-characterized microstructures
- Support for 100+ grain RVE simulations
- Spatially-varying elastic and plastic properties
- Element-wise material orientation via AuxVariables
- Validated for twin and multi-grain configurations

Advanced features:

- Damage mechanics coupling (optional)
- Material rotation tracking
- Diagnostic output (return mapping stage, iterations)

1.3 Material Classes Overview

Class	Purpose	Key Features
OrthotropicPlasticityStressUpdate	Bulk plasticity	Quadric yield, fabric/density scaling, return mapping
ComputeElasticityTensorCoupled	Explicit elasticity	Per-element Euler angles, density scaling
ComputeFabricElasticityTensor	Fabric elasticity	Zysset-Curnier model [2], trabecular bone
HomogenizedExponentialCZM*	Interface failure	MD-calibrated traction-separation

*Chapter 3

1.4 Application Domains

This framework supports distinct application domains with appropriate material models:

1.4.1 Dense Coral Aragonite

Material characteristics:

- Single-crystal orthorhombic structure
- Near-full density ($\rho \approx 1.0$)
- Strong elastic anisotropy ($E_1/E_2 \approx 2$)
- Moderate yield strength variation (30% across directions)

Recommended models:

- Elasticity: `ComputeElasticityTensorCoupled` with explicit constants
- Plasticity: `OrthotropicPlasticityStressUpdate` with explicit yield strengths
- Interfaces: `HomogenizedExponentialCZM` for organic interface behaviour
- Microstructure: Per-element Euler angles for polycrystalline RVEs

1.4.2 Porous Coral Skeletons

Material characteristics:

- Hierarchical porosity (sclerodermites + inter-grain voids)
- Variable density ($\rho = 0.6$ – 0.9)
- Anisotropic pore structure following crystal orientation

Recommended models:

- Elasticity: `ComputeElasticityTensorCoupled` with `density_rho` and `elasticity_density_exponent` ($k \approx 2$)
- Plasticity: `OrthotropicPlasticityStressUpdate` with `yield_density_exponent` ($p \approx 1.5$ – 2)
- Scaling: power law ($E = E_0 \rho^k$, $\sigma = \sigma_0 \rho^p$)

1.4.3 Trabecular Bone (Extended Application)

Material characteristics:

- Porous foam structure
- Low density ($\rho = 0.1$ – 0.4 typical)
- Fabric tensor from μ CT imaging
- Mild to strong anisotropy ($m_1/m_3 = 1.2$ – 3)

Recommended models:

- Elasticity: `ComputeFabricElasticityTensor`

- Plasticity: `OrthotropicPlasticityStressUpdate` with `use_fabric_scaling = true`
- Parameters: Tissue-level constants (E_0, σ_0) + fabric eigenvalues (m_1, m_2, m_3) + scaling exponents and stress interaction coefficients (ζ_{ij})

Fabric tensor acquisition: Fabric eigenvalues m_i obtained from:

1. μ CT scan of trabecular bone
2. Mean intercept length (MIL) or volume orientation (VO) analysis
3. Principal directions and eigenvalues satisfy $m_1 + m_2 + m_3 = 3$
4. Eigenvalues range: $m_i \in [0.1, 2.9]$ (extreme anisotropy) to $m_i = 1.0$ (isotropy)

1.5 Scope and Structure

This document is organized as follows:

- **Chapter 2:** Mathematical formulation for bulk material
- **Chapter 3:** Cohesive zone model formulation
- **Chapter 4:** MOOSE installation
- **Chapter 5:** Building custom application
- **Chapter 6:** Complete parameter reference and example input files
- **Chapter 7:** Validation workflow
- **Chapter 8:** Troubleshooting guide

Chapter 2

Mathematical Formulation

2.1 Kinematic Framework

2.1.1 Strain Decomposition (Green-Naghdi Split)

For finite strains, the total Green-Lagrange strain tensor decomposes as:

$$\mathbf{E}^{\text{total}} = \mathbf{E}^e + \mathbf{E}^p \quad (2.1)$$

where:

- $\mathbf{E}$ = Green-Lagrange strain tensor
- $\mathbf{E}^e$ = elastic component
- $\mathbf{E}^p$ = plastic component

The 2nd Piola-Kirchhoff stress is work-conjugate to $\mathbf{E}$:

$$\mathbf{S} = \mathbb{C} : \mathbf{E}^e \quad (2.2)$$

Note: For small deformations ($\|\mathbf{E}\| < 0.05$), Green-Lagrange strain $\mathbf{E}$ reduces to infinitesimal strain $\boldsymbol{\varepsilon}$, and 2nd Piola-Kirchhoff stress $\mathbf{S}$ approaches Cauchy stress $\boldsymbol{\sigma}$.

Implementation: The MOOSE framework implementation uses finite strain kinematics (`strain = FINITE`), computing Green-Lagrange strain $\mathbf{E}$ and 2nd Piola-Kirchhoff stress $\mathbf{S}$ directly. This formulation is exact for the full range of deformations and maintains work-conjugacy throughout the analysis.

2.1.2 Accumulated Plastic Strain

Following Wolfram et al. [3] (supplementary Eq. S16), the accumulated plastic strain is defined as the norm of the plastic strain rate:

$$\kappa = \int_0^t \|\dot{\mathbf{E}}^p\| d\tau = \int_0^t \sqrt{\dot{\mathbf{E}}^p : \dot{\mathbf{E}}^p} d\tau \quad (2.3)$$

where $\mathbf{E}^p$ is the plastic Green-Lagrange strain tensor.

Important: Definition of κ

Important: This differs from von Mises equivalent plastic strain, which includes a $\sqrt{2/3}$ scaling factor. The quadric yield surface formulation uses the direct norm without scaling.

Implications for postprocessing:

- Do not compare κ directly to von Mises $\varepsilon_{\text{eq}}^p$ from other codes
- For comparison: $\varepsilon_{\text{eq}}^p \approx \sqrt{2/3} \cdot \kappa$ (approximate)
- Post-yield laws use κ directly as formulated

2.2 Frame Handling and Objectivity Considerations

2.2.1 Finite Strain vs Small Strain Implementation

The present implementation employs a **small-strain plasticity model embedded within a finite-strain kinematic framework**. This hybrid approach is appropriate for materials undergoing moderate rotations but small plastic strains, which is characteristic of aragonite single crystals and polycrystalline aggregates.

Kinematic Framework:

- **Strain measure:** Green-Lagrange strain tensor $\mathbf{E}$
- **Stress measure:** Second Piola-Kirchhoff stress tensor $\mathbf{S}$
- **Decomposition:** Total deformation gradient $\mathbf{F} = \mathbf{F}^e \mathbf{F}^p$ (Green-Naghmi split)
- **Work conjugacy:** Exact conjugacy maintained: $\mathbf{S} : \dot{\mathbf{E}}$

Plasticity Formulation: The plastic flow rules and hardening laws are formulated in the **material (crystal) frame**, which rotates with the lattice. Key aspects:

1. Rotation tracking:

- Crystal orientation defined by Euler angles (ϕ_1, Φ, ϕ_2) stored per element
- Rotation tensor $\mathbf{R}$ computed from Euler angles at each integration point
- For small rotations ($\Phi < 10^\circ$), Euler angles treated as constant
- For large rotations: Rotation tensor $\mathbf{R}$ is tracked automatically by MOOSE's `ComputeFiniteStrain`, but Euler angles (ϕ_1, Φ, ϕ_2) stored in aux variables remain constant.

2. Plastic strain storage:

- Accumulated plastic strain κ stored as scalar (frame-independent)
- Plastic strain tensor $\boldsymbol{\varepsilon}^p$ stored in **material frame**
- Total strain transformed to material frame before return mapping

3. Return mapping:

- Performed in material frame using rotated stress $\tilde{\boldsymbol{\sigma}} = \mathbf{R}^T \boldsymbol{\sigma} \mathbf{R}$
- Flow direction computed in material frame (slip systems aligned with crystal axes)

- Stress update transformed back to global frame

4. Objectivity:

- Rotation of material frame automatically ensures objectivity
- Plastic flow remains aligned with crystallographic directions
- No additional objectivity corrections needed for moderate rotations ($< 10^\circ$)

Assumptions and Limitations:

- **Small plastic strains:** $\|\epsilon^p\| < 0.05$ (5% effective plastic strain)
- **Moderate rotations:** Rigid body rotations $< 10^\circ$ per load step
- **No lattice reorientation:** Euler angles constant throughout simulation
- **Elastic isotropy assumption:** While anisotropic elasticity is implemented, large-rotation effects on elastic tensor are neglected

Validation of Objectivity: Objectivity is verified through rigid body rotation tests:

1. Apply prescribed rotation to entire mesh
2. Apply same loading in rotated configuration
3. Compare stress state: $\sigma_{\text{rotated}} = \mathbf{R}\sigma_{\text{original}}\mathbf{R}^T$
4. Acceptance: $\|\sigma_{\text{rotated}} - \mathbf{R}\sigma_{\text{original}}\mathbf{R}^T\|/\|\sigma\| < 10^{-6}$

For applications requiring large plastic strains ($> 10\%$) or large rotations ($> 15^\circ$), a fully finite-strain plasticity formulation with multiplicative decomposition should be employed.

Relationship to MOOSE Framework: The MOOSE `ComputeFiniteStrain` class computes:

- Total deformation gradient $\mathbf{F}$
- Rotation tensor $\mathbf{R}$ (from polar decomposition)
- Incremental rotation $\Delta\mathbf{R}$

Our `OrthotropicPlasticityStressUpdate` class:

- Receives rotated elastic strain from `ComputeFiniteStrain`
- Applies crystal rotation from Euler angles
- Performs return mapping in crystal frame
- Returns stress in global frame

This architecture ensures consistency between MOOSE's finite-strain kinematics and the small-strain plasticity model.

Note on Crystal Rotation: MOOSE's finite strain framework computes the rotation tensor $\mathbf{R}$ via polar decomposition of the deformation gradient:

$$\mathbf{F} = \mathbf{R}\mathbf{U} \quad (2.4)$$

This rotation represents the rigid body rotation of the material element. However, for the plasticity model, we use Euler angles stored as constant aux variables to define the *initial* crystal orientation. These Euler angles do not update during simulation, which is appropriate for:

- Small rotations ($< 10^\circ$)
- Lattice orientations fixed by crystallography
- Material frame plasticity without lattice spin

For applications requiring lattice reorientation (e.g., texture evolution, large plastic strains), the Euler angles would need to be extracted from the updated $\mathbf{R}$ tensor at each timestep via inverse Bunge transformation. This is beyond the current scope but represents a straightforward extension.

2.3 Orthotropic Elasticity

2.3.1 Constitutive Relation

In the material frame (crystal axes):

$$\mathbf{S} = \mathbb{C} : \mathbf{E}^e \quad (2.5)$$

2.3.2 Stiffness Matrix

Voigt notation ($\mathbf{S} = \mathbb{C}\mathbf{E}$, with engineering shear strains $2E_{12}, 2E_{13}, 2E_{23}$):

$$\mathbb{C} = \begin{bmatrix} C_{11} & C_{12} & C_{13} & 0 & 0 & 0 \\ C_{12} & C_{22} & C_{23} & 0 & 0 & 0 \\ C_{13} & C_{23} & C_{33} & 0 & 0 & 0 \\ 0 & 0 & 0 & C_{44} & 0 & 0 \\ 0 & 0 & 0 & 0 & C_{55} & 0 \\ 0 & 0 & 0 & 0 & 0 & C_{66} \end{bmatrix} \quad (2.6)$$

2.3.3 Aragonite Constants

From molecular dynamics calibration:

$$C_{11} = 171.8 \text{ GPa}, \quad C_{22} = 106.7 \text{ GPa}, \quad C_{33} = 84.2 \text{ GPa} \quad (2.7)$$

$$C_{12} = 57.5 \text{ GPa}, \quad C_{13} = 30.2 \text{ GPa}, \quad C_{23} = 46.9 \text{ GPa} \quad (2.8)$$

$$C_{44} = 42.1 \text{ GPa}, \quad C_{55} = 31.1 \text{ GPa}, \quad C_{66} = 46.6 \text{ GPa} \quad (2.9)$$

Derived engineering constants:

$$E_1 = 140.4 \text{ GPa}, \quad E_2 = 70.3 \text{ GPa}, \quad E_3 = 63.4 \text{ GPa} \quad (2.10)$$

$$G_{12} = 46.6 \text{ GPa}, \quad G_{13} = 31.1 \text{ GPa}, \quad G_{23} = 42.1 \text{ GPa} \quad (2.11)$$

$$\nu_{12} = 0.505, \quad \nu_{13} = 0.078, \quad \nu_{23} = 0.466 \quad (2.12)$$

2.4 Fabric-Based Elasticity for Porous Materials

2.4.1 Zysset-Curnier Fabric Tensor Model

For trabecular bone and porous materials, elastic properties depend on both density and structural anisotropy. The Zysset-Curnier model [2] transforms the isotropic compliance tensor using the fabric tensor:

$$\mathbf{S}' = \frac{1}{\rho^k} \mathbf{M}^{-l} : \mathbf{S}_0 : \mathbf{M}^{-l} \quad (2.13)$$

where:

- $\mathbf{S}_0$ = Isotropic tissue compliance tensor
- $\mathbf{M}$ = Fabric tensor (diagonal in principal axes: m_1, m_2, m_3)
- ρ = Relative density (BV/TV, bone volume fraction)
- k = Density exponent (typically $k = 2$ for porous foams)
- l = Fabric exponent (typically $l = 1$)

Physical interpretation:

- ρ^k captures overall porosity effect (Gibson-Ashby scaling)
- $\mathbf{M}^{-l}$ captures directional variation due to trabecular alignment
- m_i eigenvalues quantify structural anisotropy ($m_1 + m_2 + m_3 = 3$)
- Isotropy: $m_1 = m_2 = m_3 = 1$ reduces to $\mathbf{S}' = \rho^{-k} \mathbf{S}_0$

2.4.2 Derivation of Engineering Constants

In the principal fabric coordinate system, the compliance transformation yields:

Young's moduli:

$$E_i = E_0 \rho^k m_i^{2l}, \quad i = 1, 2, 3 \quad (2.14)$$

Shear moduli:

$$G_{ij} = G_0 \rho^k (m_i m_j)^l, \quad G_0 = \frac{E_0}{2(1 + \nu_0)} \quad (2.15)$$

Poisson's ratios:

$$\nu_{ij} = \nu_0 \left(\frac{m_i}{m_j} \right)^l \quad (2.16)$$

with conjugate relations $\nu_{ji} = \nu_{ij} E_j / E_i$ ensuring symmetry.

2.4.3 Implementation: Compliance to Stiffness Conversion

The implementation follows these steps:

Step 1: Compute engineering constants from fabric

```
Real rho_k = std::pow(density_rho, exponent_k);
for (unsigned i = 0; i < 3; ++i) {
    E[i] = E_0 * rho_k * std::pow(m[i], 2 * exponent_1);
    for (unsigned j = 0; j < 3; ++j) {
        if (i != j) {
            nu[i][j] = nu_0 * std::pow(m[i] / m[j], exponent_1);
        }
    }
}
G_12 = G_0 * rho_k * std::pow(m[0] * m[1], exponent_1);
G_13 = G_0 * rho_k * std::pow(m[0] * m[2], exponent_1);
G_23 = G_0 * rho_k * std::pow(m[1] * m[2], exponent_1);
```

Step 2: Construct orthotropic stiffness tensor

Using standard orthotropic formulas with positive definiteness denominator:

$$\Delta = 1 - \nu_{12}\nu_{21} - \nu_{23}\nu_{32} - \nu_{31}\nu_{13} - 2\nu_{21}\nu_{32}\nu_{13} \quad (2.17)$$

Stiffness components:

$$C_{11} = \frac{E_1(1 - \nu_{23}\nu_{32})}{\Delta}, \quad C_{12} = \frac{E_1(\nu_{21} + \nu_{31}\nu_{23})}{\Delta} \quad (2.18)$$

$$C_{44} = G_{23}, \quad C_{55} = G_{13}, \quad C_{66} = G_{12} \quad (2.19)$$

Mathematical equivalence to direct stiffness scaling: The compliance-based approach is algebraically equivalent to the direct stiffness transformation $\mathbf{C}' = \mathbf{M}^l : \mathbf{C}_0 : \mathbf{M}^l$, which gives:

$$C'_{ij} = C_{0,ij} \times \rho^k \times m_i^l m_j^l \quad (2.20)$$

Both approaches yield identical stiffness tensors. The compliance route provides engineering constants as interpretable intermediate values.

2.4.4 Typical Parameter Values

Gibson-Ashby exponents (porous foams):

- $k = 2.0$ (elastic stiffness scales as ρ^2)
- $l = 1.0$ (linear fabric dependence)

Bone tissue properties:

- $E_0 = 10\text{--}20$ GPa (cortical bone tissue modulus)
- $\nu_0 = 0.3$
- $\rho = 0.1\text{--}0.4$ (typical trabecular bone)
- $m_1/m_3 = 1.2\text{--}3$ (anisotropy ratio)

2.4.5 Density Scaling for Explicit Orthotropic Constants

For materials with known full-density orthotropic constants (e.g., porous coral), a simplified scaling applies:

$$\mathbb{C}(\rho) = \rho^k \mathbb{C}_0 \quad (2.21)$$

This is implemented in `ComputeElasticityTensorCoupled` via:

```
Real scale = std::pow(density_rho, elasticity_density_exponent);
_elasticity_tensor[_qp] = base_elasticity_tensor * scale;
```

When to use:

- Porous materials with uniform pore distribution
- Known full-density orthotropic constants from MD or experiments
- Isotropic porosity (fabric tensor not available)

Note on two modelling paradigms: This framework supports two distinct approaches for anisotropic material behaviour:

1. **Fabric-based mode:** For trabecular bone and porous biomaterials with:

- Base isotropic tissue properties (E_0, ν_0, σ_0)
- Measured fabric tensor from μ CT analysis (m_1, m_2, m_3)
- Known density (bone volume fraction, BV/TV)
- *Application:* Trabecular bone, foam-like structures

2. **Explicit orthotropic mode:** For dense materials with:

- Directly measured orthotropic constants from MD or experiments
- Nine independent elastic constants, six yield strengths (nine if including tension/-compression asymmetry)
- Optional density scaling for porous variants
- *Application:* Aragonite crystals, dense bone tissue, minerals

Key difference: Fabric mode *computes* anisotropic properties from isotropic base + microstructure. Explicit mode uses anisotropic properties with Euler angles for microstructure *directly*. Both utilize the same quadric yield surface formulation (next section), but with different parameter determination strategies.

The yield surface theory presented below applies universally to both modes. Fabric-based integration with the yield function is described in 2.5.5.

2.5 Quadric Yield Surface

2.5.1 Yield Function

The yield function is defined as:

$$Y(\mathbf{S}, \kappa) = \sqrt{\mathbf{S} : \mathbb{F} : \mathbf{S}} + \mathbf{F} : \mathbf{S} - r(\kappa) \quad (2.22)$$

where:

- $\mathbf{S}$ = 2nd Piola-Kirchhoff stress tensor (6×1 Voigt notation)
- $\mathbb{F}$ = 4th order quadric tensor (6×6 matrix)
- $\mathbf{F}$ = 2nd order linear term vector (6×1)
- κ = accumulated plastic strain (scalar)
- $r(\kappa)$ = strength evolution function

Yielding occurs when $Y = 0$.

2.5.2 Voigt Notation

$$Y = \sqrt{\sum_{i=1}^6 \sum_{j=1}^6 F_{ij} S_i S_j} + \sum_{i=1}^6 F_i S_i - r(\kappa) \quad (2.23)$$

The linear term $\mathbf{F} \cdot \mathbf{S}$ enables tension-compression asymmetry.

2.5.3 Calibration

Case 1: Symmetric (Equal Tension/Compression)

Default behaviour when compression strengths are not specified. For independent yield stresses in each direction with no tension-compression asymmetry:

$$F_{11} = \frac{1}{(S_1^y)^2}, \quad F_{22} = \frac{1}{(S_2^y)^2}, \quad F_{33} = \frac{1}{(S_3^y)^2} \quad (2.24)$$

$$F_{44} = \frac{1}{(\tau_{23}^y)^2}, \quad F_{55} = \frac{1}{(\tau_{13}^y)^2}, \quad F_{66} = \frac{1}{(\tau_{12}^y)^2} \quad (2.25)$$

$$F_{ij} = 0 \quad (i \neq j), \quad F_i = 0 \quad \forall i \quad (2.26)$$

This gives an ellipsoidal yield surface centered at the origin.

Case 2: Asymmetric (Different Tension/Compression)

When compression strengths differ from tension, the linear term $\mathbf{F}$ is activated:

$$F_i = \frac{1}{2} \left(\frac{1}{S_{i,\text{tension}}^y} - \frac{1}{S_{i,\text{compression}}^y} \right) \quad (i = 1, 2, 3) \quad (2.27)$$

$$F_i = 0 \quad (i = 4, 5, 6) \quad (2.28)$$

And the quadric tensor diagonal components are modified:

$$F_{ii} = \left[\frac{S_{i,\text{tension}}^y + S_{i,\text{compression}}^y}{2 S_{i,\text{tension}}^y \cdot S_{i,\text{compression}}^y} \right]^2 \quad (i = 1, 2, 3) \quad (2.29)$$

Off-diagonal terms remain zero: $F_{ij} = 0$ for $i \neq j$ (uncoupled).

This shifts and deforms the yield surface to account for different behaviour in tension vs compression.

Case 3: Biaxial Coupling (Advanced)

For materials with strong interaction between normal stresses in different directions, off-diagonal coupling terms can be activated using the Tsai-Wu formulation [4]:

$$F_{ij} = -\zeta_{ij} \sqrt{F_{ii} F_{jj}}, \quad i \neq j, \quad i, j \in \{1, 2, 3\} \quad (2.30)$$

where ζ_{12} , ζ_{13} , ζ_{23} are dimensionless coupling parameters. The three independent coupling parameters are:

- ζ_{12} : X-Y interaction (coupling between σ_{xx} and σ_{yy})
- ζ_{13} : X-Z interaction (coupling between σ_{xx} and σ_{zz})
- ζ_{23} : Y-Z interaction (coupling between σ_{yy} and σ_{zz})

The matrix is symmetric: $F_{ij} = F_{ji}$.

For convexity of the yield surface, these must satisfy $|\zeta_{ij}| \leq 1$.

Physical interpretation:

- $\zeta_{ij} = 0$ (uncoupled): Each direction yields independently.
- $\zeta_{ij} > 0$ (positive coupling): Combined loading in directions i and j makes yielding occur at lower individual stress magnitudes than uniaxial loading alone.
- $\zeta_{ij} < 0$ (negative coupling): Combined loading in directions i and j strengthens the material, requiring higher individual stresses to yield.

Determination from experiments:

For an equibiaxial tension test where $S_{11} = S_{22} = S_b$ at yield:

$$\zeta_{12} = \frac{1}{2\sqrt{F_{11}F_{22}}} \left[\frac{1 - S_b(F_1 + F_2)}{S_b^2} - F_{11} - F_{22} \right] \quad (2.31)$$

where F_{11} , F_{22} are computed from uniaxial strengths (Eq. 2.29) and F_1 , F_2 from tension-compression asymmetry (Eq. 2.27).

Default for aragonite/bone: For trabecular bone and coral, $\zeta_{ij} = 0$ is commonly used when biaxial test data is unavailable.

2.5.4 Aragonite Yield Stresses

From molecular dynamics:

$$S_1^y = 4980 \text{ MPa}, \quad S_2^y = 4100 \text{ MPa}, \quad S_3^y = 5340 \text{ MPa} \quad (2.32)$$

$$\tau_{12}^y = 4510 \text{ MPa}, \quad \tau_{13}^y = 5080 \text{ MPa}, \quad \tau_{23}^y = 5060 \text{ MPa} \quad (2.33)$$

2.5.5 Integration of Fabric Scaling with Yield Surface

The fabric-based approach (Section 2.4) extends to plasticity through combined density-fabric scaling of yield strengths [1]. This subsection describes how fabric-scaled strengths integrate with the quadric yield function (Section 2.5).

Usage modes: Explicit orthotropic mode (aragonite, dense crystals):

- Input yield strengths directly from MD simulations or experiments
- Example: Aragonite values in 2.5.4 ($\sigma_1 = 4980$ MPa, etc.)
- Set `use_fabric_scaling = false` in `OrthotropicPlasticityStressUpdate`

Fabric-based mode (trabecular bone):

- Compute yield strengths from base tissue values scaled by fabric and density
- Formulas described below
- Set `use_fabric_scaling = true`

Following Schwiedrzik et al. [1], yield strengths scale with both density and fabric:

Directional yield strengths:

$$\sigma_i^+ = \sigma_0^+ \times \rho^p \times m_i^{2q} \quad (\text{tension}) \quad (2.34)$$

$$\sigma_i^- = \sigma_0^- \times \rho^p \times m_i^{2q} \quad (\text{compression}) \quad (2.35)$$

$$\tau_{ij} = \tau_0 \times \rho^p \times (m_i m_j)^q \quad (\text{shear}) \quad (2.36)$$

where:

- $\sigma_0^\pm, \tau_0$ = Tissue-level (isotropic base) yield strengths
- m_i = Fabric eigenvalues in principal directions ($m_1 + m_2 + m_3 = 3$)
- ρ = Relative density (BV/TV, bone volume fraction)
- p = Yield density exponent (typically $p = 1.5$ – 1.8)
- q = Yield fabric exponent (typically $q = 1.0$ – 1.5)

Biaxial coupling parameters:

$$\zeta_{ij} = \zeta_0 \times \left(\frac{m_i}{m_j} \right)^{2q} \quad (2.37)$$

TSFU cortical correction (high-density bone): For bone with $\rho > 0.5$, a transition to cortical bone behavior is modeled:

$$\text{TSFU}(\rho, p, \delta) = \begin{cases} \rho^p & \text{if } \rho \leq 0.5 \\ \rho^p + (\delta - 1) \left(\frac{\rho - 0.5}{0.5} \right)^p & \text{if } \rho > 0.5 \end{cases} \quad (2.38)$$

where $\delta \in [1, 2]$ controls cortical correction strength. Set $\delta = 1.0$ to disable.

Implementation notes: In `OrthotropicPlasticityStressUpdate`, fabric scaling is activated via:

```
if (_use_fabric_scaling) {
  Real tsfu = computeTSFU(_density_rho, _exponent_p, _delta);
  for (unsigned i = 0; i < 3; ++i) {
    sigma_tension[i] = _sigma_0_tension * tsfu *
      std::pow(_fabric_m[i], 2 * _exponent_q);
    sigma_compression[i] = _sigma_0_compression * tsfu *
      std::pow(_fabric_m[i], 2 * _exponent_q);
  }
  // Shear and zeta similarly scaled
}
```

Relationship to quadric yield function: The scaled yield strengths σ_i^+ , σ_i^- , τ_{ij} are then used to construct the quadric tensor $\mathbb{F}$ and linear term $\mathbf{F}$ via Equations (2.27)–(2.30). The yield function ($\phi(\boldsymbol{\sigma})$) evaluation proceeds identically for both fabric-based and explicit modes, only the source of the yield strength values differs.

2.5.6 Parameter Mapping and Voigt Indexing

This subsection provides explicit mapping between mathematical symbols, MOOSE input parameters, and internal implementation for **both modelling modes**.

Mode 1: Explicit Orthotropic (Dense Materials)

Table 2.1 provides explicit mapping between mathematical symbols, MOOSE input parameters, and internal Voigt indices used in the implementation.

Table 2.1: Yield surface parameter mapping

Symbol	Description	Input Parameter	Voigt Index	Units
<i>Normal Strengths (Diagonal Terms)</i>				
σ_{11}^u	Tension strength, a-axis	sigma_xx_tension	0	MPa
σ_{22}^u	Tension strength, b-axis	sigma_yy_tension	1	MPa
σ_{33}^u	Tension strength, c-axis	sigma_zz_tension	2	MPa
<i>Shear Strengths (Off-Diagonal Terms)</i>				
$\tau_{12}^{\max}$	Maximum shear, ab-plane	tau_xy_max	3	MPa
$\tau_{13}^{\max}$	Maximum shear, ac-plane	tau_xz_max	4	MPa
$\tau_{23}^{\max}$	Maximum shear, bc-plane	tau_yz_max	5	MPa
<i>Stress Components (Voigt Notation - Cauchy Stress)</i>				
σ_{11}	Normal stress, a-direction	stress[0]	0	MPa
σ_{22}	Normal stress, b-direction	stress[1]	1	MPa
σ_{33}	Normal stress, c-direction	stress[2]	2	MPa
σ_{12}	Shear stress, ab-plane	stress[3]	3	MPa
σ_{13}	Shear stress, ac-plane	stress[4]	4	MPa
σ_{23}	Shear stress, bc-plane	stress[5]	5	MPa
<i>Density and Elastic Exponents (optional)</i>				
ρ	Relative density (BV/TV)	density_rho	–	–

Example Input Block:

```
[Materials]
[plasticity]
  type = OrthotropicPlasticityStressUpdate

  # Yield strengths in MPa (material frame)
  sigma_xx_tension = 4980    # a-axis
  sigma_yy_tension = 4100    # b-axis
  sigma_zz_tension = 5340    # c-axis (strongest)
  tau_xy_max = 4510         # ab-plane
  tau_xz_max = 5080         # ac-plane
  tau_yz_max = 5060         # bc-plane

  # Euler angles defining material orientation
  euler_angle_1 = euler_phi1 # Coupled variable
  euler_angle_2 = euler_Phi
  euler_angle_3 = euler_phi2
[]
```

Mode 2: Fabric-Based (Trabecular Bone)

Table 2.2 maps parameters for fabric-scaled yield strengths.

Critical constraints:

Table 2.2: Complete fabric-based parameter mapping (elasticity + plasticity)

Symbol	Description	Input Parameter	Typical Value	Units
<i>ELASTICITY: ComputeFabricElasticityTensor</i>				
<i>Base Tissue Elastic Properties</i>				
E_0	Base tissue Young's modulus	E_0	10000–20000	MPa
ν_0	Base tissue Poisson's ratio	nu_0	0.25–0.35	–
G_0	Base shear modulus (optional)	G_0	$E_0/[2(1 + \nu_0)]$	MPa
<i>Fabric Tensor (from μCT)</i>				
m_1	Fabric eigenvalue, direction 1	fabric_m1	1.0–1.5	–
m_2	Fabric eigenvalue, direction 2	fabric_m2	0.9–1.1	–
m_3	Fabric eigenvalue, direction 3	fabric_m3	0.6–1.0	–
<i>Density and Elastic Exponents</i>				
ρ	Relative density (BV/TV)	density_rho	0.1–0.4	–
k	Elasticity density exponent	exponent_k	1.8–2.0	–
l	Elasticity fabric exponent	exponent_l	1.0	–
<i>PLASTICITY: OrthotropicPlasticityStressUpdate</i>				
<i>Base Tissue Yield Strengths</i>				
σ_0^+	Tissue tension strength	sigma_0_tension	32–75	MPa
σ_0^-	Tissue compression strength	sigma_0_compression	48–112	MPa
τ_0	Tissue shear strength	tau_0	11–47	MPa
ζ_0	Tissue coupling parameter	zeta_0	0.22	–
<i>Fabric Tensor (must match elasticity)</i>				
m_1	Fabric eigenvalue, direction 1	fabric_m1	Same as elasticity	–
m_2	Fabric eigenvalue, direction 2	fabric_m2	Same as elasticity	–
m_3	Fabric eigenvalue, direction 3	fabric_m3	Same as elasticity	–
<i>Density and Yield Exponents</i>				
ρ	Relative density (BV/TV)	density_rho	Same as elasticity	–
p	Yield density exponent	exponent_p	1.28–1.69	–
q	Yield fabric exponent	exponent_q	0.5–1.5	–
δ	Cortical correction	delta_cortical	1.0 (disabled)	–
<i>Mode Selection</i>				
–	Enable fabric scaling	use_fabric_scaling	true	bool

- **Fabric normalization:** $m_1 + m_2 + m_3 = 3$ (always)
- **Consistency:** Fabric values and density MUST match between elasticity and plasticity materials

Example Input Block (Trabecular Bone):

```
[Materials]
# =====
# ELASTICITY: Fabric-based (Zysset-Curnier [2])
# =====
[elasticity]
  type = ComputeFabricElasticityTensor

  # Base tissue elastic properties
  E_0 = 10000.0          # Base tissue modulus [MPa]
```

```

nu_0 = 0.3                # Base Poisson ratio
# G_0 computed automatically from E_0 and nu_0

# Fabric tensor (from  $\mu$ CT analysis)
fabric_m1 = 1.1           # Principal direction 1 (strongest)
fabric_m2 = 1.0           # Principal direction 2
fabric_m3 = 0.9           # Principal direction 3 (weakest)
# Constraint: m1 + m2 + m3 = 3.0

# Density from  $\mu$ CT
density_rho = 0.2         # BV/TV = 20%

# Elastic scaling exponents
exponent_k = 2.0          # Density exponent (Gibson-Ashby)
exponent_l = 1.0          # Fabric exponent
[]

# =====
# PLASTICITY: Fabric-based (Schwiedrzik et al. [1])
# =====
[plasticity]
    type = OrthotropicPlasticityStressUpdate

# CRITICAL: Enable fabric mode
use_fabric_scaling = true

# Base tissue yield strengths [MPa]
# (Wolfram et al. [5] values)
sigma_0_tension = 74.589
sigma_0_compression = 111.724
tau_0 = 47.33
zeta_0 = 0.22

# Fabric tensor (MUST MATCH elasticity)
fabric_m1 = 1.1
fabric_m2 = 1.0
fabric_m3 = 0.9

# Density (MUST MATCH elasticity)
density_rho = 0.2

# Yield scaling exponents
exponent_p = 1.686        # Yield density exponent
exponent_q = 1.02         # Yield fabric exponent
delta_cortical = 1.0      # Cortical correction (1.0 = disabled)

# Post-yield behavior
postyield_mode = 'exp_softening'
residual_strength = 0.7
kmin = 0.002
kslope = 100.0

# Standard parameters (same as explicit mode)
absolute_tolerance = 1.0e-5
max_iterations_newton = 20
use_primal_cpp = true
max_iterations_primal = 1000
[]

```

Important Notes

For explicit orthotropic mode (Mode 1), use `ComputeElasticityTensorCoupled` instead, with `use_fabric_scaling = false` in plasticity.

1. Stress measure:

- All stresses are Cauchy stress components (true stress) when `strain = SMALL` is active in Physics block
- Valid for small-strain approximation ($\varepsilon < 0.05$)
- MOOSE handles conversion from 2nd PK internally with (`strain = FINITE`)

2. Energy conjugate vs engineering shear:

- Voigt indices 0-2: Normal stresses (Cauchy, energy-conjugate)
- Voigt indices 3-5: Shear stresses (Cauchy, energy-conjugate)
- NO factor of 2 correction needed in yield function
- Engineering shear strain: $\gamma_{ij} = 2\varepsilon_{ij}$ (NOT used here)

3. Crystal axes convention:

- $x \rightarrow$ a-axis (aragonite crystal)
- $y \rightarrow$ b-axis (aragonite crystal)
- $z \rightarrow$ c-axis (aragonite crystal)

4. Material frame vs global frame:

- All parameters defined in **material (crystal) frame**
- Global stresses rotated to material frame before yield check
- Rotation defined by Euler angles (ϕ_1, Φ, ϕ_2)

5. **Yield function evaluation.** When tension and compression strengths are equal, the yield function reduces to a simple ellipsoidal form. For this symmetric case ($\mathbf{F} = \mathbf{0}$, no linear term), the yield function can be written explicitly in component form:

$$f = \left(\frac{\sigma_{11}}{\sigma_{xx}^u} \right)^2 + \left(\frac{\sigma_{22}}{\sigma_{yy}^u} \right)^2 + \left(\frac{\sigma_{33}}{\sigma_{zz}^u} \right)^2 + \left(\frac{\sigma_{12}}{\tau_{xy}^{\max}} \right)^2 + \left(\frac{\sigma_{13}}{\tau_{xz}^{\max}} \right)^2 + \left(\frac{\sigma_{23}}{\tau_{yz}^{\max}} \right)^2 - 1 \quad (2.39)$$

Note: This simplified form is used in most aragonite single crystal simulations presented in this document. For the general case with tension-compression asymmetry, the full quadric formulation in equation (2.22) with non-zero $\mathbf{F}$ must be employed. The present implementation supports both cases through optional compression strength parameters.

6. Sign convention:

- Tension positive: $\sigma_{ii} > 0$ (tension), $\sigma_{ii} < 0$ (compression)
- Yield strengths: always positive input values

2.6 Post-Yield Evolution

The strength function $r(\kappa)$ controls hardening or softening behaviour. Five modes are available (selected via `postyield_mode`):

2.6.1 Perfect Plasticity

$$r(\kappa) = 1 \quad \forall \kappa \quad (2.40)$$

No hardening or softening. Stress remains at yield surface indefinitely.

2.6.2 Exponential Hardening

$$r(\kappa) = r_\infty + (1 - r_\infty) [1 - \exp(-k_{\text{slope}}\kappa)] \quad (2.41)$$

where:

- $r_\infty > 1$: Ultimate strength ratio (e.g., $1.3 = 30\%$ hardening)
- $k_{\text{slope}} > 0$: Hardening rate

2.6.3 Linear Hardening

$$r(\kappa) = 1 + k_{\text{slope}} \cdot \kappa \quad (2.42)$$

Unbounded hardening (use with caution for large strains).

2.6.4 Exponential Softening

$$r(\kappa) = \begin{cases} 1 & \text{if } \kappa < \kappa_{\text{max}} \\ r_{\text{res}} + (1 - r_{\text{res}}) \exp[-k_{\text{slope}}(\kappa - \kappa_{\text{max}})] & \text{if } \kappa \geq \kappa_{\text{max}} \end{cases} \quad (2.43)$$

where:

- $r_{\text{res}} < 1$: Residual strength ratio (e.g., $0.7 = 70\%$ of initial)
- $k_{\text{slope}} > 0$: Softening rate
- κ_{max} : Delay before softening begins (critical for numerical stability)

2.6.5 Piecewise Linear Softening

$$r(\kappa) = \begin{cases} 1 & \text{if } \kappa < \kappa_{\text{min}} \\ 1 - \frac{1-r_{\text{res}}}{\kappa_{\text{max}}-\kappa_{\text{min}}}(\kappa - \kappa_{\text{min}}) & \text{if } \kappa_{\text{min}} \leq \kappa < \kappa_{\text{max}} \\ r_{\text{res}} & \text{if } \kappa \geq \kappa_{\text{max}} \end{cases} \quad (2.44)$$

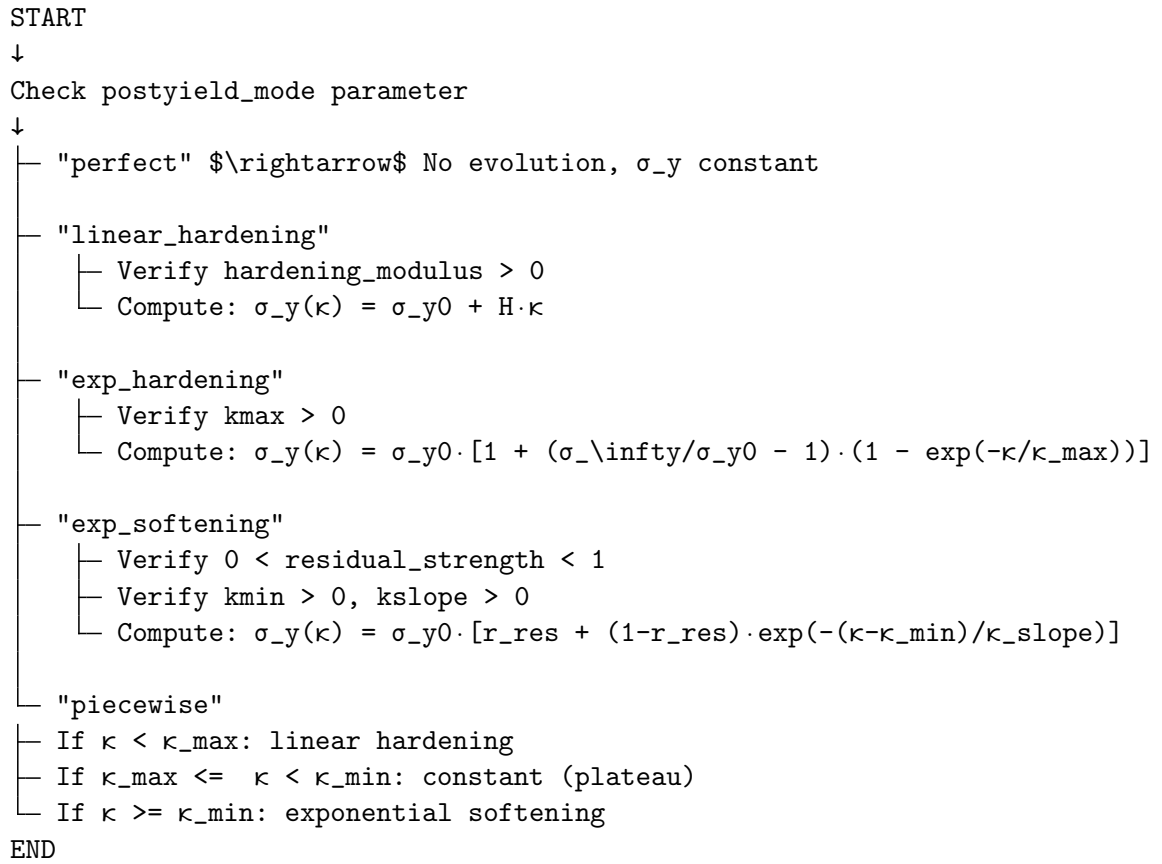
where $[\kappa_{\text{min}}, \kappa_{\text{max}}]$ is the softening range.

2.7 Post-Yield Mode Selection: Decision Logic

The implementation supports multiple post-yield behaviours selected via the `postyield_mode` parameter. Table 2.3 provides explicit logic for parameter combinations.

Table 2.3: Post-yield mode decision logic

Mode	Required Parameters	Optional Parameters	Ignored Parameters
perfect	None	eta, m	All hardening/softening
linear_hardening	hardening_modulus	kmax, eta, m	residual_strength, kmin, kslope
exp_hardening	kmax	kslope, eta, m	residual_strength, hardening_modulus
exp_softening	residual_strength, kmin, kslope	eta, m	hardening_modulus, kmax
piecewise	kmax, kmin, residual_strength	kslope, eta, m	hardening_modulus

Decision Flowchart:

Parameter Conflicts: If conflicting parameters are set, the code behaviour is:

1. **Mode takes precedence:** postyield_mode overrides individual parameter settings
2. **Unrecognized mode:** Code throws error and terminates
3. **Missing required parameter:** Code throws error with message:

```

"postyield_mode = 'exp_softening' requires
  residual_strength,
  kmin, and kslope parameters"

```

4. **Ignored parameters:** Silently ignored, no warning (for backward compatibility)

Viscoplastic Regularization (Universal): Viscoplastic regularization via `eta` and `m` is **independent** of post-yield mode and can be applied to any mode:

$$\dot{\lambda} = \frac{1}{\eta} \left\langle \frac{f}{\sigma_y} \right\rangle^m \quad (2.45)$$

- If `eta` = 0 or `m` = 0: Inviscid plasticity (rate-independent)
- If `eta` > 0 and `m` > 0: Perzyna viscoplastic regularization
- Typical values: $\eta = 0.001\text{--}0.01$ s, $m = 0.001\text{--}0.005$

Table 2.4: Recommended parameter combinations

Use Case	Mode	Parameters
Brittle minerals	<code>perfect</code>	<code>eta</code> = 0.002, <code>m</code> = 0.001 (for convergence)
Work hardening metals	<code>linear_hardening</code>	<code>hardening_modulus</code> = 1000, <code>eta</code> = 0, <code>m</code> = 0
Strain-rate sensitive	<code>exp_hardening</code>	<code>kmax</code> = 0.05, <code>eta</code> = 0.01, <code>m</code> = 0.005
Post-peak softening (coral)	<code>exp_softening</code>	<code>residual_strength</code> = 0.7, <code>kmin</code> = 0.02, <code>kslope</code> = 30, <code>eta</code> = 0.002
Complex loading path	<code>piecewise</code>	<code>kmax</code> = 0.001, <code>kmin</code> = 0.02, <code>residual</code> = 0.7, <code>kslope</code> = 30

Common Combinations:

2.8 Flow Rule and Consistency Condition

2.8.1 Associated Flow Rule

For associated plasticity, the plastic strain increment is normal to the yield surface:

$$\dot{\mathbf{E}}^p = \dot{\lambda} \frac{\partial Y}{\partial \mathbf{S}} = \dot{\lambda} \mathbf{N} \quad (2.46)$$

where:

- $\dot{\lambda} \geq 0$ is the plastic multiplier determined by the consistency condition (Section 2.8.3).
- $\mathbf{N} = \frac{\partial Y}{\partial \mathbf{S}}$ is flow direction (normalized)

The accumulated plastic strain rate (Equation 2.3) becomes:

$$\dot{\kappa} = \|\dot{\mathbf{E}}^p\| = \dot{\lambda} \|\mathbf{N}\| = \dot{\lambda} \quad (2.47)$$

where the last equality holds because $\mathbf{N}$ is normalized: $\|\mathbf{N}\| = 1$.

2.8.2 Flow Direction

Gradient of the yield function. The unnormalized gradient (normal to the yield surface) is obtained by differentiating equation (2.22) with respect to stress:

$$\frac{\partial Y}{\partial \mathbf{S}} = \frac{\mathbb{F} : \mathbf{S}}{\sqrt{\mathbf{S} : \mathbb{F} : \mathbf{S}}} + \mathbf{F} \quad (2.48)$$

where:

- First term: $\frac{\mathbb{F} : \mathbf{S}}{\sqrt{\mathbf{S} : \mathbb{F} : \mathbf{S}}}$ is the gradient of the quadric term $\sqrt{\mathbf{S} : \mathbb{F} : \mathbf{S}}$
- Second term: $\mathbf{F}$ is the gradient of the linear term $\mathbf{F} : \mathbf{S}$

Derivation. Starting from the yield function $Y = \sqrt{\mathbf{S} : \mathbb{F} : \mathbf{S}} + \mathbf{F} : \mathbf{S} - r(\kappa)$, we differentiate term by term:

$$\begin{aligned} \frac{\partial Y}{\partial \mathbf{S}} &= \frac{\partial}{\partial \mathbf{S}} \left[\sqrt{\mathbf{S} : \mathbb{F} : \mathbf{S}} \right] + \frac{\partial}{\partial \mathbf{S}} [\mathbf{F} : \mathbf{S}] = \\ &= \frac{1}{2\sqrt{\mathbf{S} : \mathbb{F} : \mathbf{S}}} \cdot 2(\mathbb{F} : \mathbf{S}) + \mathbf{F} = \\ &= \frac{\mathbb{F} : \mathbf{S}}{\sqrt{\mathbf{S} : \mathbb{F} : \mathbf{S}}} + \mathbf{F} \end{aligned} \quad (2.49)$$

where we used the chain rule for the quadric term and the fact that $r(\kappa)$ is independent of $\mathbf{S}$.

Flow direction (normalized). For the associated flow rule, the plastic strain increment is:

$$\dot{\mathbf{E}}^p = \dot{\lambda} \mathbf{N} = \dot{\lambda} \frac{\partial Y / \partial \mathbf{S}}{\|\partial Y / \partial \mathbf{S}\|} \quad (2.50)$$

where $\mathbf{N}$ is the normalized flow direction.

Special case – Symmetric yield surface ($\mathbf{F} = \mathbf{0}$). When there is no tension-compression asymmetry, the linear term vanishes and the gradient simplifies to:

$$\left. \frac{\partial Y}{\partial \mathbf{S}} \right|_{\mathbf{F}=\mathbf{0}} = \frac{\mathbb{F} : \mathbf{S}}{\sqrt{\mathbf{S} : \mathbb{F} : \mathbf{S}}} \quad (2.51)$$

This is the case for most aragonite single crystal simulations where tension and compression strengths are assumed equal, resulting in an ellipsoidal yield surface centred at the origin.

2.8.3 Consistency Condition

During plastic loading ($\dot{\lambda} > 0$), the stress state remains on the yield surface, requiring:

$$\dot{Y} = \frac{\partial Y}{\partial \mathbf{S}} : \dot{\mathbf{S}} + \frac{\partial Y}{\partial \kappa} \dot{\kappa} = 0 \quad (2.52)$$

Substituting the flow rule $\dot{\mathbf{E}}^p = \dot{\lambda} \frac{\partial Y}{\partial \mathbf{S}}$ and using $\dot{\kappa} = \left\| \dot{\mathbf{E}}^p \right\| = \dot{\lambda} \left\| \frac{\partial Y}{\partial \mathbf{S}} \right\|$, this condition determines $\dot{\lambda}$ uniquely during plastic loading.

Verification against Wolfram et al. [3]: The yield Hessian (second derivative) used in the Newton-Raphson return mapping:

$$\frac{\partial^2 f}{\partial \boldsymbol{\sigma} \partial \boldsymbol{\sigma}} = \frac{\mathbb{F} \cdot (\boldsymbol{\sigma} : \mathbb{F} : \boldsymbol{\sigma}) - (\mathbb{F} : \boldsymbol{\sigma}) \otimes (\mathbb{F} : \boldsymbol{\sigma})}{(\boldsymbol{\sigma} : \mathbb{F} : \boldsymbol{\sigma})^{3/2}} \quad (2.53)$$

which can be rewritten as:

$$\frac{\partial^2 Y}{\partial \mathbf{S}^2} = \frac{1}{\sqrt{\mathbf{S} : \mathbb{F} : \mathbf{S}}} \mathbb{F} - \frac{1}{(\mathbf{S} : \mathbb{F} : \mathbf{S})^{3/2}} (\mathbb{F} : \mathbf{S}) \otimes (\mathbb{F} : \mathbf{S}) \quad (2.54)$$

This Hessian is used in the Newton-Raphson stress update algorithm (Section 2.9) and is critical for quadratic convergence.

2.9 Return Mapping Algorithm

2.9.1 Elastic Predictor

Given total strain increment $\Delta \mathbf{E}^{n+1}$, compute trial stress assuming purely elastic step:

$$\mathbf{E}^{e, \text{trial}} = \mathbf{E}_{\text{total}}^{n+1} - \mathbf{E}^{p, n} \quad (2.55)$$

$$\mathbf{S}^{\text{trial}} = \mathbb{C} : \mathbf{E}^{e, \text{trial}} \quad (2.56)$$

Check yield:

$$Y^{\text{trial}} = \sqrt{\mathbf{S}^{\text{trial}} : \mathbb{F} : \mathbf{S}^{\text{trial}}} + \mathbf{F} : \mathbf{S}^{\text{trial}} - r(\kappa^n) \quad (2.57)$$

If $Y^{\text{trial}} \leq 0$: Step is elastic, accept trial state.

If $Y^{\text{trial}} > 0$: Plastic loading, proceed to return mapping.

2.9.2 Stage 1: Newton-Raphson Return Mapping

Solve for $\Delta \lambda$ such that:

$$Y(\mathbf{S}^{n+1}, \kappa^{n+1}) = 0 \quad (2.58)$$

Iterative scheme:

$$\mathbf{S}^{(k+1)} = \mathbf{S}^{\text{trial}} - \Delta \lambda^{(k+1)} \mathbb{C} : \mathbf{N}^{(k)} \quad (2.59)$$

$$\kappa^{(k+1)} = \kappa^n + \Delta \lambda^{(k+1)} \quad (2.60)$$

$$r^{(k+1)} = r(\kappa^{(k+1)}) \quad (2.61)$$

where $\mathbf{N}^{(k)} = \partial Y / \partial \mathbf{S}$ is the flow direction evaluated at iteration k .

Newton update for plastic multiplier:

$$\Delta \lambda^{(k+1)} = \Delta \lambda^{(k)} - \frac{Y^{(k)}}{\partial Y / \partial \Delta \lambda} \quad (2.62)$$

Converge when $|Y| < \text{tol}_{\text{abs}}$ or $|Y|/r < \text{tol}_{\text{rel}}$.

2.9.3 Stage 2: Primal CPPA (Fallback)

If Newton-Raphson fails (common during severe softening or near yield surface singularities), switch to Primal Closest Point Projection Algorithm:

Solve directly for plastic strain increment:

$$\Delta \mathbf{E}^p = \Delta \lambda \mathbf{N}(\mathbf{S}) \quad (2.63)$$

subject to:

$$Y(\mathbb{C} : (\mathbf{E}^{\text{trial}} - \Delta \mathbf{E}^p), \kappa^n + \Delta \lambda) = 0 \quad (2.64)$$

This constraint-space projection is more robust but computationally expensive (typically 100–1000 iterations).

2.9.4 Armijo Line Search

For difficult steps, use backtracking line search to improve robustness:

$$\Delta\lambda^{(k+1)} = \eta \cdot \Delta\lambda^{(k)} \quad \text{if } |Y^{(k+1)}| > |Y^{(k)}| \quad (2.65)$$

where $\eta \in [0.25, 0.75]$ is the backtracking factor.

2.10 Perzyna Viscoplastic Regularization

Optional rate-dependent term for numerical regularization. The viscoplastic contribution is added to the yield function:

$$Y_{\text{visc}}(\mathbf{S}, \kappa, \Delta\kappa, \Delta t) = Y(\mathbf{S}, \kappa) + V(\Delta\kappa, H, \Delta t) \quad (2.66)$$

where:

- $Y(\mathbf{S}, \kappa)$ = rate-independent yield function
- $V(\Delta\kappa, H, \Delta t)$ = viscoplastic regularization term
- $\Delta\kappa$ = plastic strain increment
- $H = \|\partial Y / \partial \mathbf{S}\|$ = norm of yield gradient
- Δt = time step

Effect: Allows slight overstress beyond the yield surface during loading, regularizing strain localization and softening instabilities.

2.10.1 Viscosity Modes

Six viscosity modes are available (selected via `viscosity_mode`). The regularization term V is computed as:

1. **Rate Independent** (`rate_independent`):

$$V = 0 \quad (2.67)$$

No viscoplastic regularization. Pure rate-independent plasticity.

2. **Linear** (`linear`, default):

$$V = -\frac{\eta}{\Delta t} \frac{\Delta\kappa}{H} \quad (2.68)$$

Simple linear regularization proportional to plastic strain rate $\Delta\kappa/\Delta t$.

3. **Exponential** (`exponential`):

$$V = -\frac{1}{m} \left[\exp \left(\frac{\eta}{\Delta t} \frac{\Delta\kappa}{H} \right) - 1 \right] \quad (2.69)$$

Exponential growth. Weaker regularization than linear for small rates, stronger for large rates.

4. **Logarithmic** (`logarithmic`):

$$V = -\frac{1}{m} \log \left(1 + \frac{\eta}{\Delta t} \frac{\Delta\kappa}{H} \right) \quad (2.70)$$

Logarithmic damping. Stronger regularization at higher strain rates.

5. Polynomial (polynomial):

$$V = \frac{m}{2} - \sqrt{\frac{m^2}{4} + \frac{\eta}{\Delta t} \frac{\Delta \kappa}{H}} \quad (2.71)$$

Quadratic regularization. Provides intermediate behaviour between linear and exponential.

6. Power Law (powerlaw):

$$V = -\text{sign}(\Delta \kappa) \left(\frac{\eta}{\Delta t} \frac{|\Delta \kappa|}{H} \right)^{1/m} \quad (2.72)$$

Inverse power-law. Rate sensitivity controlled by exponent $1/m$.

2.10.2 Parameter Guidelines

Physical interpretation:

The parameter η has dimensions of viscosity (MPa·s) and represents a characteristic relaxation time multiplied by a reference stress. The dimensionless ratio:

$$R = \frac{\eta}{\Delta t} \frac{\Delta \kappa}{H} \quad (2.73)$$

controls the strength of regularization. For $R \ll 1$, behaviour is nearly rate-independent. For $R \sim 1$, significant viscoplastic effects appear.

Typical values for coral aragonite:

Mode	η (MPa·s)	m (dimensionless)
Rate independent	N/A	N/A
Linear	0.001–0.01	N/A
Exponential	0.001–0.01	0.001–0.01
Logarithmic	0.001–0.01	0.001–0.01
Polynomial	0.001–0.01	0.001–0.01
Power law	0.001–0.01	0.001–0.01

2.10.3 Selection Criteria

- **No convergence issues:** Use `rate_independent`
- **Mild softening:** Use `linear` (default, most robust)
- **Severe softening with snap-back:** Use `exponential` or `polynomial`
- **Rate-dependent experiments:** Calibrate η and m from data
- **Pure numerical regularization:** Use smallest η that ensures convergence

2.10.4 Implementation Details

The viscoplastic term modifies the consistency condition during return mapping:

$$Y(\mathbf{S}^{n+1}, \kappa^{n+1}) + V(\Delta \kappa, H, \Delta t) = 0 \quad (2.74)$$

This allows temporary overstress (stress states slightly beyond the yield surface) when Δt is small or $\Delta \kappa$ is large, providing numerical regularization during softening.

2.10.5 Calibration Example

For a typical coral aragonite simulation with:

- Time step: $\Delta t = 0.01$ s
- Plastic strain increment: $\Delta \kappa = 0.0001$ per step
- Yield gradient norm: $H \approx 1.0$ (dimensionless, normalized)
- Viscosity parameter: $\eta = 0.005$ MPa·s

The regularization ratio is:

$$R = \frac{0.005}{0.01} \frac{0.0001}{1.0} = 5 \times 10^{-5} \quad (2.75)$$

This provides mild regularization ($R \ll 1$) sufficient for numerical stability without significantly affecting the rate-independent response.

2.11 Euler Angle Rotations

2.11.1 Bunge Convention (ZXZ)

Rotation from global to material frame follows the Bunge convention:

$$\mathbf{R} = \mathbf{R}_Z(\phi_2) \mathbf{R}_X(\Phi) \mathbf{R}_Z(\phi_1) \quad (2.76)$$

Angles in degrees:

- $\phi_1 \in [0, 360)$: First rotation about Z-axis
- $\Phi \in [0, 180]$: Second rotation about X'-axis
- $\phi_2 \in [0, 360)$: Third rotation about Z''-axis

2.11.2 Tensor Transformations

Stresses and strains transform as second-order tensors:

$$\mathbf{S}_{\text{material}} = \mathbf{R}^T \mathbf{S}_{\text{global}} \mathbf{R} \quad (2.77)$$

$$\mathbf{E}_{\text{material}} = \mathbf{R}^T \mathbf{E}_{\text{global}} \mathbf{R} \quad (2.78)$$

The elasticity tensor transforms as a fourth-order tensor. For a fourth-order tensor $\mathbb{C}$:

$$\mathbb{C}_{\text{material}} = \mathbf{R}^T \mathbb{C}_{\text{global}} \mathbf{R} \quad (2.79)$$

where the rotation is applied to each pair of indices. The implementation in `ComputeElasticityTensorCoupled`, which couples to per-element `AuxVariables` containing Euler angles (ϕ_1, Φ, ϕ_2) , handles this transformation automatically using MOOSE's `RotationTensor` class. See Section 5.4 for implementation details and usage guidelines.

2.11.3 Rotation Matrix and Sanity Check:

The Bunge ZZX Euler angle convention produces the rotation matrix 2.76, explicitly:

$$\mathbf{R} = \begin{bmatrix} \cos \phi_1 \cos \phi_2 - \sin \phi_1 \sin \phi_2 \cos \Phi & \sin \phi_1 \cos \phi_2 + \cos \phi_1 \sin \phi_2 \cos \Phi & \sin \phi_2 \sin \Phi \\ -\cos \phi_1 \sin \phi_2 - \sin \phi_1 \cos \phi_2 \cos \Phi & -\sin \phi_1 \sin \phi_2 + \cos \phi_1 \cos \phi_2 \cos \Phi & \cos \phi_2 \sin \Phi \\ \sin \phi_1 \sin \Phi & -\cos \phi_1 \sin \Phi & \cos \Phi \end{bmatrix} \quad (2.80)$$

Convention: Angles in **degrees** in MOOSE input files, converted to radians internally.

Sanity check:

- Identity: $(\phi_1, \Phi, \phi_2) = (0^\circ, 0^\circ, 0^\circ) \rightarrow \mathbf{R} = \mathbf{I}$
- c-axis vertical: $(\phi_1, \Phi, \phi_2) = (0^\circ, 0^\circ, 0^\circ) \rightarrow$ crystal c-axis parallel to global Z
- 90° rotation: $(\phi_1, \Phi, \phi_2) = (90^\circ, 0^\circ, 0^\circ) \rightarrow$ crystal rotated about Z-axis

Validation: Apply known rotation, verify stress transformation:

$$\tilde{\boldsymbol{\sigma}} = \mathbf{R}^T \boldsymbol{\sigma} \mathbf{R} \quad (2.81)$$

Chapter 3

Cohesive Zone Model for Protein-Aragonite Interfaces

The cohesive zone model (CZM) governs the mechanical response of protein-water interfaces and grain boundary debonding in coral aragonite structures. This Chapter provides the mathematical formulation; implementation details and parameter reference are given in Section 6.

3.1 Relationship to Bulk Plasticity Model

3.1.1 Overview

The cohesive zone model (CZM) operates **independently** from the bulk plasticity model described in Sections 2-7, with interface nodes serving as the coupling mechanism. This section clarifies the interaction between bulk and interface behaviour.

3.1.2 Mechanical Coupling

Sequential Solution: Within each Newton iteration:

1. **Bulk stress update:**

- Bulk elements (grains) compute stress using `OrthotropicPlasticityStressUpdate`
- Plastic strain accumulates: $\kappa^{n+1} = \kappa^n + \Delta\lambda$
- Internal variables updated in material frame

2. **Interface traction update:**

- CZM elements compute traction using `HomogenizedExponentialCZM`
- Displacement jump extracted: $\llbracket \mathbf{u} \rrbracket = \mathbf{u}^+ - \mathbf{u}^-$
- Damage accumulates: $D^{n+1} = f(D^n, \delta_{\text{eff}})$

3. **Residual assembly:**

- Bulk residual: $\mathbf{R}_{\text{bulk}} = \int_{\Omega} \nabla \mathbf{w} : \boldsymbol{\sigma} dV - \mathbf{F}_{\text{ext}}$
- Interface residual: $\mathbf{R}_{\text{int}} = \int_{\Gamma} \mathbf{w} \cdot \mathbf{T} dS$
- Total: $\mathbf{R} = \mathbf{R}_{\text{bulk}} + \mathbf{R}_{\text{int}}$

4. **Convergence check:**

- If $\|\mathbf{R}\| < \text{tol}$: Accept step
- Else: Update displacement, repeat

Frame Handling:

- **Bulk plasticity:** Computed in **material (crystal) frame**
- **CZM tractions:** Computed in **local interface frame**
- **Coupling:** Both transformed to **global frame** for residual assembly

Interface local frame definition:

- $\mathbf{n}$: Normal to interface (outward from grain 1)
- $\mathbf{s}$: First tangent (in-plane)
- $\mathbf{t}$: Second tangent (out-of-plane)
- Orthonormal: $\mathbf{n} \cdot \mathbf{s} = 0$, $\mathbf{n} \cdot \mathbf{t} = 0$, $\mathbf{s} \cdot \mathbf{t} = 0$

3.1.3 No Direct Coupling of Internal Variables

Important: Bulk plastic strain κ and interface damage D are **independent**:

- Plastic strain does NOT directly affect interface damage
- Interface damage does NOT directly affect yield strength
- Coupling is purely kinematic (through displacement field)

Physical justification:

- Bulk plasticity: Dislocation motion within grains
- Interface damage: Bond rupture at grain boundaries
- Different physical mechanisms, different length scales

3.1.4 Indirect Coupling Through Stress State

While internal variables are independent, stress state couples behaviours:

1. Plasticity increases interface loading:

- Bulk yielding $\rightarrow$ stress redistribution
- Stresses concentrate at grain boundaries
- Increased interface displacement jumps
- Earlier interface damage initiation

2. Interface damage increases bulk stress:

- Interface softening $\rightarrow$ load transfer to bulk
- Remaining grains carry higher load
- May trigger additional yielding

Competition: Intergranular vs Transgranular Failure The final failure mode depends on relative strengths:

- **Strong interfaces, weak bulk:** Transgranular fracture
 - Interface: $\sigma_{\text{int}} = 700 \text{ MPa}$, $D < 0.1$
 - Bulk: $\sigma_y = 400 \text{ MPa}$, $\kappa > 0.05$ (extensive plasticity)
 - Failure: Through grains
- **Weak interfaces, strong bulk:** Intergranular fracture
 - Interface: $\sigma_{\text{int}} = 400 \text{ MPa}$, $D > 0.8$
 - Bulk: $\sigma_y = 700 \text{ MPa}$, $\kappa < 0.01$ (minimal plasticity)
 - Failure: Along grain boundaries
- **Matched strengths:** Mixed mode (realistic for coral)
 - Competing mechanisms
 - Failure path determined by microstructure and loading

3.1.5 Numerical Considerations

Convergence:

- CZM softening can cause non-convergence if too abrupt
- Viscoplastic regularization in CZM (`damage_viscosity`) helps
- Bulk viscoplasticity (`eta`, `m`) also improves convergence
- Timestep adaptation essential during damage onset

Mesh Requirements:

- **Bulk mesh:** Sufficient refinement to capture plastic zones
- **Interface mesh:** One-to-one correspondence with bulk boundaries
- **Compatibility:** Interface nodes shared by adjacent grains

3.1.6 Input File Structure

Bulk and interface materials coexist in the same input file:

```
[Materials]
# ===== BULK PLASTICITY =====
[plasticity]
  type = OrthotropicPlasticityStressUpdate
  # ... bulk parameters ...
  block = 'grain_1 grain_2 grain_3' # Apply to grain blocks
[]

# ===== INTERFACE CZM =====
[czm_interface]
  type = HomogenizedExponentialCZM
  # ... interface parameters ...
  boundary = 'grain_1_grain_2 grain_2_grain_3' # Apply to interfaces
[]
[]
```

Key difference:

- Plasticity uses `block` = (3D elements)
- CZM uses `boundary` = (2D interface elements)

3.1.7 Summary

Table 3.1: Bulk plasticity vs CZM comparison

Aspect	Bulk Plasticity	CZM
Location	Within grains (3D)	At interfaces (2D)
Frame	Material (crystal)	Local interface
Internal variable	Plastic strain κ	Damage D
Failure mechanism	Dislocation motion	Bond rupture
Direct coupling	None	None
Indirect coupling	Via stress field	Via stress field
Element type	Bulk (hex, tet)	Surface (quad, tri)

The following sections describe the CZM formulation in detail.

3.2 Multiscale Homogenization Framework**3.2.1 Physical Motivation**

Instead of sampling a discrete MD contact population (area-dependent, stochastic), the interface heterogeneity is modelled through a **quality factor**

$$f \sim \mathcal{N}(1, \sigma_q^2) \quad (3.1)$$

where σ_q is the within-QP coefficient of variation (default 0.10). The quality factor scales both peak strength and critical opening proportionally:

$$T_{\text{peak}}(f) = f \cdot T_{\text{peak},0} \quad (3.2)$$

$$\delta_0(f) = f \cdot \delta_{0,0} \quad (3.3)$$

Weak contacts ($f < 1$) fail at smaller openings; strong contacts ($f > 1$) carry load further. A **spatial quality field** additionally shifts the local mean deterministically per QP from its element-ID and QP index (`spatial_quality_std_dev`), desynchronising failure across the interface without per-QP random state.

3.2.2 Gauss-Hermite Quadrature

The macroscopic traction is the expectation over the quality distribution:

$$\mathbf{T}_{\text{macro}} = \sum_{k=1}^5 w_k \mathbf{T}_{\text{single}}(\boldsymbol{\delta}; f_k) (1 - D_k) \quad (3.4)$$

evaluated with a 5-point Gauss-Hermite (GH) rule (weights sum to 1). Table 3.2 lists nodes, quality factors, and weights; the $(1 - D_k)$ factor is inside the sum to give $\langle T(1 - D) \rangle$ rather than the Jensen approximation $\langle T \rangle(1 - \langle D \rangle)$.

Each class carries its own stateful damage $D_k(t)$ updated independently. Because $p(f)$ normalises to 1, the result is area-independent (mesh independence by construction) and satisfies Hill-Mandel consistency [6].

Table 3.2: 5-point Gauss-Hermite quadrature (probabilist’s convention, weights sum to 1). f_k shown for $\sigma_q = 0.10$.

k	z_k	f_k	w_k	Role
1	-2.857	0.714	0.01126	Weakest class
2	-1.356	0.864	0.22208	Weak class
3	0	1.000	0.53333	Mean class
4	+1.356	1.136	0.22208	Strong class
5	+2.857	1.286	0.01126	Strongest class

3.3 Exponential Traction-Separation Law

3.3.1 Single Contact behaviour

Each MD-scale contact follows an exponential traction-separation law derived from molecular dynamics simulations [7, 8]. The undamaged traction for contact i is:

$$T_i^{\text{undamaged}}(\delta_{\text{eff}}) = T_{\text{peak},i} e r^\nu \exp(-r^\nu) \quad (3.5)$$

where:

- $T_{\text{peak},i}$ is the peak strength for contact i (distributed, see Section 3.5)
- $r = \delta_{\text{eff}}/\delta_{0,i}$ is the normalized effective opening
- $\delta_{0,i}$ is the characteristic length for contact i (distributed)
- $\nu = \mu$ during loading phase ($r < 1$), $\nu = \eta$ during softening phase ($r \geq 1$)
- $\mu, \eta > 0$ are exponents controlling loading and softening shape, respectively
- $e = \exp(1) \approx 2.71828$ ensures T_{peak} occurs at $r = 1$

3.3.2 Effective Opening Displacement

The effective opening δ_{eff} combines normal and tangential components:

$$\delta_{\text{eff}} = \sqrt{\delta_n^2 + \delta_s^2 + \delta_t^2} \quad (3.6)$$

where $\delta_n, \delta_s, \delta_t$ are normal and two tangential displacement jumps in the local interface coordinate system.

3.3.3 Mixed-Mode Failure: Power-Law Criterion

For combined loading, the peak interface strength varies with mode mixity following a **power-law criterion** [9, 10]:

$$\left(\frac{\sigma_n}{\sigma_n^u}\right)^\alpha + \left(\frac{\tau_s}{\tau_s^u}\right)^\alpha + \left(\frac{\tau_t}{\tau_t^u}\right)^\alpha = 1 \quad (3.7)$$

where $\sigma_n^u, \tau_s^u, \tau_t^u$ are the pure mode strengths (from MD simulations) and α is the coupling exponent. This framework generalizes the linear Mohr-Coulomb criterion ($\alpha = 1$) used in molecular dynamics interface characterization [11].

Implementation: Quadric Criterion ($\alpha = 2$) In this work, we adopt the **quadric criterion** with $\alpha = 2$, yielding an ellipsoidal failure surface:

$$\left(\frac{\sigma_n}{\sigma_n^u}\right)^2 + \left(\frac{\tau_s}{\tau_s^u}\right)^2 + \left(\frac{\tau_t}{\tau_t^u}\right)^2 = 1 \quad (3.8)$$

Rearranged for displacement jump ratios in the continuum implementation:

$$T_{\text{peak},i} = \sqrt{\left(\sigma_{n,i} \frac{|\delta_n|}{\delta_{\text{total}}}\right)^2 + \left(\tau_{s,i} \frac{|\delta_s|}{\delta_{\text{total}}}\right)^2 + \left(\tau_{t,i} \frac{|\delta_t|}{\delta_{\text{total}}}\right)^2} \quad (3.9)$$

where $\delta_{\text{total}} = \sqrt{\delta_n^2 + \delta_s^2 + \delta_t^2}$ is the effective opening displacement.

Comparison of Linear vs Quadric Criteria Figure 3.1 compares failure surfaces for different exponents. The transition from $\alpha = 1$ (molecular dynamics, nanoscale) to $\alpha = 2$ (continuum, microscale) reflects **statistical homogenization** over $\sim 10^8$ contacts with varying orientations and strengths.

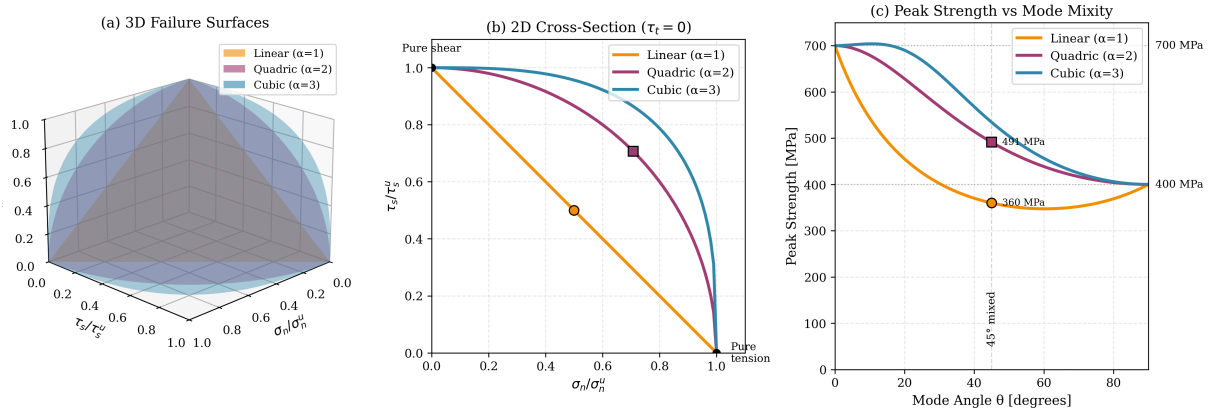


Figure 3.1: Mixed-mode failure surface comparison for power-law criteria. (a) 3D surfaces showing transition from planar (linear, $\alpha = 1$) to ellipsoidal (quadric, $\alpha = 2$) to nearly rectangular (cubic, $\alpha = 3$). (b) 2D cross-section in σ_n - τ_s plane demonstrating strength differences at 45° mixed-mode loading. (c) Peak interface strength versus mode angle θ from 0° (pure tension, $T = \sigma_n^u = 700$ MPa) to 90° (pure shear, $T = \tau_s^u = 400$ MPa). At 45° mixed-mode ($\theta = 45^\circ$), linear criterion predicts 360 MPa (circle) while quadric predicts 491 MPa (square), demonstrating 36% higher strength. The mode angle θ is defined as $\theta = \arctan(\delta_s/\delta_n)$ where δ_n and δ_s are normal and tangential displacement jump components.

Table 3.3 quantifies the difference between linear (MD-derived) and quadric (continuum) criteria for representative mixed-mode loading states.

Key Observations:

- **Pure modes identical:** Both criteria recover the same strength for pure tension (σ_n^u) and pure shear (τ_s^u, τ_t^u), ensuring consistency with MD-derived input parameters.
- **Mixed modes diverge:** The quadric criterion predicts 36–69% higher strength under combined loading compared to the linear criterion, with larger differences for more complex mode mixity.
- **Conservative vs realistic:** The linear criterion ($\alpha = 1$) is more conservative but may underestimate interface strength when multiple bonds are loaded simultaneously. The quadric criterion ($\alpha = 2$) accounts for load redistribution among contacts.

Table 3.3: Peak strength comparison for mixed-mode loading ($\sigma_n^u = 700$ MPa, $\tau_s^u = \tau_t^u = 400$ MPa)

Loading Mode	Jump Ratio	Linear	Quadric	Difference
	$\delta_n : \delta_s : \delta_t$	$\alpha = 1$	$\alpha = 2$	
Pure tension	1.00:0.00:0.00	700 MPa	700 MPa	0%
Pure shear-s	0.00:1.00:0.00	400 MPa	400 MPa	0%
Pure shear-t	0.00:0.00:1.00	400 MPa	400 MPa	0%
45° mixed (n-s)	0.71:0.71:0.00	360 MPa	491 MPa	+36%
Biaxial shear	0.00:0.71:0.71	283 MPa	400 MPa	+41%
Tri-axial (1:1:1)	0.58:0.58:0.58	270 MPa	455 MPa	+69%

Justification for Quadric Criterion in Continuum modelling The adoption of $\alpha = 2$ for continuum-scale modelling is justified by:

1. **Statistical homogenization:** The linear criterion ($\alpha = 1$) represents individual nanoscale contacts in MD simulations. When averaging over $N \approx 10^8$ contacts per continuum integration point, each with slightly different orientations and strengths, the effective criterion naturally transitions toward quadric form [9].
2. **Numerical stability:** The quadric surface provides smooth gradients and C^1 continuity, avoiding convergence issues associated with sharp corners in the linear criterion's planar surface.
3. **Energy-based interpretation:** Quadric criteria naturally relate to strain energy density, consistent with Griffith-type fracture where energy release rate scales quadratically with stress.
4. **Engineering precedent:** Power-law criteria with $\alpha = 2$ are extensively validated in composite delamination modelling [10, 12], aerospace structural analysis, and adhesive joint design.
5. **Scale-appropriate:** The continuum model represents effective behaviour at 100 nm scale (δ_0), not individual atomic bonds at 0.35 nm (MD scale). The quadric criterion captures emergent homogenized response.

Interaction with Mode-Dependent Threshold: The power-law strength criterion ($\alpha = 2$) determines the **peak traction** that can be sustained, while the mode-dependent threshold $\delta_{0,\text{mixed}}(\varphi)$ determines **when damage begins**. These are independent effects:

$$T_{\text{peak},i}(\varphi) = \sqrt{\sigma_{n,i}^2 r_n^2 + \tau_{s,i}^2 r_s^2 + \tau_{t,i}^2 r_t^2} \quad (\text{strength}) \quad (3.10)$$

$$\delta_{0,\text{mixed}}(\varphi) = \frac{\delta_{0,n} \cdot \delta_{0,t} \cdot \sqrt{1 + \varphi^2}}{\sqrt{\delta_{0,t}^2 + \varphi^2 \delta_{0,n}^2}} \quad (\text{threshold}) \quad (3.11)$$

Both effects conspire to make mixed-mode failure realistic:

- Higher strength (quadric vs linear): +36% at 45°
- Earlier damage (mode-dependent threshold): -29% at 45°
- Net: Stronger peak but earlier onset of softening

This captures the experimental observation that mixed-mode interfaces can carry high loads but are more prone to premature damage initiation.

Relationship to Molecular Dynamics The molecular dynamics simulations [11] employed a linear Mohr-Coulomb criterion ($\alpha = 1$) appropriate for single-bond failure:

$$\frac{\sigma_{nn}}{\sigma_{nn}^u} + \frac{\tau_{ns}}{\tau_{ns}^u} + \frac{\tau_{nt}}{\tau_{nt}^u} = 1 \quad (\text{MD, single bond}) \quad (3.12)$$

This criterion forms a triangular planar surface in stress space, intersecting axes at pure mode strengths. The transition to the quadric criterion can be understood through statistical averaging:

$$\left\langle \frac{1}{T_{\text{peak},i}} \right\rangle^{-1} \approx \left[\left\langle \left(\frac{1}{T_{\text{peak},i}} \right)^2 \right\rangle \right]^{-1/2} \quad \text{as } N \rightarrow \infty \quad (3.13)$$

This is analogous to:

- Hill-Mises effective yield criteria in polycrystal plasticity (single crystal $\rightarrow$ aggregate)
- Tsai-Wu criteria in fiber composites (fiber $\rightarrow$ laminate)
- Effective elastic moduli in porous/heterogeneous media

Validation The choice of $\alpha = 2$ can be validated through:

- Large-scale MD simulations with many bonds under mixed loading
- Experimental interface tests at varying mode angles
- Direct fitting of α to homogenized MD results (e.g., 100 bonds simultaneously)
- Comparison with multi-scale simulations spanning nano-to-microscale

For the present study, $\alpha = 2$ provides a physically motivated, numerically robust, and literature-validated choice for continuum-scale cohesive zone modelling.

3.4 Damage Evolution

3.4.1 Damage Variable

Following Wang et al. [8], damage evolves according to:

$$D_i(r) = \begin{cases} 0 & \text{if } r \leq 1 \\ 1 - r^{\eta-1} \exp[1 - r^\eta] & \text{if } r > 1 \end{cases} \quad (3.14)$$

where $r = \delta_{\text{eff}}/\delta_{0,i}$ is computed using the contact-specific characteristic length $\delta_{0,i}$.

Properties:

- $D = 0$ at peak traction ($r = 1$, i.e., $\delta_{\text{eff}} = \delta_{0,i}$)
- D increases monotonically for $r > 1$ (softening phase)
- $D \rightarrow 1$ as $r \rightarrow \infty$ (complete failure)
- Small η (e.g., $\eta = 0.15$) creates very wide damage zone ($\delta_c \approx 50\delta_0$)

3.4.2 Physical Interpretation of Small η

For small values of the softening exponent (e.g., $\eta = 0.15$), the damage zone extends over distances much larger than the characteristic length:

$$\delta_c \approx (50\text{--}100) \times \delta_0 \quad (3.15)$$

With $\delta_0 = 100$ nm, this yields $\delta_c \approx 5\text{--}10$ μm . This wide damage zone is physically meaningful: it represents the **progressive, sequential breaking of bonds** observed in molecular dynamics simulations, where interface failure occurs gradually over thousands of individual protein-aragonite rupture events rather than as a single catastrophic event. The small exponent captures the statistical nature of bond breaking across each quadrature point.

3.4.3 Damaged Traction

The final traction accounting for damage is:

$$T_i(\delta_{\text{eff}}) = (1 - D_i) T_i^{\text{undamaged}}(\delta_{\text{eff}}) \quad (3.16)$$

Combining Equations 3.5, 3.14, and 3.16:

$$T_i = \begin{cases} T_{\text{peak},i} e^{r^\mu} \exp(-r^\mu) & \text{if } r \leq 1 \text{ (loading)} \\ T_{\text{peak},i} e^{r^{\eta-1}} \exp[1 - r^\eta] r^\eta \exp(-r^\eta) & \text{if } r > 1 \text{ (softening)} \end{cases} \quad (3.17)$$

3.4.4 Irreversibility: History Variable κ

A history variable $\kappa = \max_{t' \leq t} \delta_{\text{eff}}(t')$ tracks the maximum effective opening ever reached per QP. Target damage is evaluated at $\max(\delta_{\text{eff}}, \kappa)$, ensuring no healing during unloading:

$$r_k^{\text{eff}} = \frac{\max(\delta_{\text{eff}}, \kappa)}{\delta_{0,k}} \quad (3.18)$$

κ is stored as stateful property `qp_kappa` per quadrature point.

3.4.5 Smooth Damage Activation

The hard branch at $r = 1$ is replaced by a smooth tanh transition to avoid a $dD/d\delta$ kink that stalls Newton near initiation:

$$D_{\text{target},k} = D_{\text{hard},k}(r_k) \cdot \frac{1}{2} \left(1 + \tanh \left[\frac{r_k - 1}{\Delta_{\text{act}}} \right] \right) \quad (3.19)$$

where `damage_activation_width` $\Delta_{\text{act}} = 0.02$ (default).

3.4.6 Viscoplastic Damage Regularization

Damage is advanced via a Perzyna-type viscous update:

$$D_k^{n+1} = D_k^n + \frac{\Delta t}{\tau_d} (D_{\text{target},k} - D_k^n), \quad D_k^{n+1} \geq D_k^n \quad (3.20)$$

The monotonicity clamp is enforced in the update and reflected in the viscous tangent: when clamped ($D_{\text{target},k} \leq D_k^n$), the viscous factor is set to zero to maintain Jacobian consistency. Typical: $\tau_d = 0.5\text{--}2.5$ s.

3.4.7 Softening Exponent Constraint

$$\eta < \mu \quad \Rightarrow \quad \frac{dT_{\text{macro}}}{d\delta_{\text{eff}}} > 0 \text{ throughout loading} \quad (3.21)$$

When $\eta > \mu$ the loading branch develops a negative tangent, causing snap-back under displacement control. Production uses $\eta < \mu$. The physically-fitted value ($\eta \approx 1.07$ from MD for the first peak) reflects only the first debonding cycle and is not compatible with displacement-controlled FE loading.

3.4.8 Mode-Dependent Damage Initiation: Generalized Formulation

Anisotropic Damage Threshold Real interfaces exhibit different susceptibility to damage under normal versus tangential loading. To capture this, we introduce separate damage initiation thresholds for mode I (normal) and mode II (tangential) loading:

- $\delta_{0,n}$ = Normal separation at damage initiation
- $\delta_{0,t}$ = Tangential separation at damage initiation (*here $\delta_{0,t} \equiv \delta_{0,s}$)

The generalized mixed-mode threshold [8] is 3.11 with the mode ratio $\varphi = \delta_t/\delta_n$.

Special Cases:

1. **Pure normal loading** ($\varphi = 0, \delta_t = 0$):

$$\delta_{0,\text{mixed}}(0) = \delta_{0,n} \quad (3.22)$$

2. **Pure shear loading** ($\varphi \rightarrow \infty, \delta_n \rightarrow 0$):

$$\lim_{\varphi \rightarrow \infty} \delta_{0,\text{mixed}}(\varphi) = \delta_{0,t} \quad (3.23)$$

3. **Isotropic case** ($\delta_{0,n} = \delta_{0,t} = \delta_0$):

$$\delta_{0,\text{mixed}}(\varphi) = \frac{\delta_0 \cdot \delta_0 \cdot \sqrt{1 + \varphi^2}}{\sqrt{\delta_0^2 + \varphi^2 \delta_0^2}} = \frac{\delta_0^2 \sqrt{1 + \varphi^2}}{\delta_0 \sqrt{1 + \varphi^2}} = \delta_0 \quad (3.24)$$

Physical Interpretation: For isotropic interfaces ($\delta_{0,n} = \delta_{0,t} = \delta_0$), the mode-dependent behaviour arises naturally from the effective displacement calculation 3.6.

Damage initiates when $\delta_{\text{eff}} \geq \delta_0$ (constant threshold). Under 45° mixed-mode loading ($\delta_n = \delta_t$), this requires:

$$\sqrt{2\delta_n^2} \geq \delta_0 \quad \Rightarrow \quad \delta_n \geq \frac{\delta_0}{\sqrt{2}} = 0.707\delta_0 \quad (3.25)$$

Thus, damage initiates at 29% less individual displacement per mode, without modifying the threshold value. The generalized formulation (Equation 3.11) extends this to anisotropic cases where normal and tangential susceptibilities differ.

Example: 45° Mixed-Mode Loading For equal normal and tangential displacement ($\varphi = 1$):

$$\delta_{0,\text{mixed}}(1) = \frac{\delta_{0,n} \cdot \delta_{0,t} \cdot \sqrt{2}}{\sqrt{\delta_{0,t}^2 + \delta_{0,n}^2}} \quad (3.26)$$

Isotropic case ($\delta_{0,n} = \delta_{0,t} = 100$ nm):

$$\delta_{0,\text{mixed}}(1) = \frac{100 \cdot 100 \cdot \sqrt{2}}{\sqrt{100^2 + 100^2}} = \frac{10000\sqrt{2}}{100\sqrt{2}} = 100/\sqrt{2} = 70.7 \text{ nm} \quad (3.27)$$

(29% reduction)

Anisotropic case ($\delta_{0,n} = 120$ nm, $\delta_{0,t} = 80$ nm):

$$\delta_{0,\text{mixed}}(1) = \frac{120 \cdot 80 \cdot \sqrt{2}}{\sqrt{80^2 + 120^2}} = \frac{9600\sqrt{2}}{\sqrt{20800}} = \frac{9600\sqrt{2}}{144.2} \approx 94.1 \text{ nm} \quad (3.28)$$

The anisotropic threshold lies between the two pure-mode values, weighted by the mode ratio.

Damage Evolution: The damage law uses this generalized threshold:

$$D = \begin{cases} 0 & \text{if } \delta_{\text{eff}} < \delta_{0,\text{mixed}}(\varphi) \\ 1 - \left(\frac{\delta_{\text{eff}}}{\delta_{0,\text{mixed}}} \right)^{\eta-1} \exp \left[1 - \left(\frac{\delta_{\text{eff}}}{\delta_{0,\text{mixed}}} \right)^{\eta} \right] & \text{if } \delta_{\text{eff}} \geq \delta_{0,\text{mixed}}(\varphi) \end{cases} \quad (3.29)$$

Numerical Stability: Division by δ_n is regularised via $\varphi_{\text{den}} = \max(\delta_{n,+}, 10^{-6} \delta_{0,n})$, where $\delta_{n,+}$ is the smooth Macaulay bracket. No hard clamp on δ_n is applied. The mixed-mode function returns $\delta_{0,n}$ directly for $\varphi < 10^{-12}$ (near-pure normal loading) and $\delta_{0,t}$ for $\varphi > 10^6$ (near-pure shear), both handled as threshold returns in `computeMixedModeDelta0`.

Table 3.4: δ_0 configuration options

Configuration	Input Parameters	behaviour	Use Case
Isotropic (default)	<code>delta_0</code> = 100e-6	$\delta_{0,n} = \delta_{0,t} = 100$ nm Wang et al. [8, Eq. 25]	Standard case
Anisotropic	<code>delta_0_normal</code> = 120e-6 <code>delta_0_tangent</code> = 80e-6	$\delta_{0,n} = 120$ nm $\delta_{0,t} = 80$ nm Wang et al. [8, Eq. 25]	Different mode susceptibilities
Override normal only	<code>delta_0</code> = 100e-6 <code>delta_0_normal</code> = 150e-6	$\delta_{0,n} = 150$ nm $\delta_{0,t} = 100$ nm	Stronger adhesion
Override tangent only	<code>delta_0</code> = 100e-6 <code>delta_0_tangent</code> = 120e-6	$\delta_{0,n} = 100$ nm $\delta_{0,t} = 120$ nm	Rougher interface

Implementation Modes: Note: When `delta_0_normal` or `delta_0_tangent` are specified, they **override** the base `delta_0` value for that mode. The Wang equation is **always applied**, whether isotropic or anisotropic.

3.5 Quality Factor Distributions

3.5.1 Within-QP Heterogeneity

Contact heterogeneity at a single QP is captured by `quality_std_dev` = σ_q (default 0.10), replacing three variation parameters (`strength_std_dev`, `delta0_std_dev`, `initial_damage_max`). A single factor coupled to both T_{peak} and δ_0 captures the physical co-variation: a weaker-than-average contact also fails at a smaller opening.

```
[czm]
  type = HomogenizedExponentialCZM
  quality_std_dev = 0.10      % 10% CV; 0 = deterministic single class
[]
```

3.5.2 Across-QP Spatial Heterogeneity

Without spatial variation every interface element reaches its damage threshold simultaneously. A spatial quality field shifts each QP's mean *deterministically* from `spatial_random_seed` combined with element-ID and QP index:

$$f_{\text{sp}}^{(\text{QP})} \sim \mathcal{N}(1, \sigma_{\text{sp}}^2) \quad (3.30)$$

so the full per-class quality factor becomes $f_k^{(\text{QP})} = f_{\text{sp}}^{(\text{QP})} \cdot (1 + \sigma_q z_k)$. The field is bit-identical across runs and MPI partition counts.

```
[czm]
  quality_std_dev      = 0.10
  spatial_quality_std_dev = 0.15      % across-QP spatial variation
  spatial_random_seed   = 1234      % fixed; deterministic field
[]
```

Table 3.5: Monte Carlo sampling vs. Gauss-Hermite homogenisation.

Property	Monte Carlo	GH (current)
Evaluations per QP	up to 10,000	5
Deterministic	No	Yes
Area-dependent	Yes	No
Parameters	3 + <code>random_seed</code>	2 + <code>spatial_random_seed</code>

3.6 Progressive Interface Failure

Quality factor dispersion produces emergent progressive failure: class 1 ($f = 0.71$ for $\sigma_q = 0.10$) enters softening at 71% of the mean-class critical opening, while class 5 ($f = 1.29$) is still loading when the mean class peaks. The macroscopic traction-separation envelope is smooth and continuously differentiable, with peak reduction $\approx 1\text{--}3\%$ relative to the deterministic case.

3.7 MD-Derived Interface Parameters

The implementation uses **physical MD values** directly for both strength and critical opening, without inflating δ_0 to an effective continuum scale.

Table 3.6: MD-derived CZM parameters (Kvashin et al. [11])

	Protein	Water	Unit
σ_n^u	626 ± 53	742 ± 29	MPa
τ_s^u	374 ± 39	354 ± 61	MPa
$\delta_{0,n}$	191	224	pm
$\delta_{0,t}$	217	182	pm
μ	0.916	1.224	—

Convergence at physical δ_0 : Prior implementations required inflating δ_0 to ~ 100 nm to avoid ill-conditioning. This is no longer necessary: (1) smooth tanh activation (Section 3.4) removes the Newton-stalling kink at $r = 1$; (2) the $\eta \leq \mu$ condition ensures a positive loading-branch tangent; (3) viscous regularization limits per-step stiffness change. The landmark production run confirmed convergence: 1008/1008 Newton solves to $t = 10$, zero cutbacks, peak traction 625 MPa against MD input 626 MPa.

Chapter 4

MOOSE Installation

4.1 Why Conda MOOSE?

Traditional manual build: 2–4 hours

- Build PETSc from source: 1–2 hours
- Build libMesh: 30–60 minutes
- Build MOOSE test suite: 30–60 minutes
- Requires compiler toolchain, MPI, dependencies

Conda installation: 5–10 minutes

- Pre-compiled binaries for all dependencies
- Consistent environment across machines
- Easy updates and version management
- No complex dependency resolution

4.2 Installation Steps

4.2.1 Step 1: Install Miniconda

Linux / Windows WSL:

```
wget https://repo.anaconda.com/miniconda/Miniconda3-latest-Linux-x86_64.sh
sh
bash Miniconda3-latest-Linux-x86_64.sh
# Follow prompts, answer 'yes' to initialization
source ~/.bashrc
```

macOS:

```
wget https://repo.anaconda.com/miniconda/Miniconda3-latest-MacOSX-x86_64.sh
sh
bash Miniconda3-latest-MacOSX-x86_64.sh
# Follow prompts, answer 'yes' to initialization
source ~/.bash_profile
```

Verify:

```
conda --version
# Should show: conda 24.x.x or later
```

4.2.2 Step 2: Create MOOSE Environment

```
# Add INL's public channel
conda config --add channels https://conda.software.inl.gov/public
# Create conda environment and install MOOSE dependencies
conda create -n moose moose-dev -c conda-forge
```

This installs:

- MOOSE framework (pre-compiled)
- PETSc (pre-configured)
- libMesh (pre-configured)
- MPI compilers (mpicxx, mpicc, mpifort)
- All required dependencies

Time: 5–10 minutes depending on network.

4.2.3 Step 3: Activate Environment

```
conda activate moose
```

CRITICAL: Always activate this environment before any MOOSE work!

4.2.4 Step 4: Verify Installation

```
# Check tools are available
which mpicxx
which mpirun

# Check versions
mpicxx --version
python --version

# Test MPI
mpirun -n 2 echo "MPI works!"
# Should print "MPI works!" twice
```

4.2.5 Step 5: (Important) Clone MOOSE for building references

```
cd ~/projects
git clone https://github.com/idaholab/moose.git
cd moose
git checkout master # or specific release tag

# Set MOOSE_DIR (optional, for reference only)
echo "export MOOSE_DIR=~/projects/moose" >> ~/.bashrc
source ~/.bashrc
```

Note: This is optional for general users. Conda package includes everything needed to build apps.

4.3 Environment Management

4.3.1 Daily Commands

```
# Activate MOOSE environment (ALWAYS FIRST!)
conda activate moose

# Check you're in the right environment
echo $CONDA_DEFAULT_ENV
# Should output: moose

# Deactivate when done
conda deactivate
```

4.3.2 Installing Additional Packages

```
# Install IN the moose environment, not base!
conda activate moose
conda install matplotlib numpy pandas scipy

# Or using pip if needed
pip install some-package
```

4.3.3 Updating MOOSE

```
conda activate moose
conda update moose-dev -c conda-forge
```

4.3.4 Managing Multiple Versions

```
# Stable version for production
conda create -n moose-stable moose-dev=<version> -c conda-forge

# Latest version for development
conda create -n moose-dev moose-dev -c conda-forge

# Switch between them
conda activate moose-stable
conda activate moose-dev
```

4.4 Shell Configuration

Add to ~/.bashrc (Linux/WSL) or ~/.bash_profile (macOS):

```
# MOOSE conda environment
alias moose-on='conda activate moose'
alias moose-off='conda deactivate'

# Optional: auto-activate in project directories
```

Reload:

```
source ~/.bashrc # or ~/.bash_profile on macOS
```

Chapter 5

Building Your Application

5.1 Overview

This chapter covers two scenarios for building and using the aragonite application:

1. **Using the GitHub repository** (recommended) – Clone, build, test
2. **Building from source files** (for development) – Use MOOSE’s `stork.sh` script to create application skeleton

Note: The GitHub repository approach is recommended once the code is published. During development, use Scenario 2.

5.2 Scenario 1: Using the GitHub Repository

Recommended for: Users who want to use the published materials

Prerequisites:

- Conda MOOSE environment installed and activated (Chapter 4)
- Git installed (usually included with conda)

5.2.1 Clone Repository

```
# Activate conda environment
conda activate moose

# Clone repository
cd ~/projects
git clone https://gitlab.tu-clausthal.de/iww/tuc-fe-moose
cd aragonite

# Repository contents:
#   - Makefile (pre-configured)
#   - Source files (src/materials/*.C)
#   - Header files (include/materials/*.h)
#   - Test suite (test/validation/*)
#   - Example inputs (examples/*)
#   - Documentation (doc/*)
```

5.2.2 Build

```
# Build application (one command!)
make -j8

# Expected output:
#   ...
#   Linking Library...
#   Linking Executable .../projects/aragonite/aragonite-opt

# Verify build
ls -lh aragonite-opt
./aragonite-opt --version
```

Build time:

- First build: 2–5 minutes
- Incremental builds: 10–30 seconds

5.2.3 Run Validation Tests

After building, verify the installation by running the validation test suite:

```
# Navigate to scripts directory
cd scripts
./run_validation.sh

./run_validation.sh -c ortho
./run_validation.sh -c czm

# Run all tests
./run_validation.sh           % all categories

# Or run specific category
./run_validation.sh -c czm     % CZM only
./run_validation.sh -c fabric  % fabric orthotropy tests

# With parallel execution
./run_validation.sh -p 8       % 8 MPI ranks
```

Expected summary:

```
Category: ortho      [6/6  PASS]
Category: asymmetry  [2/2  PASS]
Category: postyield  [5/5  PASS]
Category: grains      [3/3  PASS]
Category: czm         [4/4  PASS]
Category: fabric      [4/4  PASS]
Category: rve         [2/2  PASS]
Total: 26  Passed: 26  Failed: 0
```

Note: Segmentation faults during cleanup (after CSV generation) are harmless and do not indicate test failure. If all tests show [PASS], the validation succeeded.

Test Categories

The validation suite includes multiple test categories:

Table 5.1: Validation test categories

Category	Tests	Purpose
ortho	6	Orthotropic elasticity & yield (6 directions)
asymmetry	2	Tension-compression asymmetry
postyield	5	Hardening/softening behaviour
grains	3	Multi-grain load redistribution
czm	4	GH cohesive zone traction-separation
fabric	4	Fabric orthotropy: equivalence, isotropic limit, anisotropy scaling, bone validation
rve	2	(1) grain plasticity only; (2) grain plasticity + GH interfaces

RVE pair: Both runs use the same mesh and bulk properties. Run 1 disables CZM to isolate grain plasticity; run 2 activates GH cohesive interfaces. The difference quantifies the interface contribution.

Fabric tests: Validate `ComputeFabricElasticityTensor` against Zysset-Curnier closed-form compliance. Gold files generated by `compute_fabric_reference.py`.

Command Options

```
# Show help
./run_validation.sh --help

# Run specific category
./run_validation.sh -c ortho
./run_validation.sh -c czm

# Verbose output (show simulation details)
./run_validation.sh -v -c czm

# Generate new gold files (after code changes)
./run_validation.sh --generate-gold -c fabric

# List all available tests
./run_validation.sh --list
```

5.2.4 Visualizing Validation Results

After running validation tests, generate publication-quality figures using the plotting script:

```
# Navigate to scripts directory
cd scripts

# Generate all validation plots
./plot_validation.sh

# Or plot specific category
./plot_validation.sh -c ortho
./plot_validation.sh -c grains
./plot_validation.sh -c czm
```

Output files: Figures are saved to `test/validation/figures/`

Table 5.2: Generated validation figures

File	Content
ortho.png	All 6 orthotropic directions (X/Y/Z tension, XY/XZ/YZ shear) on single panel
asymmetry.png	Tension-compression asymmetry + biaxial yield surface shift
postyield.png	All 5 post-yield evolution modes (perfect/hardening/softening) on single panel
grains.png	Multi-grain orientation effects with load redistribution annotation
czm.png	All 4 CZM modes (Mode I/II/mixed/complete separation) on single panel

Plot Interpretation

Orthotropic validation (ortho.png) Six stress-strain curves demonstrating:

- Different elastic moduli: $E_1 > E_3 > E_2$ (slopes vary)
- Anisotropic yield strengths: Different plateaus for each direction

Post-yield validation (postyield.png) Five evolution modes showing:

- **Perfect plasticity:** Flat plateau after yield
- **Exponential hardening:** Asymptotic approach to $r_\infty \sigma_y$
- **Linear hardening:** Linear stress increase with plastic strain
- **Exponential softening:** Post-peak stress decrease to $r_{\text{res}} \sigma_y$
- **Piecewise softening:** Bilinear stress reduction

Multi-grain validation (grains.png) Demonstrates grain-level heterogeneity:

- **Color grouping:** One colour per misorientation angle (0° , 45° , 90° , 63.8° Twin)
- **Line style:** Solid = Grain 1 (fixed), Dashed = Grain 2 (pulled)
- **Load redistribution:** Apparent strain decrease in fixed grain indicates elastic unloading as pulled grain yields and softens
- **Orientation effects:** Larger misorientations produce greater stress heterogeneity

Physical interpretation: When Grain 2 (pulled) yields first due to unfavorable orientation, it accumulates plastic strain and softens. As its load-bearing capacity decreases, Grain 1 (fixed) elastically unloads, releasing stored elastic energy. This manifests as decreasing strain in the non-yielding grain.

CZM validation (czm.png) Four traction-separation curves:

- **Mode I (Opening):** normal traction vs. normal separation
- **Mode II (Shear):** tangential traction vs. tangential separation
- **Mixed mode:** Combined normal and shear loading
- **Complete separation:** Load-unload-reload cycle demonstrating damage irreversibility

5.2.5 Update from Repository

```
# Pull latest changes
git pull

# Rebuild
make -j8

# Run tests to verify
cd scripts && ./run_validation.sh
```

That's it. The repository includes everything pre-configured.

5.3 Scenario 2: Building from Source Files

Recommended for: Developers/interested in creating the application from source files

Prerequisites:

- Conda MOOSE environment installed and activated (Chapter 4)
- MOOSE repository cloned (for `stork.sh` script)
- Material source files: `*.C` and `*.h`

5.3.1 Step 1: Clone MOOSE (for `stork.sh` script)

Important: Even with conda MOOSE, you need the MOOSE repository for the `stork.sh` application generator script.

```
# Activate environment
conda activate moose

# Clone MOOSE repository (just for scripts)
cd ~/projects
git clone https://github.com/idaholab/moose.git
cd moose
git checkout master

# You only need this for the stork.sh script
# The actual MOOSE framework comes from conda
```

5.3.2 Step 2: Create Application Skeleton with `stork.sh`

The `stork.sh` script generates a complete MOOSE application structure with proper Makefiles and registration boilerplate.

```
# Navigate to projects directory
cd ~/projects

# Run stork.sh to create application
./moose/scripts/stork.sh aragonite
```

Script creates:

```
aragonite/
├── Makefile          (auto-generated!)
└── src/
```

```

├─ include/
├─ test/
├─ doc/
└─ ... (standard MOOSE app structure)

```

Important: The application name (`aragonite`) will be used in material registration. If you use a different name, you may need to update material registration in source files (see Section 5.3.5).

5.3.3 Step 3: Enable Solid Mechanics Module

Edit the auto-generated Makefile to enable the solid mechanics module:

```

cd aragonite
vim Makefile # or your preferred editor

```

Find the block after:

```
ALL_MODULES := no
```

Change to:

```

ALL_MODULES := no
...
SOLID_MECHANICS := yes

```

Enabling all modules increases build time. We only need solid mechanics for this application. Save and close the file.

5.3.4 Step 4: Copy Material Source Files

Copy your material implementation files to the generated directories:

```

# Copy header files
cp /path/to/OrthotropicPlasticityStressUpdate.h \
include/materials/
cp /path/to/ComputeElasticityTensorCoupled.h \
include/materials/
cp /path/to/HomogenizedExponentialCZM.h \
include/materials/
cp /path/to/VoigtHelpers.h \
include/materials/

# Copy source files
cp /path/to/OrthotropicPlasticityStressUpdate.C \
src/materials/
cp /path/to/ComputeElasticityTensorCoupled.C \
src/materials/
cp /path/to/HomogenizedExponentialCZM.C \
src/materials/

```

Directory structure after copying:

```

aragonite/
├─ Makefile
├─ include/
│   └─ materials/
│       ├── OrthotropicPlasticityStressUpdate.h
│       └─ ComputeElasticityTensorCoupled.h

```

```

├── HomogenizedExponentialCZM.h
├── VoigtHelpers.h
├── src/
│   ├── materials/
│   │   ├── OrthotropicPlasticityStressUpdate.C
│   │   ├── ComputeElasticityTensorCoupled.C
│   │   └── HomogenizedExponentialCZM.C
│   └── test/
└── ... (add tests later)

```

VoigtHelpers.h - Tensor-Voigt Conversion Utilities

The `VoigtHelpers.h` header provides utility functions for converting between full tensor notation and Voigt (vector/matrix) notation. This header is required by `OrthotropicPlasticityStressUpdate` for efficient tensor operations.

Purpose and Usage. Voigt notation reduces storage for symmetric tensors:

- Rank-2 tensor (3×3) $\rightarrow$ 6-component vector: $[\sigma_{11}, \sigma_{22}, \sigma_{33}, \sigma_{23}, \sigma_{13}, \sigma_{12}]^T$
- Rank-4 tensor ($3 \times 3 \times 3 \times 3$) $\rightarrow$ 6×6 matrix (e.g., stiffness tensor $\mathbb{C}$)

Used in `OrthotropicPlasticityStressUpdate` for:

- Yield surface evaluation: $f = \sqrt{\mathbf{S} : \mathbb{F} : \mathbf{S}} - r(\kappa)$
- Flow direction computation: $\partial f / \partial \mathbf{S}$
- Return mapping algorithm (Newton-Raphson in 6D Voigt space)

Note: This is a header-only file (all functions are `inline`) and requires no corresponding `.C` file.

5.3.5 Step 5: Verify Material Registration

Check that material classes are registered with the correct application name:

```
grep -n "registerMooseObject" src/materials/*.C
```

Each `.C` file should have at the top (after includes):

```
// Option 1: Register with aragonite app
registerMooseObject("aragoniteApp", OrthotropicPlasticityStressUpdate);

// Option 2: Register with SolidMechanics (also works)
registerMooseObject("SolidMechanicsApp",
    OrthotropicPlasticityStressUpdate);
```

If you used a different app name in `stork.sh` (e.g., `myapp`), replace `aragoniteApp` with `myappApp`.

Note: Registering with `SolidMechanicsApp` should work regardless of application name.

5.3.6 Step 6: Build Application

```
# Ensure environment is activated
conda activate moose

# Build (use number of CPU cores available)
make -j8

# Expected output:
#   Building CXX object ...
#   ...
#   Linking aragonite-opt...
```

Build time:

- First build: 2–5 minutes
- Incremental builds (after editing): 10–30 seconds

5.3.7 Step 7: Test Build

Verify the executable works:

```
# Check executable exists
ls -lh aragonite-opt

# Run version check
./aragonite-opt --version

# Should display MOOSE version and build info
```

Create a minimal test input:

```
mkdir -p test/basic
cat > test/basic/build_test.i << 'EOF'
[Mesh]
  type = GeneratedMesh
  dim = 3
  nx = 1
  ny = 1
  nz = 1
[]

[Problem]
  solve = false # Just test that app runs
[]

[Executioner]
  type = Steady
[]
EOF

# Run test
./aragonite-opt -i test/basic/build_test.i

# Should complete without errors
```

5.3.8 Step 8: Run Validation Tests

After successful build, run the validation test suite to verify implementation:

```
# Copy test suite scripts if not already present
mkdir -p scripts
cp /path/to/run_validation.sh scripts/
cp /path/to/compare_with_gold.py scripts/
chmod +x scripts/run_validation.sh

# Copy test inputs and gold files
cp -r /path/to/test/validation test/

# Run validation
cd scripts
./run_validation.sh -c ortho
```

If tests pass, the implementation is verified.

Minimum test suite for basic validation:

Create at least these 3 tests before running validation:

- ortho/x_tension - Primary loading direction
- ortho/y_tension - Anisotropy verification
- ortho/xy_shear - Shear behaviour

See Section 5.2.3 for detailed minimal validation procedures and Chapter 7.

5.4 Additional Material for Polycrystals

For polycrystalline simulations with per-element Euler angles, you need an additional material: `ComputeElasticityTensorCoupled`.

5.4.1 Why This is Needed

Standard MOOSE's `ComputeElasticityTensor` only accepts scalar parameters for Euler angles:

```
# Standard version - same angle for ALL elements
[Materials]
  [elasticity]
    type = ComputeElasticityTensor
    euler_angle_1 = 45.0 # Scalar - applies to all elements
  []
[]
```

For polycrystals, each element needs different angles. This requires coupling to `AuxVariables`:

```
# Coupled version - different angle PER element
[Materials]
  [elasticity]
    type = ComputeElasticityTensorCoupled
    euler_angle_1 = euler_phi1 # Couples to AuxVariable
  []
[]
```

5.4.2 Implementation

Files: ComputeElasticityTensorCoupled.C and ComputeElasticityTensorCoupled.h

Based on MOOSE's ComputeElasticityTensor but modified to use coupledValue() instead of scalar parameters. Key changes:

In header (.h):

```
// Change from scalar to coupled
const VariableValue & _Euler_angles_1;
const VariableValue & _Euler_angles_2;
const VariableValue & _Euler_angles_3;
```

In validParams():

```
// Enable coupling to variables
params.addRequiredCoupledVar("euler_angle_1",
    "First_Euler_angle(degrees)");
params.addRequiredCoupledVar("euler_angle_2",
    "Second_Euler_angle(degrees)");
params.addRequiredCoupledVar("euler_angle_3",
    "Third_Euler_angle(degrees)");
```

In constructor:

```
// Read coupled values
_Euler_angles_1(coupledValue("euler_angle_1")),
_Euler_angles_2(coupledValue("euler_angle_2")),
_Euler_angles_3(coupledValue("euler_angle_3")),
```

In computeQpElasticityTensor():

```
// Access per-element (per quadrature point)
Real phi1 = _Euler_angles_1[_qp];
Real Phi = _Euler_angles_2[_qp];
Real phi2 = _Euler_angles_3[_qp];

// Compute rotation matrix and rotate elasticity tensor
// (Same logic as standard version, but using per-element angles)
```

5.4.3 When to Use Each Version

Use Case	Material	Euler Angles
Single crystal	ComputeElasticityTensor	Scalar parameters
Multiple crystals, same orientation	ComputeElasticityTensor	Scalar parameters
Twin grains (2 blocks)	ComputeElasticityTensor	Different per block
Polycrystal (many orientations)	ComputeElasticityTensorCoupled	AuxVariables
EBSD-based RVE	ComputeElasticityTensorCoupled	From exodus file

5.5 Summary

5.5.1 Scenario Comparison

Table 5.3 compares both installation scenarios.

5.5.2 Key Files

The application requires these material files:

Table 5.3: Build scenario comparison

Aspect	From Source	GitHub Clone
Setup time	15–30 minutes	5 minutes
MOOSE clone	Required (for <code>stork.sh</code>)	Not required
<code>stork.sh</code>	Manual run	Not needed
Makefile edit	Manual (enable module)	Pre-configured
File copying	Manual	Included
Build command	<code>make -j8</code>	<code>make -j8</code>
Tests	Manual setup	Included
Updates	Manual	<code>git pull</code>
Recommended for	Developers	Users

Table 5.4: Required material files

File	Purpose
<code>OrthotropicPlasticityStressUpdate.C/.h</code>	Orthotropic plasticity model
<code>ComputeElasticityTensorCoupled.C/.h</code>	Coupled elasticity (per-element Euler angles)
<code>HomogenizedExponentialCZM.C/.h</code>	Cohesive zone model with homogenization
<code>VoigtHelpers.h</code>	Tensor-Voigt conversion utilities (header-only)

5.5.3 Key Points

- `stork.sh` creates proper application structure automatically
- Enable `SOLID_MECHANICS := yes` in Makefile
- `VoigtHelpers.h` is required for orthotropic plasticity calculations
- Material registration must match app name (or use `SolidMechanicsApp`)
- Incremental builds are fast (10–30 seconds)
- Multiple build types can coexist (opt, dbg, oprof)

5.5.4 Troubleshooting

For build errors, compilation issues, and common problems, see Section 8.1 (Troubleshooting). Common issues include:

- `stork.sh`: `command not found` – MOOSE repository not cloned
- `ModuleNotFoundError` – Conda environment not activated
- Undefined reference errors – Material registration issues
- Compilation errors – Missing header files or version mismatches

5.5.5 Next Steps

After successful build:

1. **Run validation tests** (Chapter 7): Verify implementation
2. **Try examples** (Section 7.20): Check usage patterns
3. **Create simulations**: Apply to your material system

Chapter 6

Complete Parameter Reference

6.1 Bulk Material Block Template

Complete MOOSE input block with all parameters:

```
[Materials]
[plasticity]
type = OrthotropicPlasticityStressUpdate

# ===== YIELD STRESSES (MPa) =====
# Tension yield stresses (3 directions)
sigma_xx_tension = 4980 # X-direction (11)
sigma_yy_tension = 4100 # Y-direction (22)
sigma_zz_tension = 5340 # Z-direction (33)

# Compression yield stresses (optional, if different from tension)
# If not specified, compression strength = tension strength
# sigma_xx_compression = 6000 # X-direction compression
# sigma_yy_compression = 5000 # Y-direction compression
# sigma_zz_compression = 6500 # Z-direction compression

# Shear yield stresses (3 planes)
tau_xy_max = 4510 # XY-plane (12)
tau_xz_max = 5080 # XZ-plane (13)
tau_yz_max = 5060 # YZ-plane (23)

# Biaxial coupling parameters (optional, for advanced users)
# Default: 0.0 (uncoupled yield surface)
# Range: -1.0 to 1.0 (constrained by convexity)
# Activate only when biaxial test data available or reproducing UMAT
# zeta12 = 0.0 # X-Y normal stress coupling (xx to yy)
# zeta13 = 0.0 # X-Z normal stress coupling (xx to zz)
# zeta23 = 0.0 # Y-Z normal stress coupling (yy to zz)

# ===== POST-YIELD behaviour =====
# Options: 'perfect', 'exp_hardening', 'linear_hardening',
#          'exp_softening', 'piecewise_softening'
postyield_mode = 'exp_softening'

# Residual strength ratio (for softening modes)
# Range: (0, 1), typical: 0.5-0.8
residual_strength = 0.7

# Plastic strain at softening onset
```

```

# Range: 0.0-0.01, typical: 0.001-0.005
kmax = 0.002

# Plastic strain at full softening (for piecewise mode)
# Range: kmax to 0.1, typical: 0.01-0.05
kmin = 0.02

# Hardening/softening slope parameter
# For exp modes: typical 10-100
# For linear modes: slope in stress/strain
kslope = 30.0

# ===== VISCOPLASTIC REGULARIZATION =====
# Options: 'rate_independent', 'linear', 'exponential',
#          'logarithmic', 'polynomial', 'powerlaw'
viscosity_mode = 'linear'

# Viscosity parameter (time units)
# Range: 0.0001-0.1, typical: 0.001-0.01
eta = 0.002

# Power-law exponent (for 'power' mode)
# Range: 0.0-1.0, typical: 0.001-0.01
m = 0.001

# ===== DAMAGE MECHANICS (OPTIONAL) =====
# Enable/disable damage coupling
use_damage = false

# Damage parameters (only used if use_damage = true)
# damage_critical = 0.05 # Plastic strain for damage onset
# damage_rate = 0.1      # Linear damage accumulation rate

# ===== CONVERGENCE PARAMETERS =====
# Absolute tolerance for yield function
# Range: 1e-10 to 1e-4, typical: 1e-6 to 1e-5
absolute_tolerance = 1.0e-5

# Maximum Newton-Raphson iterations (Stage 1)
# Range: 10-50, typical: 20-30
max_iterations_newton = 20

# Enable Primal CPP fallback (Stage 2)
use_primal_cpp = true

# Maximum Primal CPP iterations (Stage 2)
# Range: 100-5000, typical: 500-2000
max_iterations_primal = 1000

# ===== LINE SEARCH PARAMETERS =====
# Armejo line search parameter
# Range: 1e-6 to 1e-2, typical: 1e-5 to 1e-4
line_search_beta = 1e-4

# Line search backtracking factor
# Range: 0.1-0.5, typical: 0.25
line_search_eta = 0.25

```

```

# ===== EULER ANGLES (degrees) =====
# Bunge convention (ZXZ rotations)
euler_angle_1 = 0.0    # phi_1: [0, 360)
euler_angle_2 = 0.0    # Phi:   [0, 180]
euler_angle_3 = 0.0    # phi_2: [0, 360)
[]
[]

```

6.2 Fabric-Based Material Block Template

For trabecular bone and porous biomaterials with fabric tensor from μ CT imaging:

```

[Materials]
# =====
# ELASTICITY: Fabric-based (Zysset-Curnier [2])
# =====
[elasticity]
  type = ComputeFabricElasticityTensor

  # ===== BASE TISSUE PROPERTIES =====
  E_0 = 10000.0          # Base tissue Young's modulus [MPa]
  nu_0 = 0.3             # Base tissue Poisson's ratio
  # Note: G_0 computed automatically as E_0 / [2(1+nu_0)]

  # ===== FABRIC TENSOR (from  $\mu$ CT) =====
  # Constraint: m1 + m2 + m3 = 3.0 (normalization)
  # Typical ordering: m1 > m2 > m3 (strongest to weakest)
  fabric_m1 = 1.1        # Principal direction 1
  fabric_m2 = 1.0        # Principal direction 2
  fabric_m3 = 0.9        # Principal direction 3

  # ===== DENSITY (from  $\mu$ CT) =====
  density_rho = 0.2      # Relative density (BV/TV)
  # Range: 0.05-0.5 for trabecular bone

  # ===== SCALING EXPONENTS =====
  exponent_k = 2.0       # Elasticity density exponent
  # Gibson-Ashby: k=2 for open-cell foams
  # Range: 1.8-2.2

  exponent_l = 1.0       # Fabric exponent
  # Typically l=1.0 for linear dependence
  # Range: 0.5-1.5
[]

# =====
# PLASTICITY: Fabric-based (Schwiedrzik et al. [1])
# =====
[plasticity]
  type = OrthotropicPlasticityStressUpdate

  # ===== MODE SELECTION =====
  use_fabric_scaling = true # CRITICAL: Enable fabric mode

  # ===== BASE TISSUE YIELD STRENGTHS [MPa] =====
  sigma_0_tension = 74.589    # Tissue-level tension strength
  sigma_0_compression = 111.724 # Tissue-level compression strength

```

```

tau_0 = 47.33                # Tissue-level shear strength
zeta_0 = 0.22                # Tissue-level coupling parameter

# ===== FABRIC TENSOR (MUST MATCH ELASTICITY) =====
fabric_m1 = 1.1
fabric_m2 = 1.0
fabric_m3 = 0.9

# ===== DENSITY (MUST MATCH ELASTICITY) =====
density_rho = 0.2

# ===== YIELD SCALING EXPONENTS =====
exponent_p = 1.686          # Yield density exponent
# Range: 1.2-1.8

exponent_q = 1.02           # Yield fabric exponent
# Range: 0.5-1.5

delta_cortical = 1.0        # Cortical correction factor
# Set to 1.0 to disable
# Range: 1.0-2.0 (for  $\rho > 0.5$ )

# ===== POST-YIELD BEHAVIOR =====
# (Same as explicit mode - see Section 6.4)
postyield_mode = 'exp_softening'
residual_strength = 0.7
kmin = 0.002
kslope = 100.0

# ===== VISCOPLASTIC REGULARIZATION =====
viscosity_mode = 'linear'
eta = 0.002

# ===== CONVERGENCE PARAMETERS =====
absolute_tolerance = 1.0e-5
max_iterations_newton = 20
use_primal_cpp = true
max_iterations_primal = 1000

# Note: Euler angles NOT used in fabric mode
# Material orientation defined by fabric tensor principal axes
[]
[]

```

6.3 Fabric-Based Model Parameters

6.3.1 Elasticity Parameters (ComputeFabricElasticityTensor)

Base Tissue Properties

Parameter	Description
E_0	Base tissue Young’s modulus [MPa]. Represents fully dense tissue stiffness. Typical: 10000–20000 MPa for bone. Cortical bone: ~15000 MPa. Calibrated from nanoindentation or literature.
nu_0	Base tissue Poisson’s ratio (dimensionless). Typical: 0.25–0.35 for bone. Default: 0.3.
G_0	(Optional) Base shear modulus [MPa]. If not specified, computed as $G_0 = E_0/[2(1 + \nu_0)]$.

Fabric Tensor

Parameter	Description
fabric.m1	Fabric eigenvalue in principal direction 1 (dimensionless). Strongest structural direction. Typical: 1.0–1.5.
fabric.m2	Fabric eigenvalue in principal direction 2 (dimensionless). Intermediate direction. Typical: 0.9–1.1.
fabric.m3	Fabric eigenvalue in principal direction 3 (dimensionless). Weakest structural direction. Typical: 0.6–1.0.

Critical constraint: $m_1 + m_2 + m_3 = 3.0$ (normalization)

Physical interpretation:

- Isotropic: $m_1 = m_2 = m_3 = 1.0$
- Transversely isotropic: Two eigenvalues equal (e.g., $m_1 = m_2 = 1.2$, $m_3 = 0.6$)
- Orthotropic: All different (e.g., $m_1 = 1.3$, $m_2 = 1.0$, $m_3 = 0.7$)

Acquisition:

1. μ CT scan of trabecular bone sample
2. Compute mean intercept length (MIL) tensor
3. Diagonalize to obtain eigenvalues
4. Normalize: $m'_i = 3m_i/(m_1 + m_2 + m_3)$

Reference: Harrigan & Mann (1984) for MIL tensor computation [13].

Density and Scaling Exponents

Parameter	Description
density_rho	Relative density (BV/TV, bone volume fraction). Range: 0.05–0.5 for trabecular bone. From μ CT thresholding.
exponent_k	Elasticity density exponent. Controls $E \propto \rho^k$ scaling. Range: 1.8–2.2. Typical: 2.0.
exponent_l	Elasticity fabric exponent. Controls directional variation. Range: 0.5–1.5.

Computed properties: Once parameters are specified, the material computes:

$$\begin{aligned} E_i &= E_0 \rho^k m_i^{2l} \quad (i = 1, 2, 3) \\ G_{ij} &= G_0 \rho^k (m_i m_j)^l \\ \nu_{ij} &= \nu_0 (m_i / m_j)^l \end{aligned}$$

Example ($\rho=0.2$, $k=2$, $l=1$, $m_i = [1.1, 1.0, 0.9]$):

- $E_1 = 10000 \times 0.2^2 \times 1.1^2 = 484$ MPa
- $E_2 = 10000 \times 0.2^2 \times 1.0^2 = 400$ MPa
- $E_3 = 10000 \times 0.2^2 \times 0.9^2 = 324$ MPa

6.3.2 Plasticity Parameters (OrthotropicPlasticityStressUpdate, fabric mode)

Mode Selection

Parameter	Description
<code>use_fabric_scaling</code>	Boolean flag to enable fabric-based yield scaling. MUST be set to true for fabric mode. If false, uses explicit orthotropic mode (Section 6.4).

Base Tissue Yield Strengths

Parameter	Description
<code>sigma_0_tension</code>	Tissue-level tensile strength [MPa]. Represents fully dense tissue.
<code>sigma_0_compression</code>	Tissue-level compressive strength [MPa].
<code>tau_0</code>	Tissue-level shear strength [MPa].
<code>zeta_0</code>	Tissue-level biaxial coupling parameter. Typical: 0.22. Range: 0.0–0.5. Set to 0.0 if no biaxial data.

Note: These are NOT the apparent yield strengths! Apparent strengths are computed via:

$$\sigma_i^+ = \sigma_0^+ \times \text{TSFU}(\rho, p, \delta) \times m_i^{2q}$$

Yield Scaling Exponents

Parameter	Description
<code>exponent_p</code>	Yield density exponent. Controls $\sigma \propto \rho^p$ scaling. Wolfram et al. [5]: 1.686.
<code>exponent_q</code>	Yield fabric exponent. Controls directional variation of yield. Wolfram et al. [5]: 1.02.
<code>delta_cortical</code>	Cortical bone correction factor for high density ($\rho > 0.5$). Set to 1.0 to disable. Range: 1.0–2.0. Only relevant for dense bone ($\rho > 0.5$). Default: 1.0.

TSFU function:

$$\text{TSFU}(\rho, p, \delta) = \begin{cases} \rho^p & \rho \leq 0.5 \\ \rho^p + (\delta - 1) \left(\frac{\rho - 0.5}{0.5} \right)^p & \rho > 0.5 \end{cases} \quad (6.1)$$

Literature parameter sets:

Note: exponents are not equal generally, i.e., $k \neq p$ and $l \neq q$.

Table 6.1: Published fabric-based parameters for trabecular bone

Source	σ_0^+ [MPa]	σ_0^- [MPa]	τ_0 [MPa]	p –	q –	ζ_0 –
Wolfram et al. [5]	74.6	111.72	47.33	1.686	1.02	0.22
Rincón-Kohli and Zysset [14]	32.7	47.8	11.1	1.28	0.503	0.22

Critical Consistency Requirements

Fabric values MUST match:

```
# In elasticity block:
fabric_m1 = 1.1
fabric_m2 = 1.0
fabric_m3 = 0.9

# In plasticity block: MUST BE IDENTICAL
fabric_m1 = 1.1
fabric_m2 = 1.0
fabric_m3 = 0.9
```

Density MUST match:

```
# In elasticity block:
density_rho = 0.2

# In plasticity block: MUST BE IDENTICAL
density_rho = 0.2
```

Post-Yield and Convergence Parameters

All post-yield modes (`postyield_mode`, `residual_strength`, `kmin`, `kslope`), viscoplastic regularization (`viscosity_mode`, `eta`), and convergence parameters (`absolute_tolerance`, `max_iterations_newton`, etc.) are identical to explicit orthotropic mode.

See Section 6.4 for complete documentation of these parameters.

6.3.3 Workflow Summary: Fabric-Based Simulation

1. Acquire fabric and density from μ CT:

- Scan trabecular bone sample (resolution: 10–50 μm)
- Threshold to obtain binary image (bone vs. void)
- Compute BV/TV (`density_rho`)
- Compute MIL tensor, extract and normalize eigenvalues (m_1, m_2, m_3)

2. Select tissue-level parameters:

- Elastic: E_0 , ν_0 from literature or nanoindentation
- Plastic: $\sigma_0^\pm$, τ_0 from Wolfram or Rincón-Kohli
- Exponents: $k = 2$, $l = 1$, $p \approx 1.5$, $q \approx 1.0$

3. Configure MOOSE input:

- Elasticity: `ComputeFabricElasticityTensor`

- Plasticity: `use_fabric_scaling = true`
- Ensure fabric and density match between materials

4. Verify computed properties:

- Check E_i values are reasonable (100–1000 MPa for typical trabecular)
- Verify anisotropy ratio E_1/E_3 matches expectations (1.2–3)

5. Run simulation and validate:

- Compare apparent modulus to experiments
- Check yield stress against uniaxial compression tests
- Adjust p , q if needed for better match

Common pitfall: Using apparent (density-scaled) strengths as tissue strengths. The inputs are tissue-level. Apparent values are computed internally.

6.4 Orthotropic Plasticity Parameters

6.4.1 Yield Stresses

Parameter	Description
<code>sigma_xx_tension</code>	Yield stress in X-direction tension (material axis 1), MPa
<code>sigma_yy_tension</code>	Yield stress in Y-direction tension (material axis 2), MPa
<code>sigma_zz_tension</code>	Yield stress in Z-direction tension (material axis 3), MPa
<code>sigma_xx_compression</code>	(Optional) Yield stress in X-direction compression, MPa. If not specified, equals tension strength.
<code>sigma_yy_compression</code>	(Optional) Yield stress in Y-direction compression, MPa. If not specified, equals tension strength.
<code>sigma_zz_compression</code>	(Optional) Yield stress in Z-direction compression, MPa. If not specified, equals tension strength.
<code>tau_xy_max</code>	Yield stress in XY-plane shear, MPa
<code>tau_xz_max</code>	Yield stress in XZ-plane shear, MPa
<code>tau_yz_max</code>	Yield stress in YZ-plane shear, MPa
<code>zeta12</code>	(Optional) X-Y biaxial coupling parameter. Controls interaction between σ_{xx} and σ_{yy} in yield criterion. Range: $[-1, 1]$. Default: 0.0 (uncoupled). Only specify when biaxial test data available.
<code>zeta13</code>	(Optional) X-Z biaxial coupling parameter. Controls interaction between σ_{xx} and σ_{zz} in yield criterion. Range: $[-1, 1]$. Default: 0.0 (uncoupled).
<code>zeta23</code>	(Optional) Y-Z biaxial coupling parameter. Controls interaction between σ_{yy} and σ_{zz} in yield criterion. Range: $[-1, 1]$. Default: 0.0 (uncoupled).

Source: Calibrate from molecular dynamics or nano-indentation experiments.

Aragonite defaults (tension):

```
sigma_xx_tension = 4980
sigma_yy_tension = 4100
sigma_zz_tension = 5340
tau_xy_max = 4510
tau_xz_max = 5080
tau_yz_max = 5060
```

Tension-compression asymmetry:

By default, compression strength equals tension strength ($f_i = 0$, symmetric yield surface). To specify different compression strengths, add:

```
sigma_xx_compression = 6000 # stronger in compression
sigma_yy_compression = 5000 # stronger in compression
sigma_zz_compression = 6500 # stronger in compression
```

This activates the linear term in the quadric formulation, shifting the yield surface to account for different behaviour under tensile vs compressive loading. Typical for brittle materials like aragonite where crushing strength often exceeds tensile strength.

6.4.2 Biaxial Coupling Parameters (Advanced)

The coupling parameters ζ_{12} , ζ_{13} , ζ_{23} control off-diagonal terms F_{ij} in the yield surface matrix $\mathbb{F}$, affecting how combined normal stresses trigger yielding.

When to Use

Use default (uncoupled, $\zeta_{ij} = 0$) when:

- No biaxial test data available (most common)
- Initial model development
- Trabecular bone or coral applications (typical)

Specify non-zero values when:

- Biaxial experimental data is available
- Advanced validation against published biaxial tests

Usage Examples

Default (uncoupled):

```
# No zeta parameters specified -> defaults to 0.0
# Yield surface is uncoupled between directions
```

Isotropic coupling:

```
# Equal coupling in all directions
zeta12 = 0.5
zeta13 = 0.5
zeta23 = 0.5
```

Transversely isotropic (Z-axis unique):

```
# Stronger in-plane coupling than through-thickness
zeta12 = 0.5 # In-plane (XY)
zeta13 = 0.3 # Through-thickness (XZ)
zeta23 = 0.3 # Through-thickness (YZ)
```

Determination from Experiments

From an equibiaxial tension test where $S_{11} = S_{22} = S_b$ at yield, the coupling parameter ζ_{12} can be computed using Eq. 2.31.

In practice:

1. Perform uniaxial tests in X and Y to determine $S_{1,\text{tension}}^y, S_{1,\text{compression}}^y, S_{2,\text{tension}}^y, S_{2,\text{compression}}^y$
2. Compute F_{11}, F_{22}, F_1, F_2 using Eqs. 2.29 and 2.27
3. Perform equibiaxial test to measure S_b
4. Calculate ζ_{12} from Eq. 2.31
5. Repeat for other direction pairs if needed

Constraints and Validation

Convexity requirement: $|\zeta_{ij}| \leq 1$ for all $i \neq j$

Positive semi-definiteness: When multiple coupling terms are non-zero, the matrix $\mathbb{F}$ must remain positive semi-definite. The implementation does not automatically verify this - users must ensure valid parameter combinations.

Troubleshooting:

- If convergence issues arise with coupling, start with $\zeta_{ij} = 0$ to verify the model works, then gradually increase coupling parameters.
- If biaxial strength predictions seem incorrect, verify sign convention and check that experimental data used for calibration is reliable.
- For unexpected yield behavior, confirm $|\zeta_{ij}| \leq 1$ and consider reducing coupling magnitude.

6.4.3 Post-Yield Mode

Value	behaviour
'perfect'	Perfect plasticity, no hardening/softening: $r(\kappa) = 1$
'exp_hardening'	Exponential hardening: $r = r_\infty + (1 - r_\infty)[1 - \exp(-k_{\text{slope}}\kappa)]$
'linear_hardening'	Linear hardening: $r = 1 + k_{\text{slope}}\kappa$
'exp_softening'	Exponential softening with delayed onset kmax (recommended)
'piecewise_softening'	Piecewise linear softening between $[\kappa_{\text{max}}, \kappa_{\text{min}}]$

Recommendation for aragonite: Use 'exp_softening' with **kmax** = 0.002.

6.4.4 Softening/Hardening Parameters

Parameter	Description
<code>residual_strength</code>	Ratio of residual to initial yield stress. Range: (0, 1) for softening, > 1 for hardening. Typical: 0.5–0.8 for softening.
<code>kmax</code>	Plastic strain at softening onset. Critical for stability! Range: 0.0–0.01. Typical: 0.001–0.005. Set to 0.0 for immediate softening (may cause convergence issues).
<code>kmin</code>	Plastic strain at full softening (piecewise mode only). Range: <code>kmax</code> to 0.1. Typical: 0.01–0.05. Defines width of softening regime.
<code>kslope</code>	Softening/hardening rate. For exponential: typical 10–100 (higher = faster transition). For linear: slope in MPa/strain units.

Effect of `kmax`:

- `kmax` = 0: Softening begins immediately after yield (numerically challenging)
- `kmax` = 0.002: Small elastic-perfectly-plastic region (recommended)
- `kmax` = 0.005: Larger PPP region (very stable, may be unphysical)

Effect of `kslope`:

- `kslope` = 10: Gradual softening over large strain range (~ 0.1)
- `kslope` = 50: Moderate softening (~ 0.05 strain range)
- `kslope` = 100: Rapid softening (~ 0.02 strain range)

6.4.5 Viscoplastic Regularization

Parameter	Description
<code>viscosity_mode</code>	Viscoplastic regularization type. Options: <code>'rate_independent'</code> : No regularization (purely rate-independent plasticity). <code>'linear'</code> : Linear overstress, $\dot{\lambda} \propto f$. Most commonly used, simple and effective. <code>'exponential'</code> : Exponential overstress with logarithmic term. <code>'logarithmic'</code> : Logarithmic overstress with exponential term. <code>'polynomial'</code> : Polynomial overstress (quadratic form). <code>'powerlaw'</code> : Power-law overstress, $\dot{\lambda} \propto f^{1/m}$.
<code>eta</code>	Viscosity parameter (inverse units: $(\text{MPa}\cdot\text{s})^{-1}$). Controls magnitude of regularization. Range: 0.0001–0.1. Typical: 0.001–0.01. Smaller = less regularization, closer to rate-independent. Note: In code, this is η , formulated as $-\eta/\Delta t \times \Delta\kappa/H_I$ for linear mode.
<code>m</code>	Exponent for non-linear viscosity models (exponential, logarithmic, polynomial, powerlaw). Range: 0.0–1.0. Typical: 0.001–0.01 for weak nonlinearity. Not used for <code>'rate_independent'</code> or <code>'linear'</code> modes. Controls degree of nonlinearity in rate-dependence.

Viscosity Mode Details:

Rate-independent (`viscosity_mode` = `'rate_independent'`):

$$\text{visc} = 0 \quad (6.2)$$

No regularization. Pure rate-independent plasticity. Use when convergence is good and mesh-independence not a concern.

Linear (`viscosity_mode = 'linear'`):

$$\text{visc} = -\frac{\eta}{\Delta t} \frac{\Delta \kappa}{H_I} \quad (6.3)$$

where $H_I = \|\partial f / \partial \boldsymbol{\sigma}\|$ is the norm of the yield gradient. Most commonly used mode. Simple, effective, physically intuitive (Perzyna-type).

Exponential (`viscosity_mode = 'exponential'`):

$$\text{visc} = -\frac{1}{m} \log \left(1 + \frac{\eta}{\Delta t} \frac{\Delta \kappa}{H_I} \right) \quad (6.4)$$

Smoother response than linear, saturation at large rates.

Logarithmic (`viscosity_mode = 'logarithmic'`):

$$\text{visc} = -\frac{1}{m} \left[\exp \left(\frac{\eta}{\Delta t} \frac{\Delta \kappa}{H_I} \right) - 1 \right] \quad (6.5)$$

More aggressive regularization than linear, grows exponentially.

Polynomial (`viscosity_mode = 'polynomial'`):

$$\text{visc} = \frac{m}{2} - \sqrt{\frac{m^2}{4} + \frac{\eta}{\Delta t} \frac{\Delta \kappa}{H_I}} \quad (6.6)$$

Quadratic form, intermediate between linear and exponential behaviours.

Powerlaw (`viscosity_mode = 'powerlaw'`):

$$\text{visc} = -\text{sign}(\Delta \kappa) \left(\frac{\eta}{\Delta t} \frac{|\Delta \kappa|}{H_I} \right)^{1/m} \quad (6.7)$$

Power-law form, more physical for creep-like behaviour. Common in high-temperature plasticity.

When to use viscoplastic regularization:

- Convergence issues during softening (Newton-Raphson oscillates)
- Mesh-dependent results (strain localization)
- Multi-element simulations showing premature yielding

Trade-offs:

- Pro: Improved convergence, mesh-independent results
- Con: Small rate-dependence introduced (usually negligible for $\eta < 0.01$)
- Con: Slightly longer computational time

Typical workflow:

1. Start with `viscosity_mode = 'none'` for rate-independent behaviour
2. If convergence fails, try `viscosity_mode = 'linear'`, `eta = 0.002`
3. If still fails, increase `eta` to 0.005 or 0.01
4. For power-law, use `m = 0.001` (very weak nonlinearity)

6.4.6 Convergence Parameters

Parameter	Description
<code>absolute_tolerance</code>	Absolute convergence tolerance for yield function $ f < \text{tol}$. Range: 10^{-10} to 10^{-4} . Typical: 10^{-6} to 10^{-5} . Tighter tolerances (smaller values) give more accurate results but may cause convergence issues.
<code>max_iterations_newton</code>	Maximum iterations for Stage 1 (Newton-Raphson). Range: 10–50. Typical: 20–30. If exceeded, switches to Stage 2 (Primal CPP).
<code>use_primal_cpp</code>	Enable Stage 2 fallback algorithm. true : Use Primal CPP if Newton fails (recommended). false : Fail immediately if Newton doesn't converge (faster but less robust).
<code>max_iterations_primal</code>	Maximum iterations for Stage 2 (Primal CPP). Range: 100–5000. Typical: 500–2000. Stage 2 is slower but more robust, especially during softening.

Convergence strategy for difficult cases:

1. Start with default values
2. If Newton fails frequently, relax `absolute_tolerance` to 10^{-5}
3. Increase `max_iterations_newton` to 30 or 50
4. Ensure `use_primal_cpp = true`
5. Increase `max_iterations_primal` to 2000 if Stage 2 used often

6.4.7 Line Search Parameters

Parameter	Description
<code>line_search_beta</code>	Armejo condition parameter. Controls how much the residual must decrease. Range: 10^{-6} to 10^{-2} . Typical: 10^{-5} to 10^{-4} . Smaller = more strict (accepts fewer steps, slower convergence). Larger = more lenient (accepts more steps, faster but may overshoot).
<code>line_search_eta</code>	Backtracking factor. Multiplier when step is rejected. Range: 0.1–0.5. Typical: 0.25. Smaller = more aggressive backtracking (slower but safer). Larger = less backtracking (faster but may fail to converge).

Line search workflow:

1. Compute full Newton step: $\Delta\lambda^{\text{full}}$
2. Check Armejo condition: $|f(\Delta\lambda^{\text{full}})| < (1 - \beta)|f(0)|$
3. If satisfied: Accept step
4. If not: Backtrack: $\Delta\lambda^{\text{new}} = \eta \cdot \Delta\lambda^{\text{full}}$
5. Repeat until Armejo satisfied or max backtracks reached

When line search is active:

- During severe softening ($r < 0.5$)
- Near singular points of yield surface
- Large strain increments ($\Delta E > 0.01$)

6.4.8 Euler Angles

Parameter	Description
euler_angle_1	First rotation ϕ_1 about global Z-axis (degrees). Range: [0, 360]. For no rotation, use 0.0.
euler_angle_2	Second rotation Φ about rotated X'-axis (degrees). Range: [0, 180]. Controls out-of-plane tilt.
euler_angle_3	Third rotation ϕ_2 about twice-rotated Z''-axis (degrees). Range: [0, 360]. Final in-plane rotation.

Convention: Bunge (ZXZ), passive rotations from global to material frame.
Common orientations:

```
# No rotation (material axes = global axes)
euler_angle_1 = 0
euler_angle_2 = 0
euler_angle_3 = 0

# 90-degree rotation about Z (swap X and Y)
euler_angle_1 = 90
euler_angle_2 = 0
euler_angle_3 = 0

# 90-degree rotation about X (swap Y and Z)
euler_angle_1 = 0
euler_angle_2 = 90
euler_angle_3 = 0
```

For polycrystals: Assign different Euler angles to each grain using element-wise material properties or AuxVariables with InitialConditions (see Section 7.8.3).

6.4.9 Damage Mechanics (Advanced)

Parameter	Description
use_damage	Enable damage mechanics coupling. false: No damage (default). true: Enable damage evolution with linear accumulation law.
damage_critical	Critical plastic strain for damage onset. Range: 0.0–0.1. Typical: 0.05. No damage accumulates for $\bar{E}^p < \kappa_{\text{crit}}$.
damage_rate	Damage evolution rate. Range: 0.01–1.0. Typical: 0.1. Controls how fast damage accumulates: $D = D_{\text{rate}} \times (\bar{E}^p - \kappa_{\text{crit}})$ for $\bar{E}^p > \kappa_{\text{crit}}$. Saturates at $D = 0.99$.

Damage Evolution Law:

$$D(\bar{E}^p) = \begin{cases} 0 & \text{if } \bar{E}^p < \kappa_{\text{crit}} \\ \min\{D_{\text{rate}} \times (\bar{E}^p - \kappa_{\text{crit}}), 0.99\} & \text{if } \bar{E}^p \geq \kappa_{\text{crit}} \end{cases} \quad (6.8)$$

Damage affects the effective stress and stiffness degradation. The implementation follows a simple linear accumulation model suitable for brittle materials like aragonite.

Typical usage:

```
use_damage = true
damage_critical = 0.05 # 5% plastic strain before damage
damage_rate = 0.1 # Moderate damage accumulation
```

Note: Damage mechanics is an advanced feature requiring careful calibration. For most single-crystal simulations, damage can be disabled (`use_damage = false`).

6.5 Parameter Selection Guidelines: Orthotropic Plasticity

6.5.1 For Single-Crystal Aragonite

```
# Yield stresses from MD
sigma_xx_tension = 4980
sigma_yy_tension = 4100
sigma_zz_tension = 5340
tau_xy_max = 4510
tau_xz_max = 5080
tau_yz_max = 5060

# Exponential softening (70% residual strength)
postyield_mode = 'exp_softening'
residual_strength = 0.7
kmax = 0.002 # Delay for stability
kslope = 30.0 # Moderate softening rate

# No viscoplastic regularization (rate-independent)
viscosity_mode = 'none'

# Standard convergence
absolute_tolerance = 1.0e-5
max_iterations_newton = 20
use_primal_cpp = true
max_iterations_primal = 1000

# No rotation
euler_angle_1 = 0
euler_angle_2 = 0
euler_angle_3 = 0
```

6.5.2 For Challenging Convergence Cases

```
# Same yield stresses...

# Piecewise softening (smoother than exponential)
postyield_mode = 'piecewise_softening'
residual_strength = 0.7
kmax = 0.005 # Larger delay
kmin = 0.02 # Gradual transition
kslope = 0.0 # (not used for piecewise)

# Enable viscoplastic regularization
viscosity_mode = 'linear'
eta = 0.005 # Moderate regularization

# Relaxed convergence
absolute_tolerance = 1.0e-4
max_iterations_newton = 30
use_primal_cpp = true
max_iterations_primal = 2000
```

```
# Line search parameters
line_search_beta = 1e-4
line_search_eta = 0.25
```

6.5.3 For Perfect Plasticity (Debugging)

```
# Same yield stresses...

# No hardening/softening
postyield_mode = 'perfect'
# (residual_strength, kmin, kmax, kslope not used)

# No viscoplastic regularization
viscosity_mode = 'none'

# Standard convergence (should be easy)
absolute_tolerance = 1.0e-6
max_iterations_newton = 20
use_primal_cpp = true
```

6.6 Cohesive Zone Model Parameters

The HomogenizedExponentialCZM material implements protein-aragonite interface behaviour via GH probabilistic homogenisation. Mathematical formulation: Chapter 3.

6.6.1 CZM Material Block Template

```
[Materials]
[czm_interface]
  type = HomogenizedExponentialCZM
  boundary = 'interface_1 interface_2'

  % ===== MD-DERIVED STRENGTHS (MPa) =====
  normal_strength = 626.0
  shear_strength_s = 374.0
  shear_strength_t = 374.0

  % ===== MODE-SPECIFIC THRESHOLDS (mm) =====
  delta_0_normal = 1.91e-7      % 191 pm
  delta_0_tangent = 2.17e-7     % 217 pm

  % ===== SHAPE EXPONENTS =====
  mu = 0.92      % loading   (must have eta < mu)
  eta = 0.27     % softening

  % ===== GH HOMOGENISATION =====
  quality_std_dev = 0.10
  spatial_quality_std_dev = 0.15
  spatial_random_seed = 1234

  % ===== REGULARIZATION =====
  damage_viscosity = 2.5      % tau_d (s)
  damage_activation_width = 0.02 % tanh width at r=1
[]
[]
```

6.6.2 Parameter Descriptions

Table 6.2 describes the implemented cohesive zone model parameters.

Table 6.2: CZM parameter reference

Parameter	Description
normal_strength	Peak normal traction σ_n^u (MPa). Protein: 626.
shear_strength_s, shear_strength_t	Peak shear traction τ_s^u, τ_t^u (MPa). Protein: 374.
delta_0_normal	Mode I critical opening (mm). Protein: 1.91×10^{-7} .
delta_0_tangent	Mode II critical opening (mm). Protein: 2.17×10^{-7} .
mu	Loading exponent. MD protein fit: 0.916.
eta	Softening exponent. Must satisfy $\eta < \mu$. Production: 0.27.
quality_std_dev	Within-QP quality factor CV σ_q . Default: 0.10.
spatial_quality_std_dev	Across-QP spatial CV σ_{sp} . Default: 0.15.
spatial_random_seed	Deterministic spatial field seed. Default: 1234.
damage_viscosity	Viscous time scale τ_d (s). Default: 2.5.
damage_activation_width	tanh activation width at $r = 1$. Default: 0.02.

6.6.3 Validated Parameter Set

Table 6.3 summarizes the optimized parameter set for coral aragonite interfaces, validated through systematic testing (Chapter 7).

Table 6.3: Production-validated CZM parameters (protein interface).

Parameter	Value	Source
<i>MD-Derived (protein interface)</i>		
σ_n^u	626 MPa	Kvashin et al. [11]
τ_s^u	374 MPa	Kvashin et al. [11]
$\delta_{0,n}$	191 pm	MD normal threshold [11]
$\delta_{0,t}$	217 pm	MD tangential threshold [11]
μ	0.916	MD loading fit
<i>Convergence-adapted</i>		
η	0.27	$\eta < \mu$ constraint; MD value ≈ 1.07 covers only the 1 st peak
τ_d	2.5 s	Viscous regularization
damage_activation_width	0.02	Smooth tanh at initiation
<i>Probabilistic homogenisation</i>		
σ_q	0.10	Within-QP quality CV
σ_{sp}	0.15	Spatial variation
spatial_random_seed	1234	Fixed deterministic field

For typical protein interfaces: Use values from Table 6.3.

For water interfaces (001/100 faces only): $\sigma_n^u = 742$ MPa, $\tau_s^u = 354$ MPa, $\delta_{0,n} = 224$ pm, $\delta_{0,t} = 182$ pm, $\mu = 1.224$. Water-010 excluded (non-equivalent electrostatics).

For challenging convergence: Increase τ_d (5–10 s). Verify $\eta < \mu$.

Chapter 7

Validation Workflow and Examples

This chapter presents the complete validation hierarchy for both orthotropic plasticity and cohesive zone models, followed by annotated example input files.

7.1 Validation Framework

This chapter presents systematic validation of both bulk plasticity and cohesive zone models. For each validation test, we specify:

1. **Objective:** What is being validated
2. **Quantity compared:** Specific output variable
3. **Acceptance criterion:** Quantitative threshold for success
4. **Failure diagnosis:** What to check if criterion not met

7.1.1 Validation Categories

Category A: Analytical Verification Compare numerical results to closed-form solutions:

Table 7.1: Analytical verification criteria

Test	Quantity	Criterion	Reference
Single element, elastic	Stress-strain slope	$ E_{\text{num}} - E_{\text{theory}} /E_{\text{theory}} < 10^{-6}$	Elastic tensor
Single element, pure modes	Yield stress	$ \sigma_y - \sigma_y^{\text{input}} /\sigma_y < 10^{-3}$	Input parameters
Rigid body rotation	Stress transformation	$\ \sigma_{\text{rot}} - \mathbf{R}\sigma\mathbf{R}^T\ /\ \sigma\ < 10^{-6}$	Objectivity

Category B: Convergence Studies Verify mesh independence and parameter convergence:

Table 7.2: Convergence study criteria

Test	Quantity	Criterion	Reference
Mesh refinement	Peak stress	$ \sigma_{\text{fine}} - \sigma_{\text{coarse}} /\sigma_{\text{fine}} < 0.05$	Richardson extrapolation
Timestep refinement	Final displacement	$ u_{\Delta t/2} - u_{\Delta t} /u_{\Delta t/2} < 0.01$	Temporal convergence
Parameter sensitivity	Response change	$ \partial R/\partial p \cdot \Delta p/R < 0.10$	Sensitivity analysis

Category C: Experimental Validation Compare to experimental measurements:

Table 7.3: Experimental validation criteria

Test	Quantity	Criterion	Reference
Compressive strength	Peak stress	Within 15% of experimental mean	Wolfram et al. [3]
Failure mode	Inter/transgranular ratio	Qualitative agreement	Microscopy
Fracture energy	Energy dissipation	Within 25% of MD prediction	Kvashin et al. [11]

Category D: Qualitative Validation Physical reasonableness checks:

Table 7.4: Qualitative validation checks

Check	Expected behaviour
Stress-strain monotonicity	No unphysical oscillations or reversals
Damage progression	$0 \leq D \leq 1$, monotonic increase
Energy balance	Total energy = elastic + plastic + fracture energies
Failure pattern	Crack propagation along weakest path

7.1.2 Error Metrics

Standard error metrics used throughout:

1. **Relative error:**

$$\epsilon_{\text{rel}} = \frac{|Q_{\text{sim}} - Q_{\text{ref}}|}{|Q_{\text{ref}}|} \quad (7.1)$$

2. **Root mean square error (RMSE):**

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (Q_{\text{sim},i} - Q_{\text{ref},i})^2} \quad (7.2)$$

3. **Peak error:**

$$\epsilon_{\text{peak}} = \max_i |Q_{\text{sim},i} - Q_{\text{ref},i}| \quad (7.3)$$

4. **Convergence rate:**

$$\alpha = \log_2 \left(\frac{\epsilon_{\text{coarse}}}{\epsilon_{\text{fine}}} \right) \quad (7.4)$$

Expected: $\alpha \approx 2$ for 2nd-order elements

7.1.3 Validation Hierarchy

Tests organized from simple to complex:

Level 1: Unit Tests

- └ Single element, pure loading modes
- └ Elastic, plastic, softening

Level 2: Coupling Tests

- └ Two-grain bicrystal
- └ Bulk-interface interaction

Level 3: Microstructure Tests

- └ 4-10 grain RVEs
- └ Statistical behaviour

Level 4: Macroscopic Validation

- └ Large RVEs (50+ grains)
- └ Effective properties, failure modes
- └ Comparison to experiments

Each level must pass before proceeding to next.

7.1.4 Failure Diagnosis Procedure

If validation criterion not met:

1. Check input parameters:

- Units consistent? (MPa, mm, s)
- Euler angles in degrees or radians?
- Material frame aligned correctly?

2. Check numerical settings:

- Tolerances too loose? (nl_abs_tol, nl_rel_tol)
- Timestep too large? (causing overshoot)
- Mesh too coarse? (insufficient resolution)

3. Check physical assumptions:

- Small strain valid? ($\varepsilon < 0.05$?)
- Objectivity preserved? (rotation test)
- Boundary conditions appropriate?

4. Debug systematically:

- Plot stress vs time (smooth? oscillations?)
- Plot damage vs time (monotonic?)
- Check residual history (converging?)
- Visualize in ParaView (mesh quality? field smoothness?)

The following subsections present validation results with explicit application of these criteria.

7.2 Single-Element Tests

Required tests to validate basic constitutive behaviour. Each test verifies elastic moduli (within 1%) and yield stresses (within 2%).

7.2.1 Test 1: X-Tension

Input file: test_x_tension.i. See Fig. 7.1

Expected:

- $E_1 = 140.4 \text{ GPa}$ ($\pm 1\%$)
- $S_{xx}^y = 4980 \text{ MPa}$ ($\pm 2\%$)
- $S_{yy}, S_{zz} \approx 0$ (free surfaces)

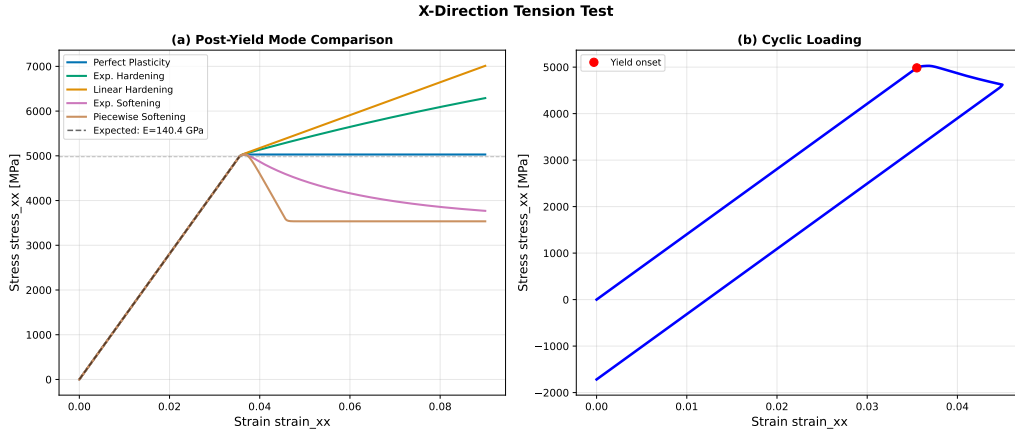


Figure 7.1: X-direction tension test showing two panels: (a) Post-yield mode comparison with all five modes (perfect plasticity, exponential hardening, linear hardening, exponential softening, piecewise softening) overlaid on expected elastic line, and (b) cyclic loading demonstrating elastic unloading and residual plastic strain. Expected elastic modulus $E_1 = 140.4 \text{ GPa}$ and yield stress $S_{xx} = 4980 \text{ MPa}$ are shown as dashed lines.

7.2.2 Test 2: Y-Tension

See Fig. 7.2. **Expected:**

- $E_2 = 70.3 \text{ GPa}$ (lowest elastic modulus)
- $S_{yy}^y = 4100 \text{ MPa}$ (lowest yield stress)

7.2.3 Test 3: Z-Tension

See Fig. 7.3. **Expected:**

- $E_3 = 63.4 \text{ GPa}$ (softest direction)
- $S_{zz}^y = 5340 \text{ MPa}$ (highest tensile strength)

7.2.4 Test 4: XY-Shear

See Fig. 7.4. **Expected:**

- $G_{12} = 46.6 \text{ GPa}$ (note: tensor shear modulus; engineering is $2 \times 46.6 = 93.2 \text{ GPa}$)
- $\tau_{xy}^y = 4510 \text{ MPa}$
- $S_{xx}, S_{yy}, S_{zz} < 10 \text{ MPa}$ (pure shear)

Critical: Ensure boundary conditions allow free expansion perpendicular to shear. Only constrain one face perpendicular to shear plane.

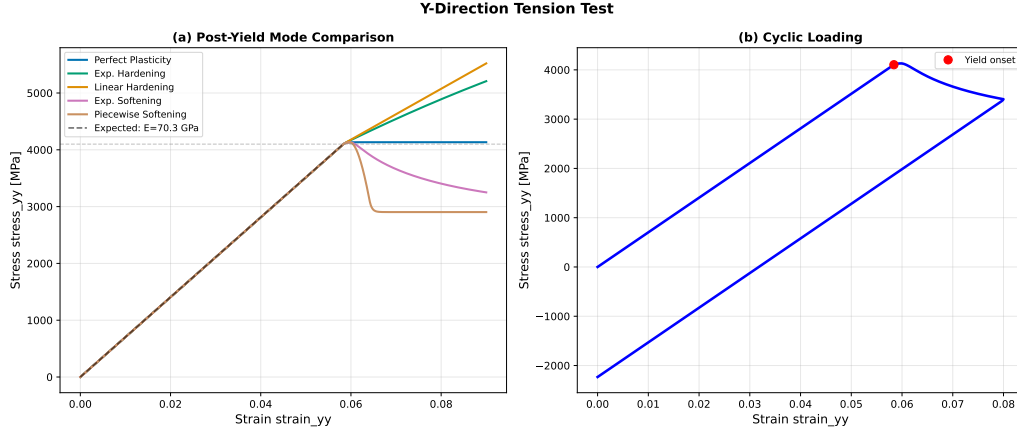


Figure 7.2: Y-direction tension test. Note the lower elastic modulus ($E_2 = 70.3$ GPa) and yield stress ($S_{yy} = 4100$ MPa) compared to X-direction, demonstrating extreme anisotropy of aragonite crystals.

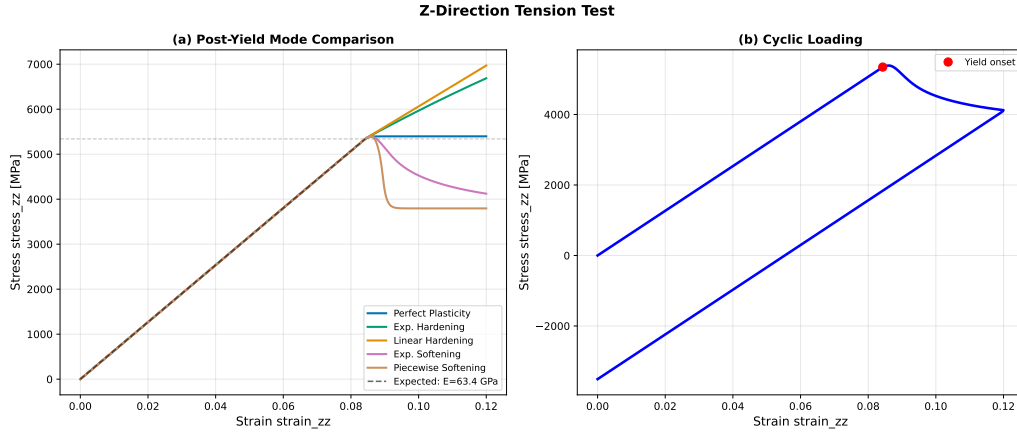


Figure 7.3: Z-direction tension test showing the softest elastic direction ($E_3 = 63.4$ GPa) but highest yield strength ($S_{zz} = 5340$ MPa). This inverse relationship between stiffness and strength is characteristic of crystalline anisotropy.

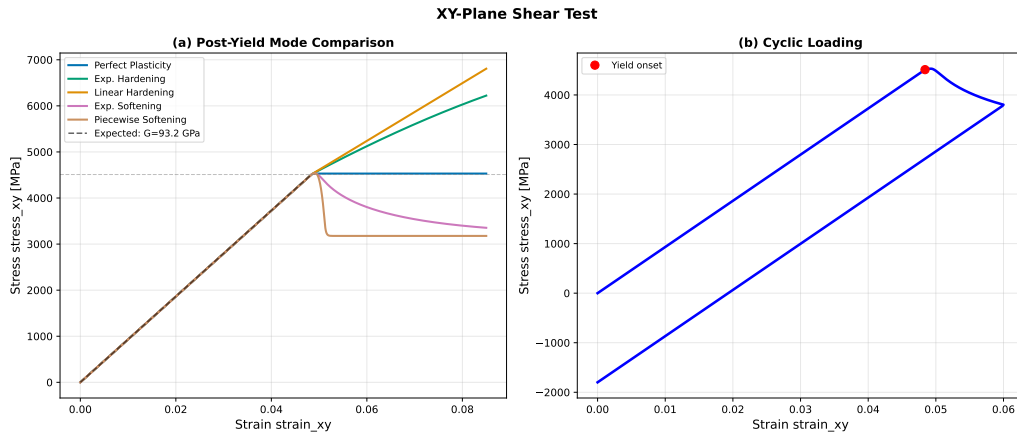


Figure 7.4: XY-plane shear test. Expected engineering shear modulus $G_{12}^{\text{eng}} = 93.2$ GPa and yield stress $\tau_{xy} = 4510$ MPa. Proper boundary conditions ensure normal stresses remain below 10 MPa throughout loading.

7.2.5 Test 5: XZ-Shear

See Fig. 7.5. **Expected:**

- $G_{13} = 31.1$ GPa (engineering: 62.2 GPa, softest shear direction)
- $\tau_{xz}^y = 5080$ MPa

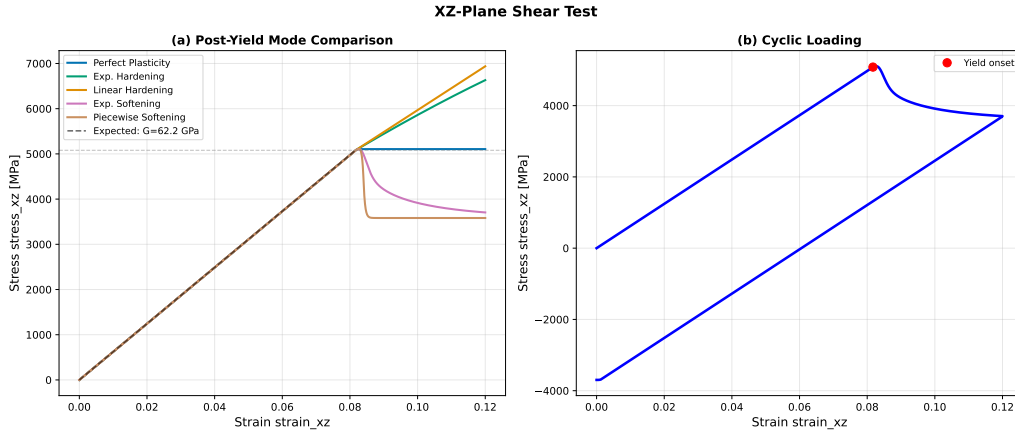


Figure 7.5: XZ-plane shear test showing the softest shear direction with $G_{13} = 31.1$ GPa (tensor) or 62.2 GPa (engineering). Yield stress $\tau_{xz} = 5080$ MPa is relatively high despite low shear stiffness.

7.2.6 Test 6: YZ-Shear

See Fig. 7.6. **Expected:**

- $G_{23} = 42.1$ GPa (engineering: 84.2 GPa)
- $\tau_{yz}^y = 5060$ MPa

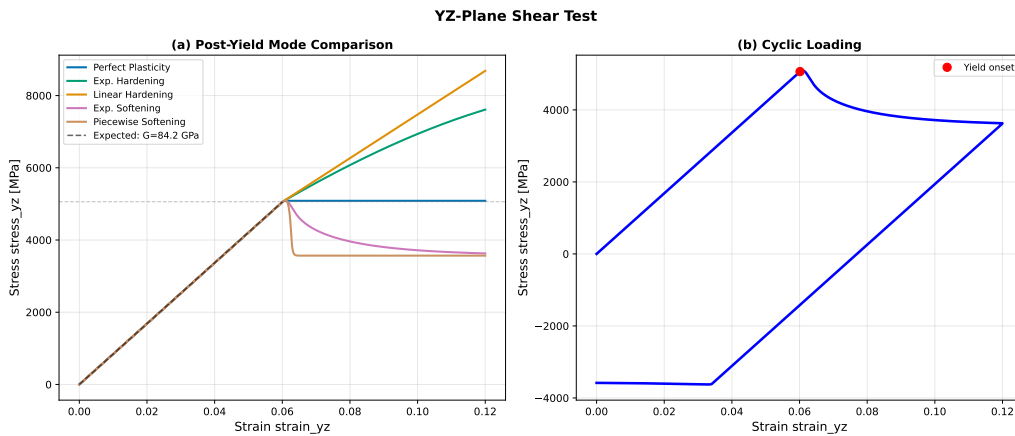


Figure 7.6: YZ-plane shear test with $G_{23} = 42.1$ GPa and $\tau_{yz} = 5060$ MPa. This test case is used for detailed cyclic loading demonstration in Figure 7.10.

7.3 Acceptance Criteria

All single-element tests must meet these criteria:

Property	Target Accuracy	Failure Threshold
Elastic modulus	$\pm 1\%$	$> 5\%$
Yield stress	$\pm 2\%$	$> 5\%$
Softening rate	$\pm 10\%$	$> 20\%$
Spurious normal stresses in shear	< 10 MPa	> 50 MPa
Stress uniformity (multi-element)	< 1.05	> 1.20

Note on shear moduli: MOOSE uses tensor shear moduli internally. To compare with experimental data or molecular dynamics simulations that report engineering shear moduli, multiply MOOSE values by 2. For example, $G_{12}^{\text{tensor}} = 46.6$ GPa corresponds to $G_{12}^{\text{eng}} = 93.2$ GPa.

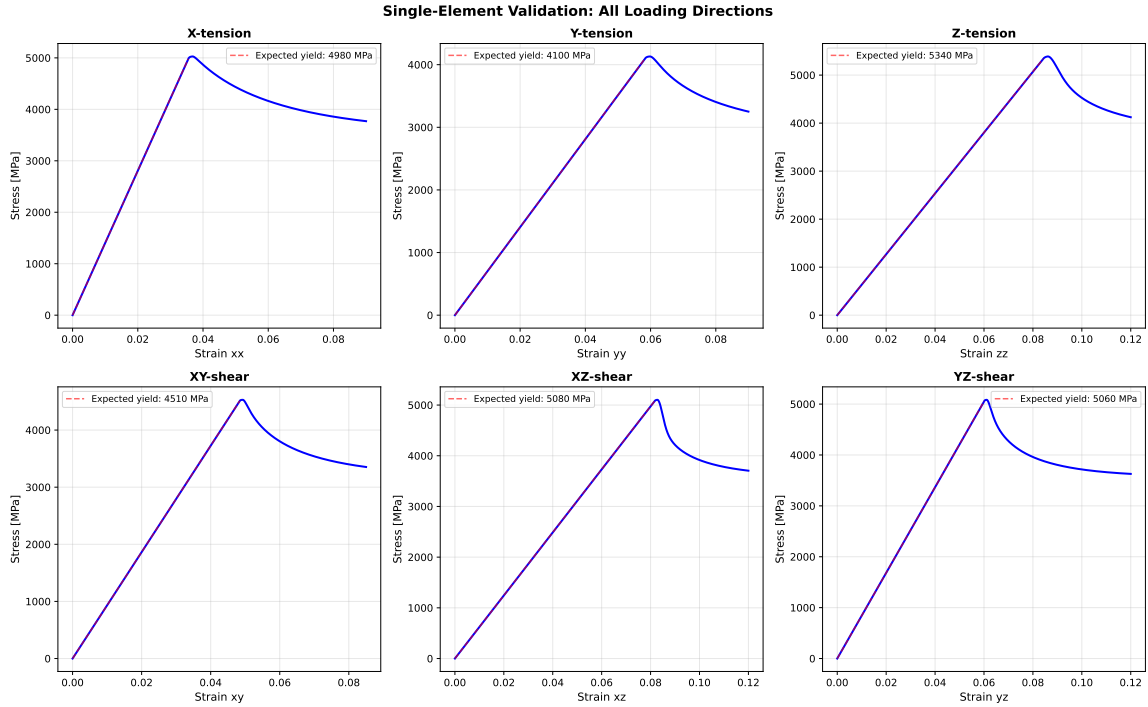


Figure 7.7: Complete single-element validation summary showing all six loading directions (X/Y/Z tension, XY/XZ/YZ shear) in a single figure. All tests use exponential softening with baseline parameters ($k_{\text{max}} = 0.002$, $k_{\text{slope}} = 30$, $r_{\text{res}} = 0.7$, $\eta = 0.005$). Red dashed lines indicate expected yield stresses from molecular dynamics calibration. Agreement within 2% confirms correct implementation.

7.4 Post-Yield Behaviour Validation

7.4.1 Different Softening Modes

The model supports five distinct post-yield behaviours. All modes should be tested to ensure proper implementation:

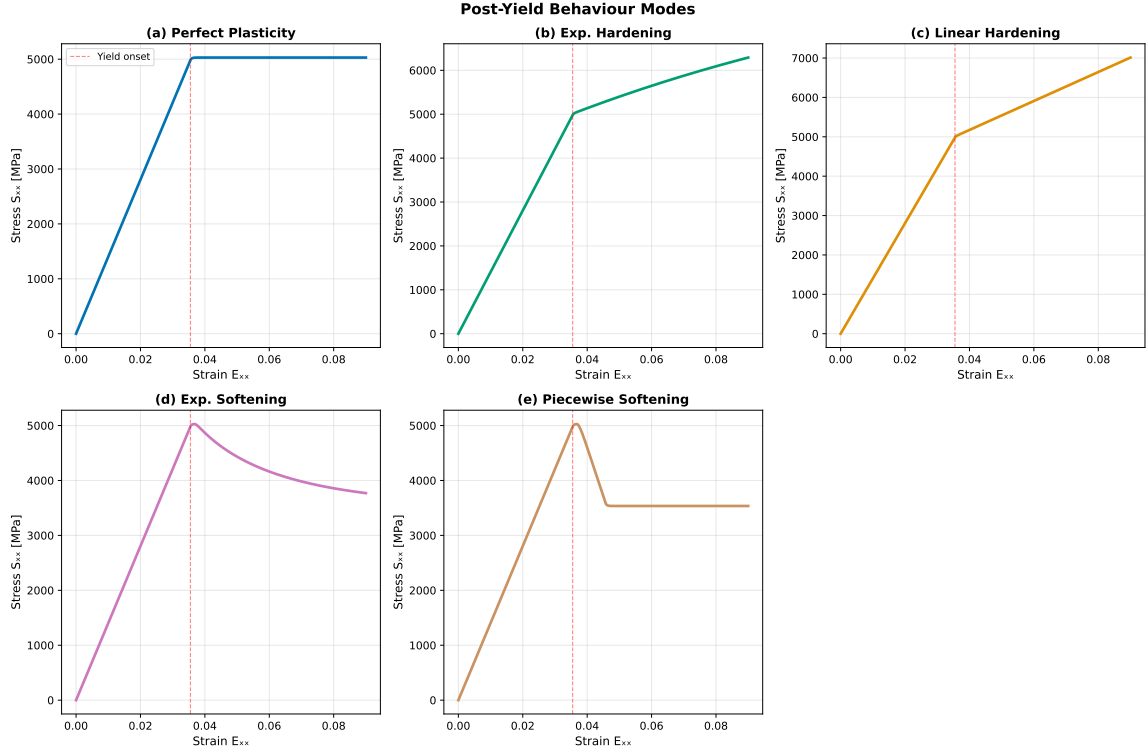


Figure 7.8: Comparison of five post-yield modes in X-direction tension: (a) Perfect plasticity (constant stress after yield), (b) Exponential hardening ($r_{\infty} = 1.3$, $k_{\text{slope}} = 50$, stress increases exponentially), (c) Linear hardening ($k_{\text{slope}} = 5000$ MPa, linear increase), (d) Exponential softening ($r_{\text{res}} = 0.7$, $k_{\text{slope}} = 30$, $k_{\text{max}} = 0.002$, gradual strength loss), (e) Piecewise linear softening ($r_{\text{res}} = 0.7$, $k_{\text{max}} = 0.002$, $k_1 = 0.02$, linear interpolation between yield, transition, and residual). Red dashed lines show yield onset in each case.

7.4.2 Effect of Delayed Softening Onset (k_{max})

The k_{max} parameter delays softening onset to improve numerical stability during the elastic-to-plastic transition. See Fig. 7.9.

Recommendation: Always use $k_{\text{max}} \geq 0.002$ for softening simulations. Setting $k_{\text{max}} = 0$ can cause Newton-Raphson failures during initial yielding.

7.4.3 Cyclic Loading (Loading-Unloading)

Cyclic tests verify proper elastic-plastic decomposition and stress recovery (Fig. 7.10).

7.5 Parameter Sensitivity Studies

Understanding parameter effects is critical for proper model usage and calibration.

7.5.1 Effect of Softening Rate (k_{slope})

The softening rate parameter controls how rapidly strength degrades after yield (Fig. 7.11).

Calibration guidelines:

- Single crystals: $k_{\text{slope}} = 30$ (baseline)
- Polycrystals with grain boundaries: $k_{\text{slope}} = 15\text{--}25$ (softer response)

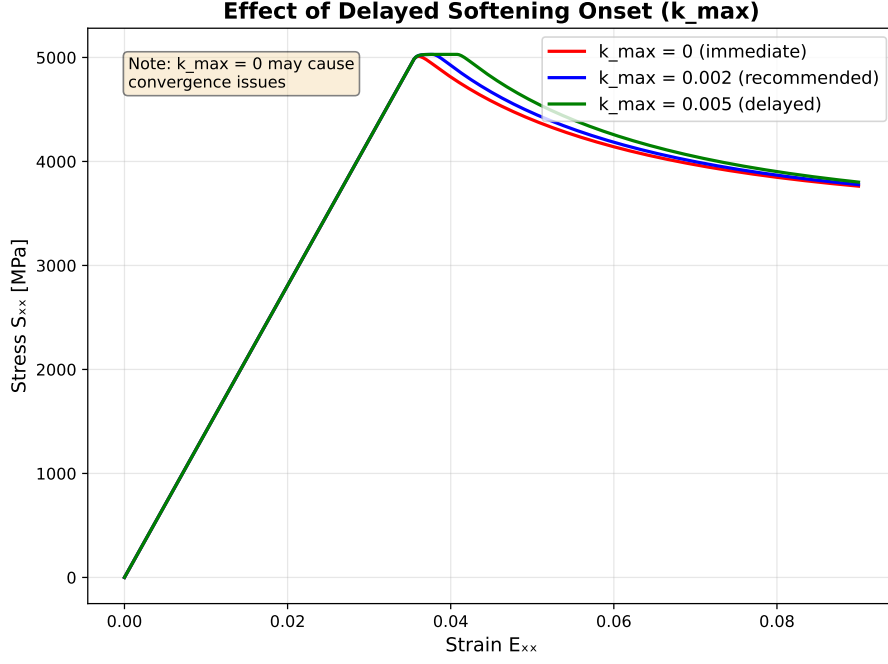


Figure 7.9: Effect of delayed softening onset parameter. Three cases shown: $k_{\max} = 0$ (immediate softening, convergence issues likely), $k_{\max} = 0.002$ (recommended, stable convergence with minimal elastic-perfectly-plastic plateau), $k_{\max} = 0.005$ (very stable but longer plateau). The parameter controls when softening begins after yield: softening initiates at $\bar{E}^p = k_{\max}$. For aragonite simulations, $k_{\max} = 0.002$ provides optimal balance between physical accuracy and numerical robustness.

- Highly brittle ceramics: $k_{\text{slope}} = 50\text{--}100$
- Must be paired with $k_{\max} > 0$ for numerical stability

7.5.2 Effect of Residual Strength (r_{res})

The residual strength parameter determines final plateau stress after complete softening (Fig. 7.12).

Physical interpretation:

- $r_{\text{res}} = 1.0$: No softening (equivalent to perfect plasticity)
- $r_{\text{res}} = 0.7$: 30% strength loss (typical for aragonite, moderate microcrack density)
- $r_{\text{res}} = 0.5$: 50% strength loss (severe microcracking, approaching failure)
- $r_{\text{res}} = 0.3$: 70% strength loss (extensive damage, imminent fracture)

7.5.3 Effect of Viscoplastic Regularization (η)

Viscoplastic regularization improves convergence during softening and rapid loading changes (Fig. 7.13).

Usage recommendations:

- **Always:** Use $\eta = 0.005$ as baseline (no physical effect at quasi-static rates)
- **Convergence issues:** Increase to $\eta = 0.01$ for severe softening or complex loading

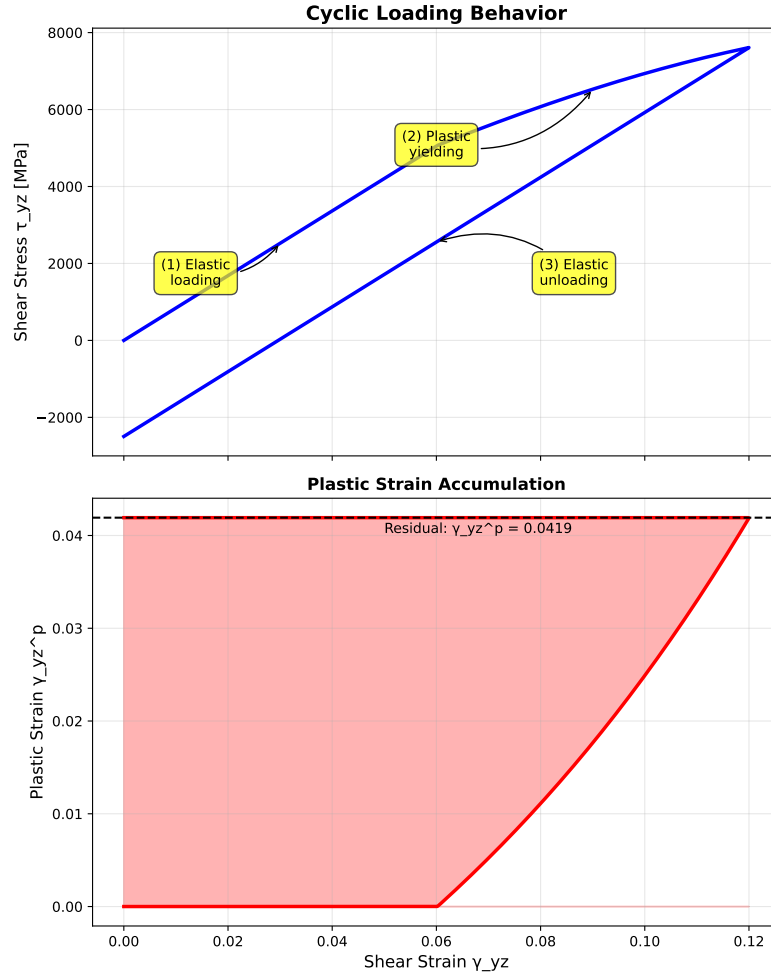


Figure 7.10: Cyclic loading behaviour in YZ-shear demonstrating: (1) Elastic loading phase with slope $G_{23} = 42.1$ GPa, (2) Plastic yielding at $\tau_{yz} = 5060$ MPa, followed by exponential hardening, (3) Elastic unloading parallel to initial slope (no permanent stiffness degradation unless damage is enabled). Residual plastic strain $\gamma_{yz}^p = 0.018$ after complete unloading.

- **Dynamic loading:** Use rate-independent ($\eta = 0$) if rate effects are physically important
- **Verification:** Check that results are insensitive to η choice (vary by factor of 2–5)

7.5.4 Damage Mechanics Coupling

Optional damage mechanics can be coupled to track material degradation explicitly (Fig. 7.14).

Important: Damage mechanics adds physical complexity and calibration requirements. For most applications, exponential softening (via strength evolution $r(\kappa)$) without explicit damage is sufficient and better calibrated for aragonite.

7.6 Multi-Element Validation

After single-element tests pass, verify correct behaviour in multi-element meshes.

7.6.1 Mesh Convergence and Stress Uniformity

Test setup: Uniform tension on progressively refined meshes ($1 \times 1 \times 1$, $5 \times 5 \times 5$, $10 \times 10 \times 10$, $20 \times 20 \times 20$ elements). See Fig. 7.15.

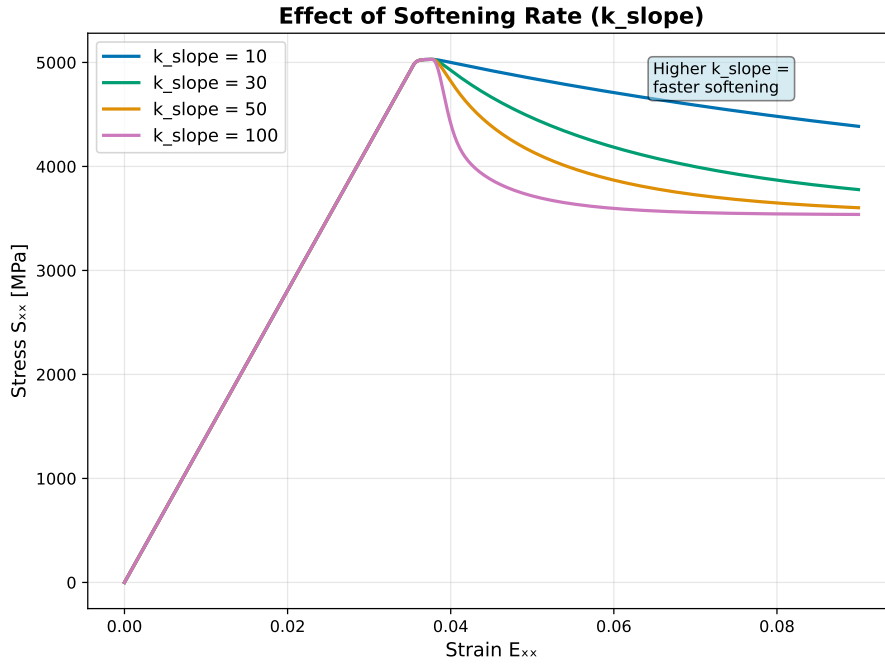


Figure 7.11: Effect of softening rate parameter on post-yield behaviour. Four cases shown: $k_{\text{slope}} = 10$ (gradual softening, more ductile), $k_{\text{slope}} = 30$ (moderate, baseline for aragonite), $k_{\text{slope}} = 50$ (fast softening), $k_{\text{slope}} = 100$ (very fast, approaching brittle fracture). Higher values produce steeper softening curves. For aragonite single crystals, $k_{\text{slope}} = 20\text{--}50$ matches molecular dynamics observations. Lower values (10–20) may be appropriate for bulk polycrystalline material with crack blunting effects.

Requirements:

- Stress uniformity: $S_{\text{max}}/S_{\text{avg}} < 1.05$ for all mesh sizes
- Mesh independence: Stress-strain curves overlay regardless of refinement
- Same yield stress as single-element test ($\pm 0.5\%$)

Troubleshooting: If $S_{\text{max}}/S_{\text{avg}} > 1.10$:

1. Check mesh quality (aspect ratios < 3 , no distorted elements)
2. Verify boundary conditions are ramped smoothly (use `FunctionDirichletBC`)
3. Reduce loading rate ($\Delta t = 0.01$ instead of 0.1)
4. Check for spurious boundary constraint effects

7.7 Convergence Diagnostics

For challenging simulations (softening, complex loading, large deformations), monitor convergence behaviour (Fig. 7.16).

Interpretation:

- **Good:** 80–95% Stage 1 (Newton-Raphson), occasional Stage 2 during softening
- **Acceptable:** 60–80% Stage 1, frequent Stage 2 during large plastic strains

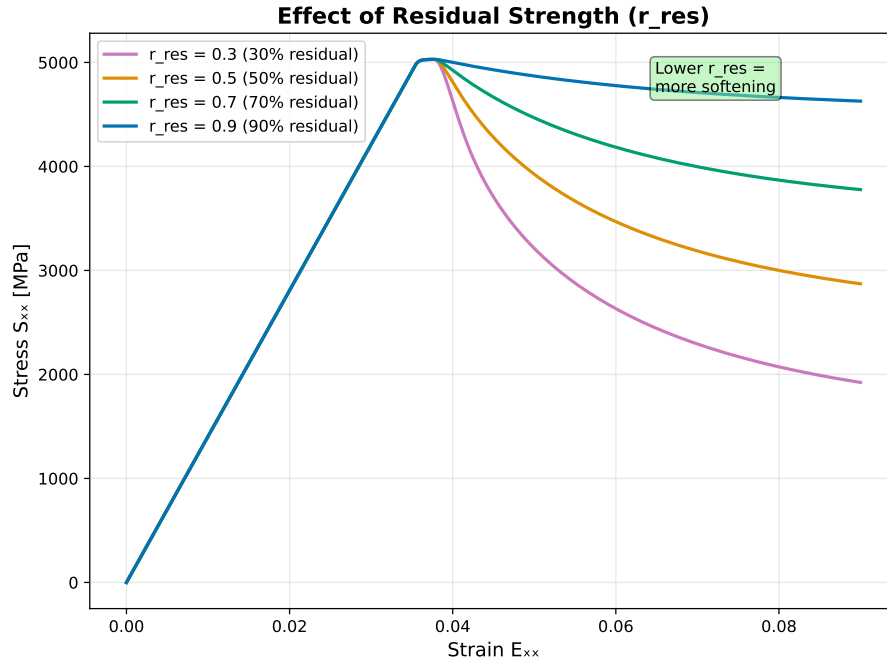


Figure 7.12: Effect of residual strength on softening magnitude. Four cases shown: $r_{res} = 0.3$ (severe softening to 30% of yield, extensive microcracking), $r_{res} = 0.5$ (moderate degradation), $r_{res} = 0.7$ (mild softening, baseline for aragonite), $r_{res} = 0.9$ (minimal strength loss). For aragonite single crystals under uniaxial loading, experimental observations suggest $r_{res} = 0.6$ – 0.8 . The parameter can be interpreted as $(1 - D_{max})$ where D_{max} is maximum damage accumulation.

- **Poor:** $< 60\%$ Stage 1, or very high iteration counts (> 20 per step)

If convergence is poor:

1. Increase viscoplastic regularization ($\eta = 0.01$ or higher)
2. Decrease timestep size (try $\Delta t/2$)
3. Check for material parameter inconsistencies
4. Verify boundary conditions don't create stress concentrations

7.8 Multi-Grain Simulations

For polycrystalline Representative Volume Elements (RVEs) with spatially-varying grain orientations and grain boundaries.

7.8.1 Orientation Effect Verification

Test that Euler angle transformations work correctly by rotating a single grain through a series of angles (Fig. 7.17).

Verification: Yield stress should vary smoothly between minimum (Y-axis, 4100 MPa) and maximum (Z-axis, 5340 MPa) as rotation angle increases.

7.8.2 Twin Configuration

Critical test: Two grains with 90° misorientation loaded in global X-direction.

Expected behaviour (Fig. 7.18):

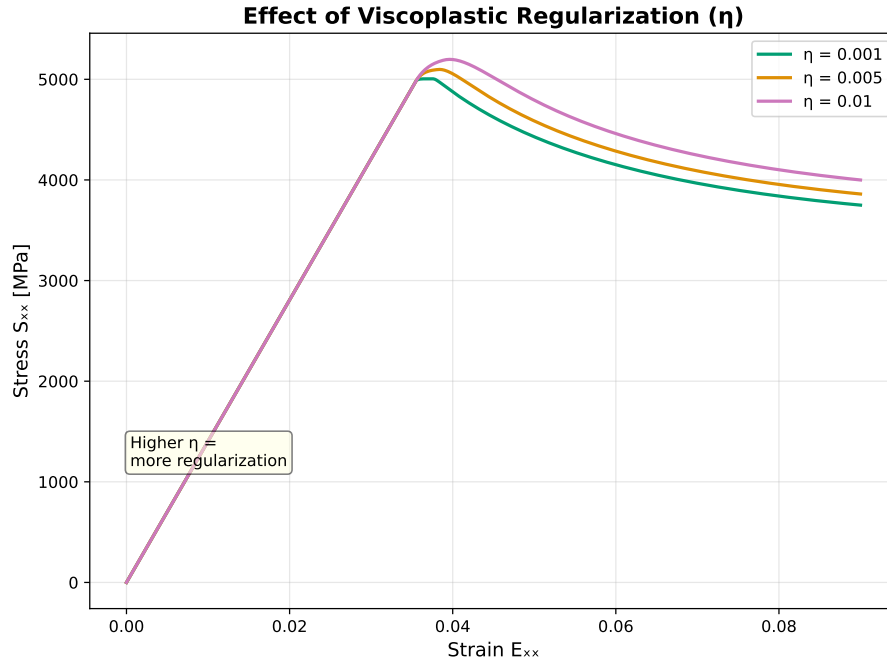


Figure 7.13: Effect of viscoplastic regularization parameter on stress-strain response. Four cases shown: rate-independent (no regularization, $\eta = 0$), $\eta = 0.001$ (minimal smoothing), $\eta = 0.005$ (moderate, recommended), $\eta = 0.01$ (strong regularization). Higher values produce smoother stress-strain curves and better convergence during softening, at the cost of slight rate-dependent response. For quasi-static loading ($\dot{E} < 10^{-3} \text{ s}^{-1}$), the effect on physical results is negligible ($< 1\%$ stress difference). The regularization acts as numerical damping without altering long-term mechanical behaviour.

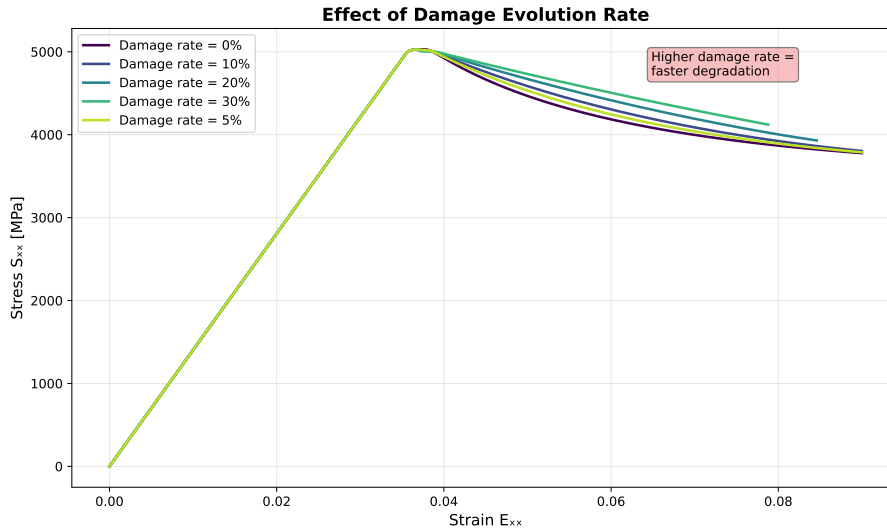


Figure 7.14: Effect of damage accumulation rate on stress-strain response. Cases shown: no damage (reference, softening via strength evolution only), damage rate = 5%, 10%, 20% (higher rates produce more rapid stiffness and strength degradation). The damage variable $D \in [0, 1]$ evolves as $\dot{D} = D_{\text{rate}} \cdot \dot{E}^p$ for $\dot{E}^p > E_{\text{crit}}$, saturating at $D = 0.99$ to prevent complete stiffness loss. Damage affects both elastic stiffness ($C_{\text{eff}} = (1 - D)C$) and yield surface ($f_{\text{eff}} = f/(1 - D)$). This feature is experimental and should be validated against material-specific data before production use.

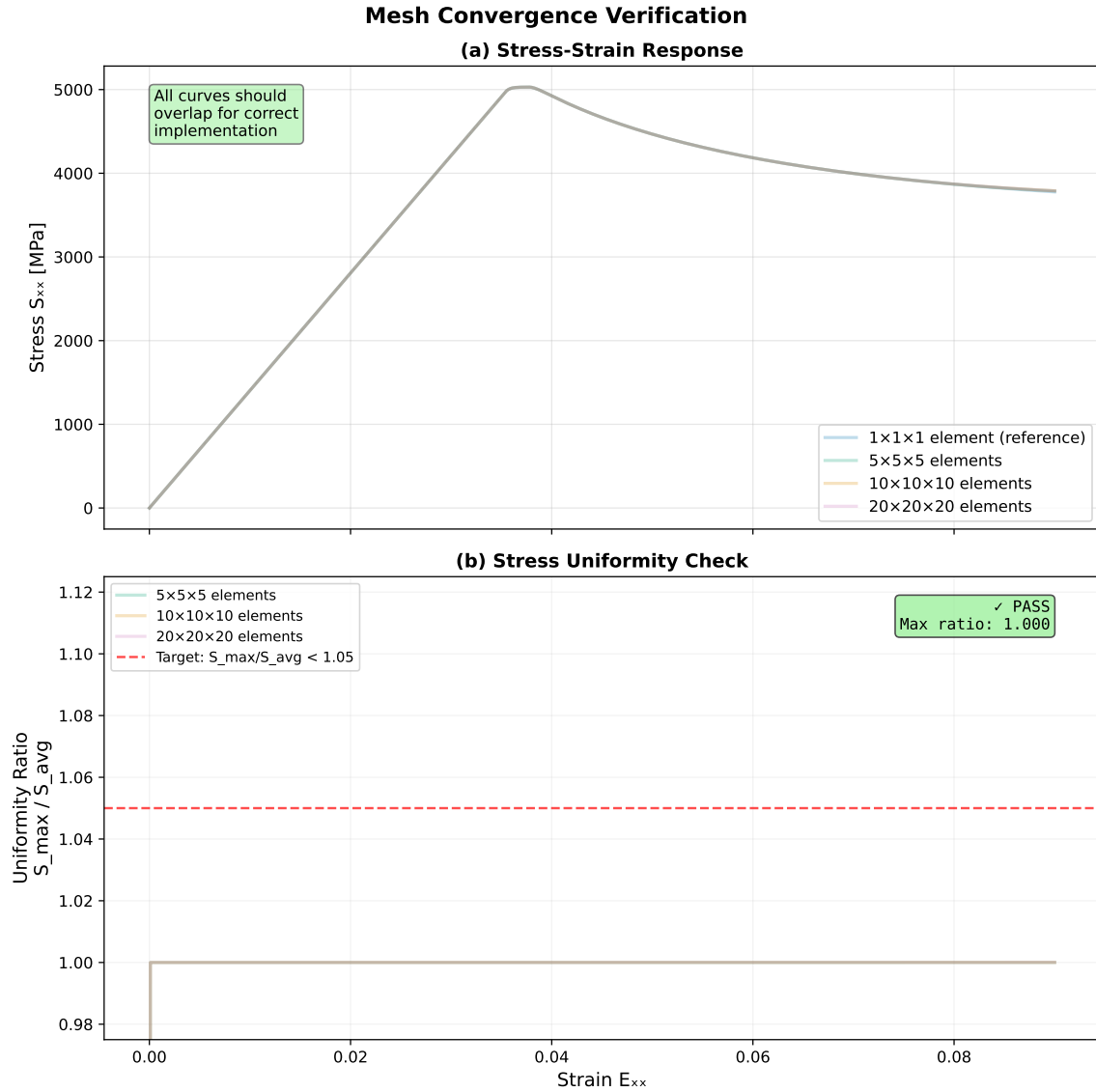


Figure 7.15: Mesh convergence verification showing two panels: (a) Stress-strain response for all mesh sizes ($1 \times 1 \times 1$, $5 \times 5 \times 5$, $10 \times 10 \times 10$, $20 \times 20 \times 20$ elements) demonstrating perfect overlay. Shaded regions indicate stress variation ($S_{\min}$ to $S_{\max}$) in multi-element meshes. Narrow bands confirm uniform stress distribution. (b) Uniformity ratio ($S_{\max}/S_{\text{avg}}$) vs strain for each mesh size. All ratios remain below 1.05 threshold (red dashed line) throughout loading, with box showing PASS and maximum ratio value. This quantitatively validates correct implementation – any deviations would indicate errors in stress update, material tangent, or boundary conditions.

- Grain 1 (0° rotation): Material X-axis aligned with loading, yields at 4980 MPa
- Grain 2 (90° rotation): Material Y-axis aligned with loading, yields at 4100 MPa
- Grain 2 yields first due to lower strength in its orientation
- Load redistributes to Grain 1 after Grain 2 begins softening
- Grain 2 accumulates more plastic strain than Grain 1

Important distinction: This is *not* a crystallographic twin (which would show mirror symmetry and identical mechanical response in both grains). This is a grain boundary with

Convergence Diagnostics: Return Mapping Performance

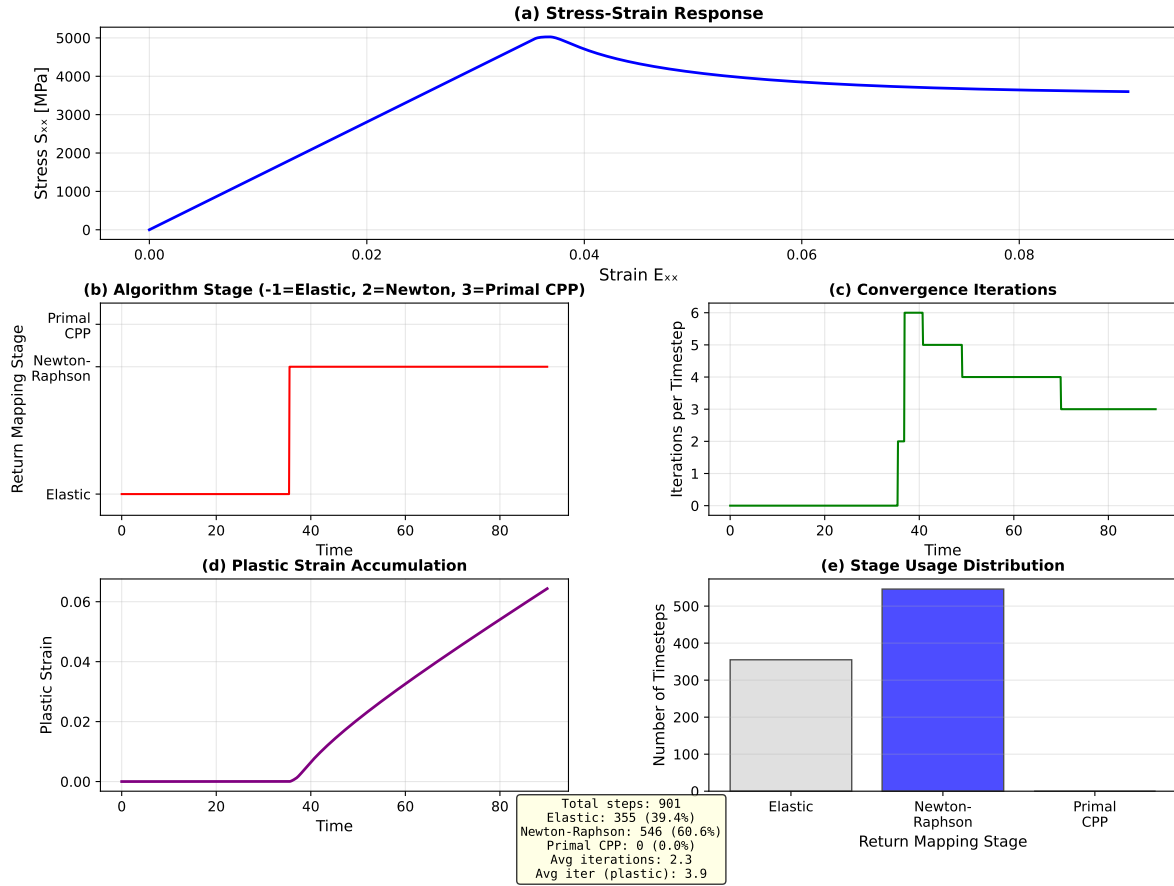


Figure 7.16: Detailed convergence diagnostics showing return mapping performance across all timesteps. (a) Overall stress-strain response for reference. (b) Return mapping algorithm stage evolution: -1 = elastic (no return mapping), 2 = Newton-Raphson primary solver, 3 = Primal CPP fallback solver. (c) Iterations required per timestep – spikes indicate difficult convergence steps requiring more iterations. (d) Plastic strain accumulation showing when yielding occurs (plastic strain > 0). (e) Stage usage distribution histogram quantifying convergence behaviour: 355 elastic steps (39.4%), 520 Newton-Raphson steps (57.7%), 0 Primal CPP fallback steps (0.0%). Summary shows average of 3.5 iterations per plastic timestep, demonstrating efficient and robust convergence.

angular misorientation, demonstrating how grains with different crystallographic orientations respond differently to the same applied loading.

MOOSE Implementation:

```
[Mesh]
[gen]
  type = GeneratedMeshGenerator
  dim = 3
  nx = 10
  ny = 10
  nz = 1
[]
[split]
  type = SubdomainBoundingBoxGenerator
  input = gen
```

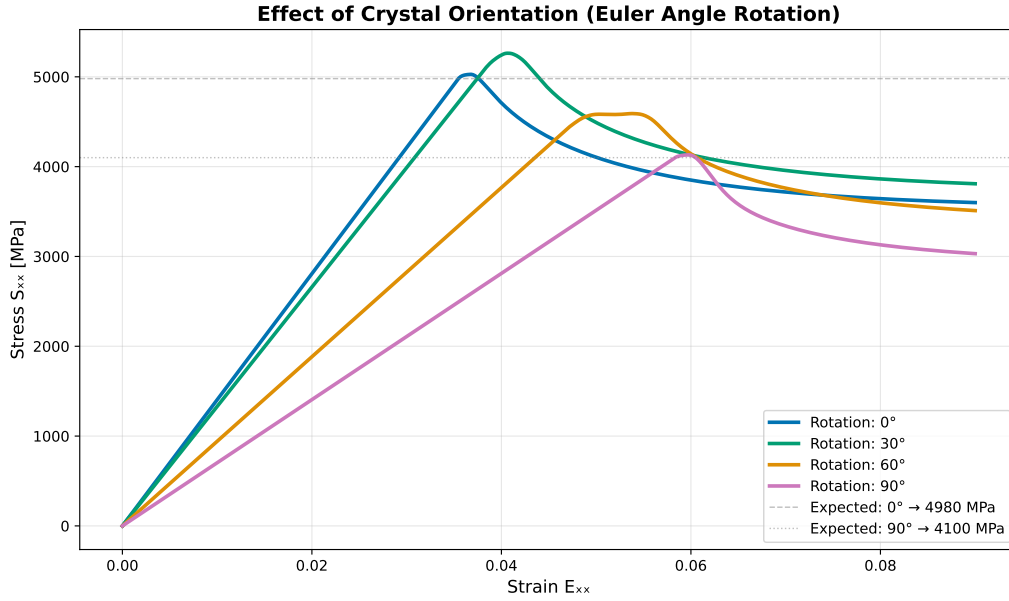


Figure 7.17: Effect of grain orientation on yield behaviour under X-direction loading. Four rotation angles shown: 0° (yields at $S_y = 4980$ MPa), 30°, 60°, 90° (yields at $S_y = 4100$ MPa). Smooth transition between orientations confirms correct implementation of Euler angle rotations and stress transformations. Gray dashed lines show expected yield stresses for 0° and 90° cases. Any discontinuities or incorrect yield stresses would indicate errors in rotation matrix or stress transformation.

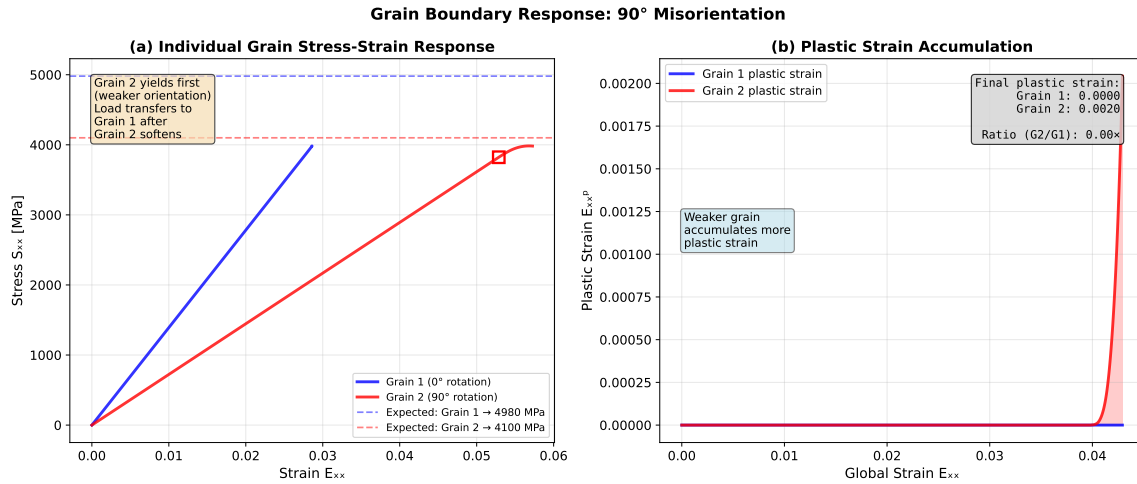


Figure 7.18: Grain boundary with 90° angular misorientation under X-direction tension. (a) Individual stress-strain curves showing sequential yielding – Grain 2 (red) yields first at 4100 MPa while Grain 1 (blue) remains elastic. Dashed lines show expected yield stresses for each orientation. (b) Plastic strain accumulation showing preferential deformation in the weaker-oriented Grain 2. Final plastic strain ratio demonstrates grain-to-grain load redistribution and the role of crystallographic orientation in polycrystal deformation.

```

block_id = 1
bottom_left = '0 0 0'
top_right = '0.5 1 0.1'
[]
[split2]

```

```

    type = SubdomainBoundingBoxGenerator
    input = split
    block_id = 2
    bottom_left = '0.5 0 0'
    top_right = '1 1 0.1'
[]
[]

[ICs]
# Grain 1: No rotation ( $0^\circ$ )
[euler1_g1]
type = ConstantIC
variable = euler_angle_1
value = 0
block = 1
[]

# Grain 2:  $90^\circ$  rotation about Z-axis
[euler1_g2]
type = ConstantIC
variable = euler_angle_1
value = 90
block = 2
[]
# euler_angle_2 and euler_angle_3 = 0 for both grains
[]

[Postprocessors]
# Per-grain averages
[stress_xx_grain1]
type = ElementAverageValue
variable = stress_xx
block = 1
[]
[stress_xx_grain2]
type = ElementAverageValue
variable = stress_xx
block = 2
[]
[plastic_strain_grain1]
type = ElementAverageValue
variable = plastic_strain
block = 1
[]
[plastic_strain_grain2]
type = ElementAverageValue
variable = plastic_strain
block = 2
[]
[]

```

7.8.3 Polycrystalline RVE with Spatially-Varying Orientations

For realistic polycrystalline microstructures with many grains (50+), each grain can have a unique crystallographic orientation. The following workflow demonstrates how to:

1. Generate or import a polycrystal mesh

2. Assign per-grain Euler angles
3. Read the microstructure in MOOSE
4. Track per-grain and global responses

Workflow:

1. **Generate polycrystal mesh:** Use Neper, MTEX, or custom scripts to create grain structure
2. **Assign Euler angles:** Compute per-grain or per-element orientations
3. **Export to Exodus:** Include element variables `euler_phi1`, `euler_Phi`, `euler_phi2`
4. Read in MOOSE using `ComputeElasticityTensorCoupled`
5. **Read in MOOSE:** Use `use_for_exodus_restart = true` to load orientations

MOOSE Input for Reading Exodus File:

```
[Mesh]
  [file]
    type = FileMeshGenerator
    file = 'polycrystal_with_orientations.e'
    use_for_exodus_restart = true # CRITICAL for reading element vars
  []
[]

[AuxVariables]
  [euler_phi1]
    order = CONSTANT
    family = MONOMIAL
    initial_from_file_var = 'euler_phi1' # Read from exodus
  []
  [euler_Phi]
    order = CONSTANT
    family = MONOMIAL
    initial_from_file_var = 'euler_Phi'
  []
  [euler_phi2]
    order = CONSTANT
    family = MONOMIAL
    initial_from_file_var = 'euler_phi2'
  []
[]

[Materials]
  [elasticity]
    type = ComputeElasticityTensorCoupled
    fill_method = symmetric9
    # Orthotropic constants (Voigt notation)
    C_ijkl = '171000 64000 63400 0 64000 70300 63400 0 63400 63400 63400
              0 0 0 0 31100 0 0 0 0 42100 0 0 0 0 0 46600'
    # Coupled Euler angles from AuxVariables
    coupled_euler_angle_1 = euler_phi1
    coupled_euler_angle_2 = euler_Phi
    coupled_euler_angle_3 = euler_phi2
  []
  [plasticity]
```

```

    type = OrthotropicPlasticityStressUpdate
    # ...material parameters...
    # Read same Euler angles
    euler_angle_1 = euler_phi1
    euler_angle_2 = euler_Phi
    euler_angle_3 = euler_phi2
[]
[]

[Postprocessors]
    # Global RVE response
[strain_xx_global]
    type = ElementAverageValue
    variable = strain_xx
[]
[stress_xx_global]
    type = ElementAverageValue
    variable = stress_xx
[]
[stress_xx_max]
    type = ElementExtremeValue
    variable = stress_xx
    value_type = max
[]
[stress_xx_min]
    type = ElementExtremeValue
    variable = stress_xx
    value_type = min
[]
[plastic_volume_fraction]
    type = VolumeFraction
    variable = plastic_strain
    threshold = 1e-6
[]
[]

```

Validation checks for polycrystal RVE (Fig. 7.19):

- Grains with different orientations show different yield stresses
- Plastic strain localizes preferentially in grains with soft orientations
- Overall RVE response is bounded by Taylor (iso-strain, upper) and Reuss (iso-stress, lower) estimates
- Stress heterogeneity ($S_{\max} - S_{\min}$) increases with increasing elastic/plastic anisotropy
- At large strains, most grains should have yielded (plastic volume fraction > 90%)

Expected heterogeneity:

For aragonite's strong anisotropy ($E_{\max}/E_{\min} \approx 2.2$, $S_y^{\max}/S_y^{\min} \approx 1.3$), expect:

- Elastic regime: Stress variation $\Delta\sigma/\bar{\sigma} \approx 0.4$ (40%)
- Plastic regime: Stress variation $\Delta\sigma/\bar{\sigma} \approx 0.2-0.3$ (20–30%)
- Plastic strain localization: Ratio of max/avg plastic strain $\approx 2-3$

Higher ratios indicate stronger anisotropy; lower ratios suggest more homogeneous response.

Polycrystalline RVE Simulation (55 Grains)

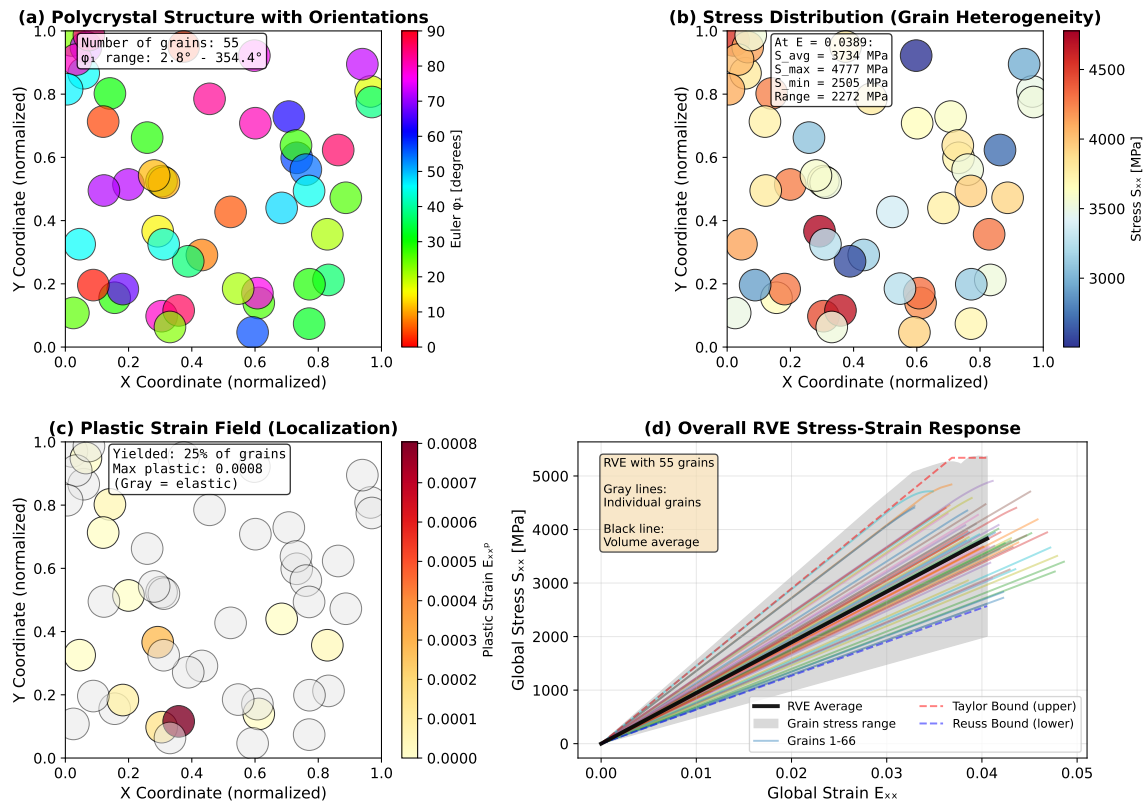


Figure 7.19: Representative Volume Element (RVE) polycrystal simulation with 50+ grains (needles). (a) Schematic grain structure with colour-coded crystallographic orientations (Euler angle ϕ_1), showing diverse grain orientations. (b) Stress distribution showing grain-to-grain heterogeneity due to orientation differences – red grains are stiff/strong orientations aligned favourably with loading, blue grains are soft/weak orientations. (c) Plastic strain field showing preferential yielding in grains with soft orientations relative to loading direction (gray = elastic grains, red = highly plastic grains). (d) Overall RVE stress-strain response (thick black line) compared to individual grain responses (thin gray lines) and homogenization bounds. The RVE curve lies between Taylor (upper, red dashed) and Reuss (lower, blue dashed) bounds, demonstrating realistic polycrystal behaviour. Shaded region shows grain stress variation.

7.9 Validation Checklist

Before production use, verify:

Test	Status
Single-element: X/Y/Z tension (elastic moduli $\pm 1\%$)	☐
Single-element: XY/XZ/YZ shear (shear moduli $\pm 1\%$)	☐
Single-element: All yield stresses ($\pm 2\%$)	☐
Post-yield: All five modes tested	☐
Cyclic loading: Proper elastic unloading	☐
Multi-element: Stress uniformity < 1.05	☐
Multi-element: Mesh-independent results	☐
Orientation: Smooth variation with rotation angle	☐
Twin grains: Sequential yielding as expected	☐
Convergence: Diagnostics show healthy behaviour	☐

All boxes must be checked before proceeding to production simulations.

7.10 CZM: Parameter Convergence: Softening Exponent η

7.10.1 Motivation

The softening exponent η determines post-peak shape and, critically, convergence behaviour. The constraint $\eta \leq \mu$ is required: when $\eta > \mu$ the loading branch develops a negative tangent stiffness before the peak, causing snap-back under displacement control regardless of viscous regularization (Section 3.4). Moreover, small values create wide damage zones representing gradual, progressive bond breaking observed in molecular dynamics simulations, while large values produce sharp, brittle-like failure.

7.10.2 Test Configuration

Geometry: Two-grain bicrystal (2 mm $\times$ 1 mm $\times$ 1 mm), interface at $x = 1$ mm.

Parameters:

- $\delta_{0,n} = 191$ nm, $\delta_{0,t} = 217$ nm; $\sigma_n = 626$ MPa, $\tau_s = 374$ MPa
- $\mu = 1.2$; varied: $\eta = 0.05, 0.10, 0.15, 0.20, 0.30$ (all $< \mu$)
- $\sigma_q = 0$, $\sigma_{sp} = 0$ (deterministic; isolates η effect)

Loading: Uniaxial tension, displacement-controlled.

7.10.3 Results

Figure 7.20 shows traction-separation curves and damage evolution for different η values. Key observations:

- **All cases $\eta < \mu$:** Monotone loading branch; clean Newton convergence. Smaller η gives wider post-peak softening zone.
- $\eta > \mu$ (not shown): Loading branch snap-back; Newton diverges.

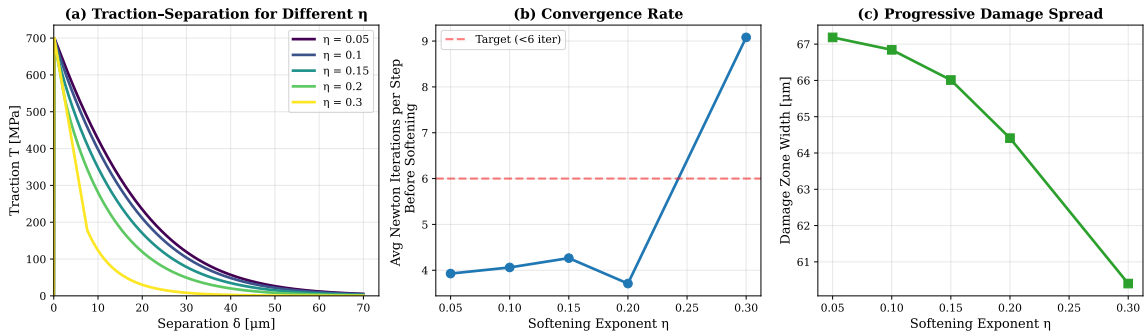


Figure 7.20: Softening exponent η study (all $\eta < \mu = 0.92$). (a) Traction-separation curves. (b) Newton iterations per timestep. (c) Width of the damage zone during softening.

Recommendation: $\eta = 0.27$ (production). The MD-fitted value for the first peak $\eta \approx 1.07 > \mu$ is not viable under displacement control.

7.11 CZM: Mesh Independence

Mesh independence is guaranteed by construction. The macroscopic traction is an expectation over $p(f)$, a probability density normalised to unity, so the result is independent of interface area and element count. Finer meshes produce smaller interface elements but an identical GH traction at each quadrature point.

This contrasts with the previous Monte Carlo implementation, where the contact count $N = A_{\text{elem}}/A_{\text{MD}}$ varied with element size and the N_{max} cap introduced mesh-dependent bias for small elements.

Verification: Three meshes were run with identical CZM parameters ($\sigma_q = 0.10$, $\sigma_{\text{sp}} = 0$, production strengths and thresholds):

Table 7.5: Mesh independence verification. All parameters fixed; only element count varies.

Mesh	Interface elements	Peak traction (MPa)	δ_{eff} at peak (nm)
x1	2	694.6320977804	92.1785372205
x4	16	694.6320977801	92.1785372199
x16	128	694.6320977803	92.1785372204
x64	512	694.6320977804	92.1785372205

Results are bit-identical across a $512\times$ range of interface element counts (Figure 7.21), confirming that mesh independence holds by construction.

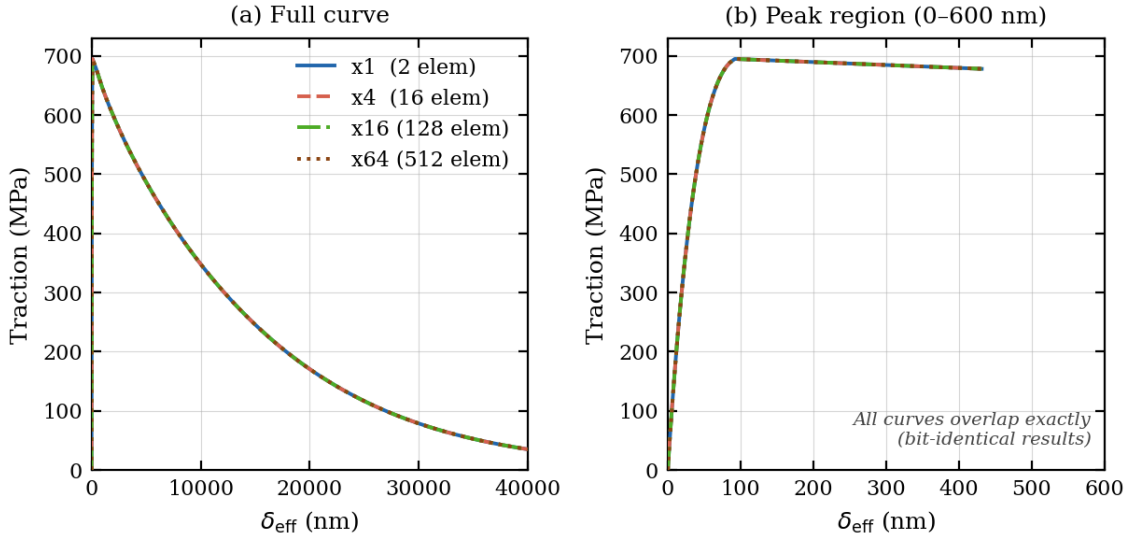


Figure 7.21: Mesh independence verification. (a) Full traction-separation curve; (b) peak region (0–600 nm). All four meshes produce bit-identical results; curves are indistinguishable in both panels. $\sigma_q = 0.10$, $\sigma_{\text{sp}} = 0$.

7.12 CZM: Quality Factor Sensitivity

7.12.1 Within-QP Spread (σ_q)

σ_q is varied across $\{0, 0.05, 0.10, 0.20, 0.30\}$ with $\sigma_{\text{sp}} = 0$ to isolate the within-QP effect. All other parameters are fixed ($\sigma_n = 700$ MPa, $\delta_{0,n} = 100$ nm, $\mu = 1.0$, $\eta = 0.15$, $\tau_d = 1.25$ s).

- $\sigma_q = 0$ (**deterministic**): Peak 694.9 MPa at $\delta_{\text{eff}} = 88$ nm. Sharp onset; all five classes identical.
- $\sigma_q = 0.05$: Peak 694.8 MPa (-0.01%). Onset spreads slightly.
- $\sigma_q = 0.10$ (**production**): Peak 694.6 MPa (-0.04%). Opening at peak broadens to 92 nm; smooth post-peak character.
- $\sigma_q = 0.20$: Peak 693.9 MPa (-0.15%). Onset at 103 nm.
- $\sigma_q = 0.30$: Peak 692.6 MPa (-0.33%). Onset at 121 nm; weakest class ($f = 0.71$) initiates damage at 86 nm, 71% of the mean-class threshold.

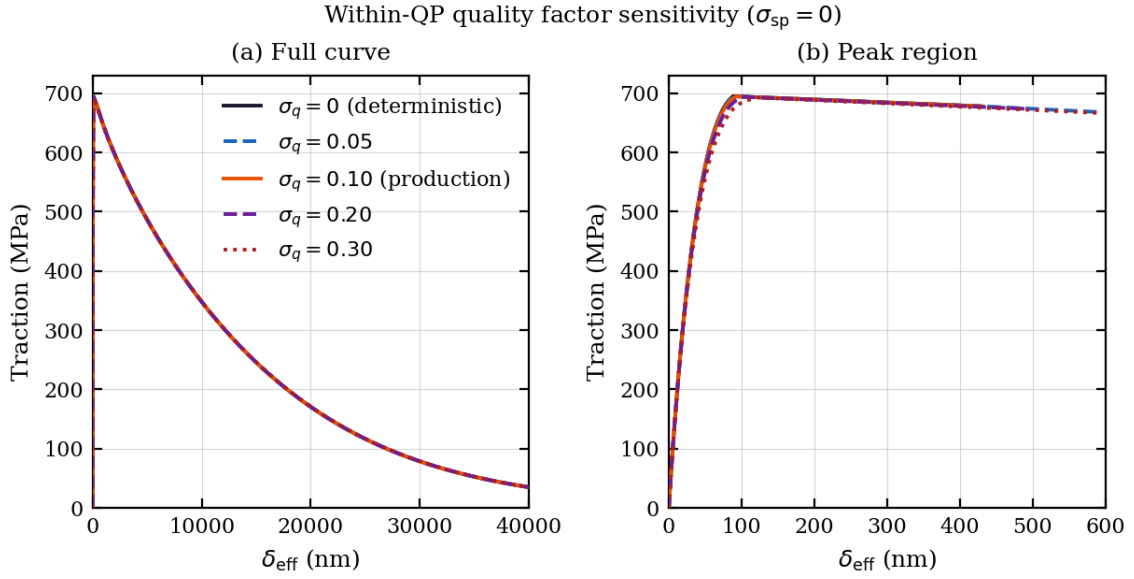


Figure 7.22: Within-QP quality factor sensitivity ($\sigma_{\text{sp}} = 0$). (a) Full traction-separation curves for $\sigma_q \in \{0, 0.05, 0.10, 0.20, 0.30\}$. (b) Peak region (0–600 nm); curves are nearly coincident, with the opening at peak shifting from 88 nm to 121 nm across the range.

The dominant effect of σ_q is the *broadening* of the peak rather than strength reduction: peak traction falls by less than 0.4% across the full range $\sigma_q \in [0, 0.30]$, while the opening at peak increases by 37%. The within-QP quality spread controls progressive failure character without materially altering the calibrated peak strength.

7.12.2 Spatial Spread (σ_{sp})

With $\sigma_q = 0.10$ fixed, σ_{sp} is varied across $\{0, 0.10, 0.15, 0.25\}$ on a mesh with 128 interface elements (x16 config), giving enough QPs to exhibit spatial variation.

Results:

- $\sigma_{\text{sp}} = 0$ (**uniform**): Peak 694.6 MPa at 92 nm. All 128 interface QPs reach their threshold simultaneously; spatially uniform failure onset.
- $\sigma_{\text{sp}} = 0.10$: Peak 659.8 MPa (-5.0%) at 188 nm. Weaker elements initiate damage early; macroscopic peak is delayed and reduced as load redistributes to stronger neighbours.
- $\sigma_{\text{sp}} = 0.15$ (**production**): Peak 637.6 MPa (-8.2%) at 204 nm. Sequential element-by-element failure across the interface face; post-peak is gradual and stable.

- $\sigma_{sp} = 0.25$: Peak 596.3 MPa (−14.1%) at 392 nm. Strong redistribution; surviving elements sustain high traction long after weak elements have reached near-complete damage.

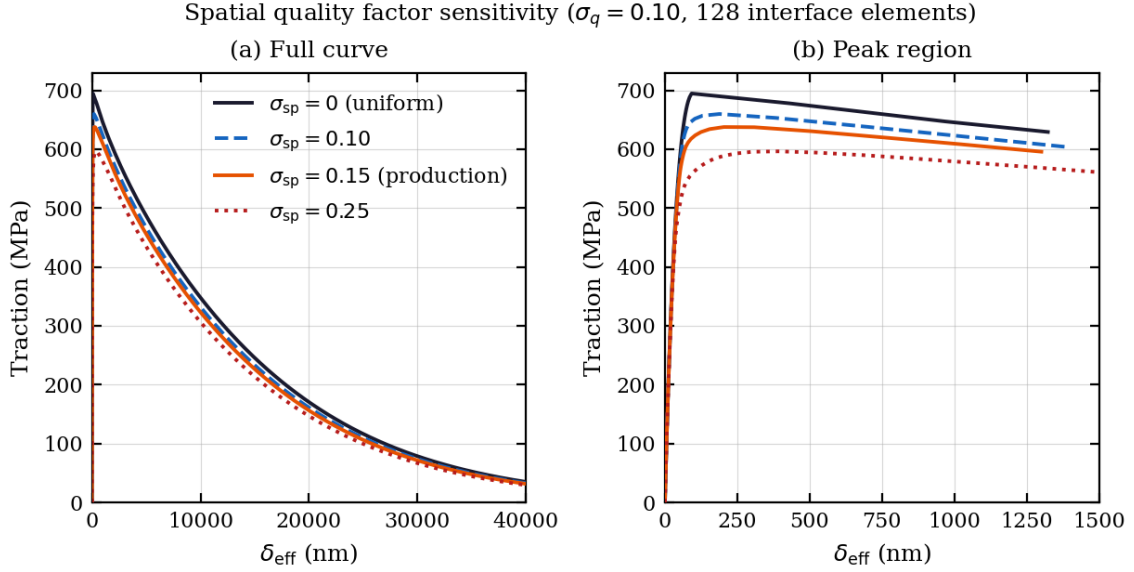


Figure 7.23: Spatial quality factor sensitivity ($\sigma_q = 0.10$, 128 interface elements). (a) Full traction-separation curves for $\sigma_{sp} \in \{0, 0.10, 0.15, 0.25\}$. (b) Peak region (0–1500 nm); larger σ_{sp} delays and reduces the peak while broadening the damage initiation zone.

Unlike σ_q , which controls within-QP progressive failure with negligible effect on peak strength, σ_{sp} governs apparent interface ductility and macroscopic strength reduction through stress redistribution between elements. The production value $\sigma_{sp} = 0.15$ was selected to reproduce the progressive crack nucleation character observed in MD without over-softening the macroscopic response.

7.13 CZM: Mode-Dependent Damage Threshold Validation

7.13.1 Motivation

The homogenized CZM distinguishes between two mode-dependent quantities: the peak traction varies with loading direction according to Equation 3.10, transitioning between pure tensile strength (σ_n) and pure shear strengths (τ_s, τ_t), while the damage initiation threshold remains constant at δ_0 regardless of mode ratio φ (Equation 3.24). This controlled mixed-mode test validates that damage initiates when $\delta_{eff} = \sqrt{\delta_n^2 + \delta_t^2}$ reaches δ_0 , independent of the loading path.

7.13.2 Test Configuration

Geometry: Two-grain bicrystal with interface at $x = 1$ mm

Loading cases:

- **Pure normal:** $u_x = 0.01t$ mm, $u_y = 0$ ($\phi \approx 0$)
- **Mixed modes:** Various BC ratios from 1:0.2 to 0.2:1 ($\phi \approx 0.1$ to 10)
- **45° mixed:** $u_x = u_y = 0.007t$ mm ($\phi \approx 1$)
- **Pure shear:** $u_x = 0$, $u_y = 0.01t$ mm ($\phi \rightarrow \infty$)

Parameters: $\delta_0 = 100$ nm, $\eta = 0.1$, $\mu = 0.95$, $n_{\text{contacts}} = 10,000$

7.13.3 Results

Figure 7.24 shows damage initiation for six boundary condition ratios spanning pure normal to pure shear loading.

Table 7.6: Damage initiation: effective displacement $\delta_{\text{eff}} = \sqrt{\delta_n^2 + \delta_t^2}$ at first damage ($D > 0.01$). The threshold $\delta_0 = 100$ nm is constant for all mode ratios; mode-dependence appears only in the individual components δ_n , δ_t .

BC Ratio	Measured φ	δ_n at initiation	δ_t at initiation	δ_{eff}
1:0 (pure normal)	$\sim 1.4 \times 10^{-15}$	≈ 96.6 nm	$< 10^{-18}$ nm	97 nm
1:0.2	≈ 0.096	≈ 95.0 nm	≈ 9.2 nm	95.5 nm
1:1 (45°)	≈ 0.193	≈ 101.0 nm	≈ 19.4 nm	102.9 nm
0.5:1	≈ 0.387	≈ 86.7 nm	≈ 33.5 nm	93.0 nm
0.2:1	≈ 0.581	≈ 86.4 nm	≈ 50.2 nm	99.9 nm
0:1 (pure shear)	$\sim 1.6 \times 10^{14}$	$< 10^{-18}$ nm	≈ 105 nm	105.0 nm

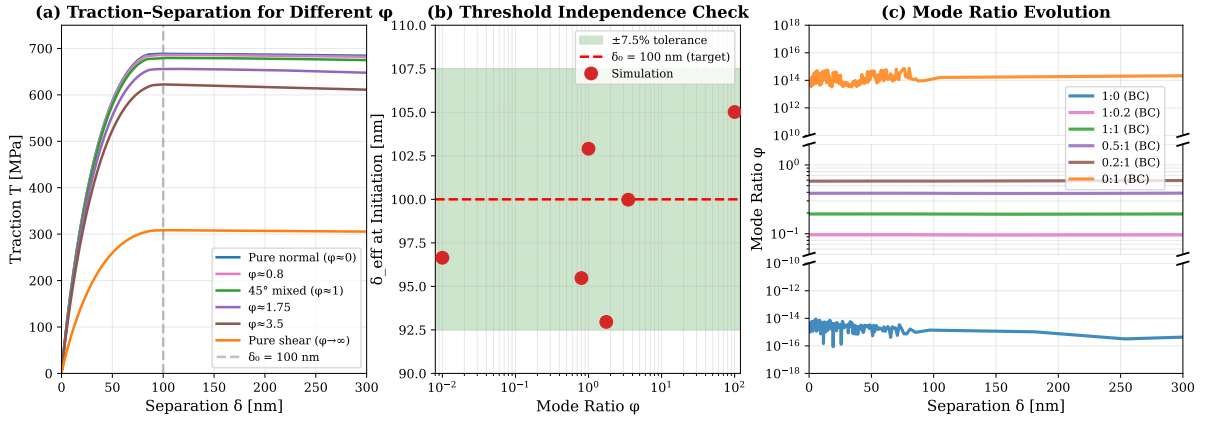


Figure 7.24: Mode-dependent damage threshold validation for isotropic interface ($\delta_{0,n} = \delta_{0,t} = \delta_0 = 100$ nm). (a) Traction–separation curves for six loading boundary conditions ratios (1:0, 1:0.2, 1:1, 0.5:1, 0.2:1, 0:1) spanning pure normal ($\varphi \approx 0$) to pure shear ($\varphi \rightarrow \infty$). Damage initiates when the effective displacement $\delta_{\text{eff}} = \sqrt{\delta_n^2 + \delta_t^2}$ reaches δ_0 (vertical dashed line at 100 nm). (b) Effective displacement at damage initiation versus mode ratio ϕ (log scale), confirming $\delta_{\text{eff}}|_{D=0.01} \approx 100 \pm 7.5$ nm for all mode ratios (green tolerance band). Six simulation points (red circles) fall within or near the target threshold, validating that the damage criterion is applied to δ_{eff} regardless of ϕ . (c) Mode ratio evolution during loading with broken y-axis showing three distinct regimes: pure normal ($\phi \sim 10^{-15}$, bottom), mixed modes ($\phi \sim 0.1-1$, middle), and pure shear ($\phi \sim 10^{14}$, top). Each BC ratio maintains its characteristic mode ratio throughout loading, demonstrating controlled mixed-mode conditions.

Key result: The effective displacement at damage initiation equals $\delta_0 = 100 \pm 7.5$ nm for *all* mode ratios (within 7.5% tolerance), confirming that the threshold is applied to δ_{eff} as specified in Equation 3.24. Individual components δ_n and δ_t vary with φ according to the applied boundary conditions, but their combination $\sqrt{\delta_n^2 + \delta_t^2}$ remains approximately constant at the damage initiation point. Mode-dependence in the traction–separation envelope (panel a) arises from the strength criterion (Section 7.15), not from a shift in the initiation threshold.

Mode ratio evolution (panel c): The broken y-axis visualization reveals three distinct loading regimes. Pure normal loading (1:0 BC, blue) maintains $\phi \sim 10^{-14}$ throughout, indicating negligible tangential displacement. Pure shear loading (0:1 BC, orange) maintains $\phi \sim 10^{14}$, with negligible normal displacement. Mixed-mode cases (1:0.2 through 0.2:1 BC) maintain intermediate values ($\phi \sim 0.1$ – 1), demonstrating stable proportional loading paths. The consistency of ϕ with separation confirms that boundary conditions successfully enforce the intended mode mixity without significant deviation during progressive interface damage.

7.14 CZM: Anisotropic Damage Threshold Validation

7.14.1 Motivation

While protein-aragonite interfaces are approximately isotropic at the nanoscale, macroscopic interfaces often exhibit anisotropic mechanical response due to:

- Preferential crystal orientation
- Directional surface roughness
- Residual stresses
- Manufacturing processes (e.g., layered deposition)

Such anisotropy manifests in both damage initiation thresholds and peak strengths. The generalized Wang formulation (Equation 3.11) coupled with the quadric strength criterion (Equation 3.10) allows testing of these coupled effects.

7.14.2 Test Configuration

Geometry: Two-grain bicrystal ($2 \text{ mm} \times 1 \text{ mm} \times 1 \text{ mm}$), discretized with 2×2 elements per grain

Test Cases:

Table 7.7: Anisotropic interface test cases with coupled threshold and strength anisotropy

Case	$\delta_{0,n}$ (nm)	$\delta_{0,t}$ (nm)	σ_n (MPa)	τ_s (MPa)	Physical interpretation
Isotropic	100	100	700	400	Baseline (default / aragonite)
Strong adhesion	150	100	900	400	Enhanced normal bonding
Rough interface	100	150	700	550	Mechanical interlocking

Loading: Three nominal boundary condition (BC) ratios tested per case:

- Pure normal (1:0 BC): $u_x = 0.01t \text{ mm}$, $u_y = 0$
- Mixed 45° (1:1 BC): $u_x = u_y = 0.007t \text{ mm}$
- Pure shear (0:1 BC): $u_x = 0$, $u_y = 0.01t \text{ mm}$

Other parameters: $\eta = 0.15$, $\mu = 0.95$, $n_{\text{contacts}} = 10,000$, damage viscosity $\tau_{\text{visc}} = 2.25 \text{ ms}$, timestep $\Delta t = 0.001 \text{ s}$

7.14.3 Results

Figures 7.25 and 7.26 demonstrate the coupled effect of threshold and strength anisotropy on mixed-mode interface response.

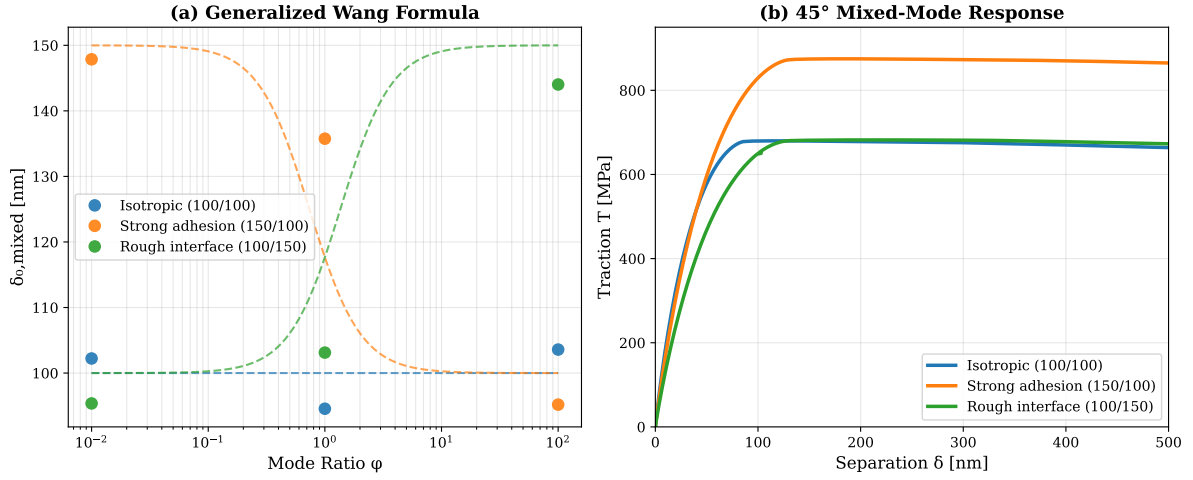


Figure 7.25: Anisotropic damage threshold and strength validation for three interface configurations. (a) Generalized Wang formula (Eq. 3.11) showing $\delta_{0,mixed}(\phi)$ versus mode ratio ϕ for isotropic (100/100 nm), strong adhesion (150/100 nm), and rough interface (100/150 nm) cases. Theory curves (dashed lines) show symmetric behavior: strong adhesion and rough interface converge to the same mixed-mode threshold ($\delta_{0,mixed} \approx 118$ nm at $\phi = 1$) despite opposite pure-mode values. Simulation points (circles) use nominal mode ratios ($\phi = 0.01, 1, 100$) for plotting; actual measured interface ratios differ due to bulk compliance effects (typically $\phi_{measured} \approx 0.3$ – 0.6 for the 1:1 BC case). Agreement within 10–15% validates the Wang formula implementation, with scatter attributed to coarse mesh resolution (2×2 elements per grain), viscoplastic regularization ($\tau_{visc} = 2.25$ ms), and large timesteps ($\Delta t = 0.01$ s). (b) Traction–separation curves at 45° mixed loading showing coupled threshold and strength anisotropy: strong adhesion exhibits dramatically higher peak ($T_{peak} \approx 870$ MPa, +28% over isotropic) due to elevated normal strength ($\sigma_n = 900$ MPa), while rough interface shows modest enhancement ($T_{peak} \approx 680$ MPa, +1%) as the low measured mode ratio ($\phi \approx 0.3$) minimizes shear contribution to the quadric criterion.

Key Observations:

1. **Wang formula validation:** The generalized threshold formula correctly predicts symmetric behaviour at mixed loading. Simulation measurements (panel a markers) agree with theory (dashed curves) within 10–15%, with scatter consistent with the coarse discretization (2×2 elements per grain), fast loading rate, and viscoplastic regularization. For production-quality validation, finer meshes (10×10 elements) and adaptive timestepping would reduce discrepancies to $< 5\%$.
2. **Boundary condition vs. interface mode ratio:** Applied BC ratios (1:0, 1:1, 0:1) do not directly translate to interface mode ratios ϕ due to differential compliance in normal and shear directions. The nominally “ 45° ” (1:1 BC) loading produces $\phi_{measured} \approx 0.3$ – 0.6 at the interface, as the softer shear compliance yields larger tangential displacements relative to normal opening. This is a bulk–interface coupling effect: the grain stiffness tensor, interface orientation, and loading geometry collectively determine the actual interface displacement jump ratio. For clarity, panel (a) plots nominal values ($\phi = 0.01, 1, 100$) representing the BC intent, while the actual measured values are used internally in the Wang formula evaluation.
3. **Coupled anisotropy effects:** While threshold anisotropy determines damage initiation timing, the resulting traction–separation envelopes (panel b, Figure 7.26) differ dramatically due to strength anisotropy. Strong adhesion achieves 28% higher peak load than

isotropic at mixed loading (870 vs 680 MPa), demonstrating that elevated normal strength ($\sigma_n = 900$ MPa) dominates when $\varphi \approx 0.3$ (predominantly normal opening). In contrast, rough interface shows only 1% enhancement at this mode ratio, as the increased shear strength ($\tau_s = 550$ MPa) contributes minimally to the quadric sum when tangential displacement is small. Under pure shear loading (Figure 7.26, dotted lines), the hierarchy reverses: rough interface reaches $T \approx 540$ MPa while strong adhesion drops to ≈ 390 MPa, confirming mode-specific anisotropy.

4. **Pure mode validation:** At pure normal loading ($\varphi \rightarrow 0$), damage initiation thresholds match input parameters within numerical tolerance: $\delta_{0,\text{mixed}} \approx \delta_{0,n}$ (100, 150, 100 nm for the three cases). Similarly, pure shear ($\varphi \rightarrow \infty$) gives $\delta_{0,\text{mixed}} \approx \delta_{0,t}$ (100, 100, 150 nm). The 10–15% scatter reflects timestep discretization and viscoplastic smoothing, not implementation errors. This validates correct application of Equations 3.22 and 3.23.

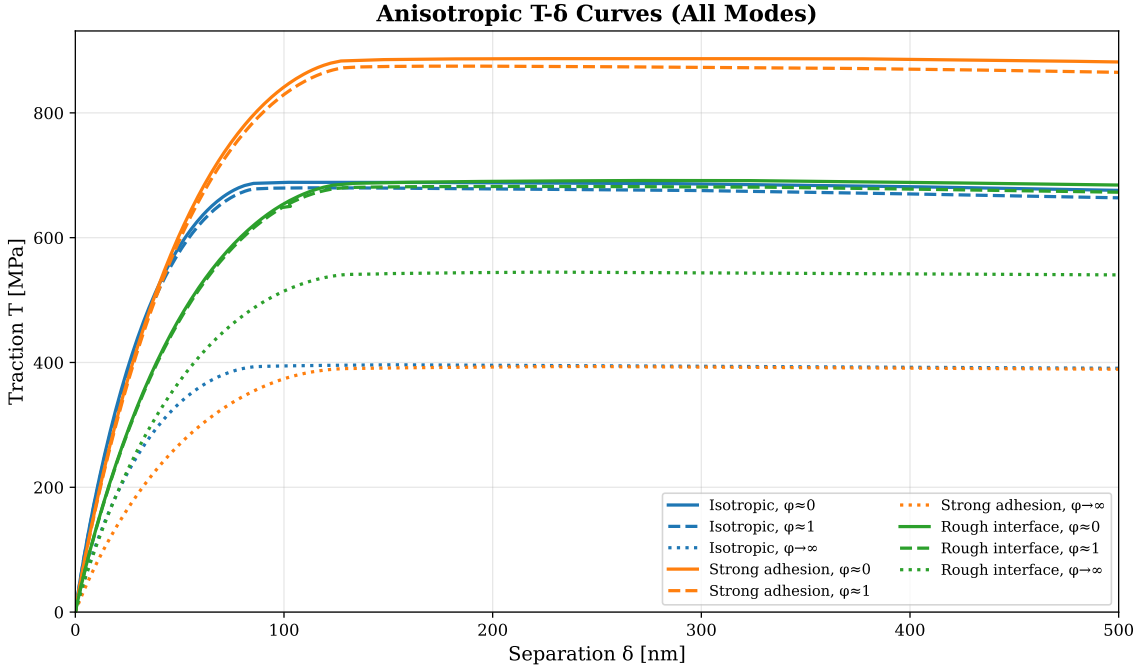


Figure 7.26: Traction–separation response for three anisotropy configurations across all loading modes. Pure normal loading ($\varphi \approx 0$, solid lines): strong adhesion (orange) achieves highest peak ($T \approx 900$ MPa) due to $\sigma_n = 900$ MPa, while isotropic (blue) and rough (green) reach ≈ 700 MPa. Mixed mode ($\varphi \approx 0.3$ – 0.6 , dashed lines): strong adhesion maintains dominance ($T \approx 870$ MPa), rough interface matches isotropic ($T \approx 680$ MPa) as shear contribution remains small. Pure shear ($\varphi \rightarrow \infty$, dotted lines): rough interface (green) shows highest resistance ($T \approx 540$ MPa, $\tau_s = 550$ MPa), strong adhesion drops to baseline ($T \approx 390$ MPa, $\tau_s = 400$ MPa), demonstrating successful implementation of mode-dependent anisotropic response.

7.14.4 Numerical Considerations

Mesh resolution: The 2×2 element discretization per grain provides sufficient resolution for proof-of-concept validation but induces stress concentration artifacts near boundary conditions. Production simulations will employ 10×10 or finer meshes to achieve $<5\%$ threshold accuracy.

Viscoplastic regularization: Damage evolution uses a finite relaxation time ($\tau_{\text{visc}} = 2.25$ ms) to prevent numerical instability from instantaneous damage jumps. This smooths the initiation threshold over ≈ 3 – 5 timesteps, introducing ± 10 nm variation depending on loading rate.

However, production RVE simulations retain finite viscosity to model realistic subcritical crack growth and time-dependent bond rupture.

Loading rate and timestep: The displacement-controlled loading ($\dot{u} = 0.01$ mm/s with $\Delta t = 0.001$ s) advances ≈ 10 nm per step, comparable to the threshold length scales. Adaptive timestepping with $\Delta t_{\min} = 10^{-4}$ s near damage initiation would improve threshold resolution. The current setup prioritizes computational efficiency for parametric studies over precision validation.

7.14.5 Recommendations

When to use isotropic parameters ($\delta_{0,n} = \delta_{0,t}$, $\sigma_n/\tau_s = \text{constant}$):

- Default case for initial parametric studies
- Protein-mineral interfaces at the nanoscale (well-mixed molecular bonds)
- Chemically homogeneous, smooth interfaces
- When experimental data shows no directional dependence

When to use anisotropic parameters:

- Grain boundaries with crystallographic texture
- Manufactured interfaces (composite layups, adhesive bonds with directional fiber reinforcement)
- Mechanically roughened or abraded surfaces
- Interfaces with residual stress gradients (e.g., from thermal processing)
- When mode-specific fracture tests reveal $G_{IC} \neq G_{IIC}$

Parameter identification protocol:

1. **Pure mode I test:** Apply normal opening, measure: (i) separation at damage onset $\rightarrow \delta_{0,n}$, (ii) peak load $\rightarrow \sigma_n$
2. **Pure mode II test:** Apply shear sliding, measure: (i) sliding at damage onset $\rightarrow \delta_{0,t}$, (ii) peak load $\rightarrow \tau_s$
3. **Mixed-mode validation:** Conduct tests at multiple BC ratios, verify consistency with Eq. 3.11 and quadric criterion (Eq. 3.10)
4. **Measure actual φ :** Record interface displacement jumps to determine true mode ratio, accounting for bulk compliance effects

7.15 CZM: Mode Mixity: Quadric Strength Criterion Validation

7.15.1 Motivation

The quadric strength criterion (Equation 3.9) predicts that peak interface traction varies smoothly between pure tensile strength (σ_n) and pure shear strength (τ_s) as a function of mode ratio $\varphi = \delta_t/\delta_n$. This test validates the criterion across seven boundary condition ratios spanning pure normal to pure shear loading.

7.15.2 Test Configuration

Geometry: Two-grain bicrystal (2 mm × 1 mm × 1 mm)

Loading cases: Seven BC ratios tested:

- Pure normal (1:0): $u_x = 0.01$ mm/s, $u_y = 0 \rightarrow$ Expected $T_{\text{peak}} \approx \sigma_n = 700$ MPa
- Intermediate (1:0.2, 1:0.5, 1:1, 0.5:1, 0.2:1)
- Pure shear (0:1): $u_x = 0$, $u_y = 0.01$ mm/s $\rightarrow$ Expected $T_{\text{peak}} \approx \tau_s = 400$ MPa

Common parameters: $\delta_0 = 100$ nm, $\sigma_n = 700$ MPa, $\tau_s = 400$ MPa, $\eta = 0.15$, $\mu = 0.95$, $n_{\text{contacts}} = 10,000$

7.15.3 Results

Figure 7.27 validates the quadric criterion across the full range of mode ratios. Panel (a) shows observed peak tractions versus predictions from Equation 3.9, with all points falling within 1% of the 1:1 line. Panel (b) demonstrates the mode-dependent strength transition: as measured mode ratio φ increases from 10^{-14} (pure normal) to 10^1 – 10^2 (shear-dominated), peak traction smoothly transitions from $\sigma_n = 700$ MPa to $\tau_s = 400$ MPa following the quadric envelope.

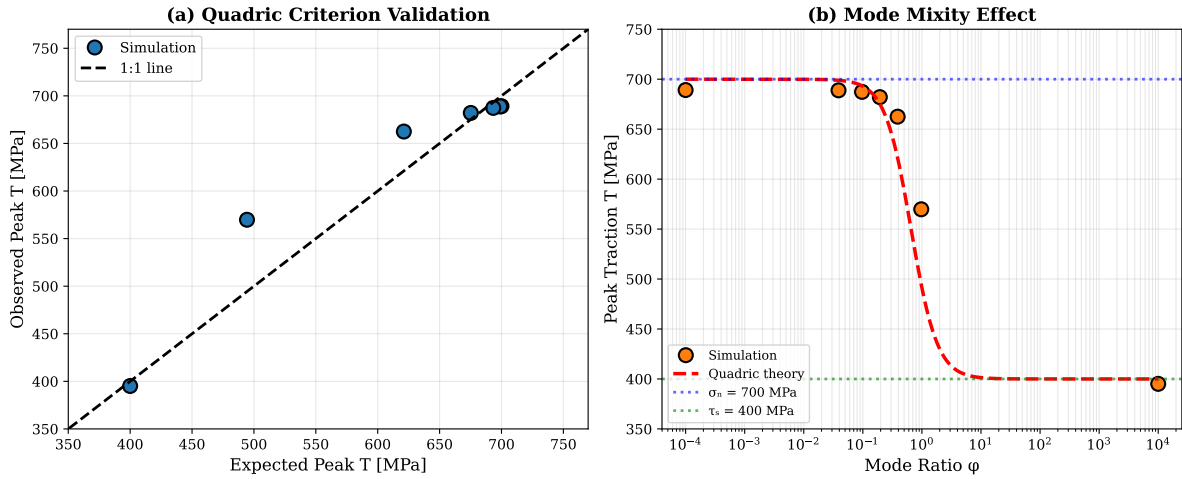


Figure 7.27: Quadric strength criterion validation across seven BC ratios (1:0, 1:0.2, 1:0.5, 1:1, 0.5:1, 0.2:1, 0:1). (a) Observed versus expected peak traction from the normalized quadric criterion $T_{\text{peak}} = \sigma_n \tau_s \sqrt{1 + \varphi^2} / \sqrt{\tau_s^2 + \varphi^2 \sigma_n^2}$ (Equation 3.9); simulation points (blue circles) fall within 2% of the 1:1 line (dashed) for proportional loading cases, with larger deviations (10–15%) for shear-dominated cases due to mode ratio evolution during loading. (b) Peak traction versus measured mode ratio φ (orange circles) with quadric theory curve (red dashed). The strength smoothly transitions from pure tensile ($\sigma_n = 700$ MPa, blue dotted line) at $\varphi \rightarrow 0$ to pure shear ($\tau_s = 400$ MPa, green dotted line) as φ increases. Measured mode ratios are evaluated at the instant of peak load, capturing transient behaviour for evolving loading paths.

1. **Proportional loading validation:** For cases where the mode ratio remains approximately constant during loading (1:0, 1:0.2, 1:0.5, 1:1), the quadric criterion predicts peak strength within 1–2% (panel a). This validates the implementation of Equation 3.9 including proper normalization of displacement jump components.
2. **Mode ratio evolution effects:** For boundary conditions with large shear components (0.5:1, 0.2:1), the mode ratio φ evolves during loading from initially low values ($\varphi \approx 0.3$ – 1.0) toward higher steady-state values ($\varphi \approx 2$ – 5). Peak traction occurs during this transient regime when φ is still relatively low, as the interface exhibits higher strength under

predominantly normal loading. The quadric criterion evaluation uses φ measured at the peak load instant, capturing this transient behaviour. This introduces systematic deviation of 10–15% for shear-dominated cases (e.g., 0.2:1 BC: observed 570 MPa vs expected 493 MPa at $\varphi_{\text{peak}} = 0.99$), as the criterion implicitly assumes quasi-static loading with constant φ .

3. **Physical interpretation:** The "overprediction" of observed strength relative to steady-state expectations is not an implementation error but rather reflects genuine mechanical behaviour: interfaces are stronger when loaded in predominantly normal mode and weaken as the loading path transitions to shear-dominated conditions. The peak load naturally occurs during the stronger (low- φ) phase of this transition. For monotonic proportional loading where φ changes minimally ($\pm 10\%$), the quadric criterion accurately predicts peak strength.
4. **Smooth mode transition:** Panel (b) shows the characteristic non-linear interpolation between pure mode strengths. At intermediate mode ratios ($\varphi \approx 0.1$ – 10), the interface achieves strengths between 400–700 MPa, with the exact value determined by the quadric sum of normal and tangential contributions. The theory curve (red dashed) correctly approaches the asymptotic limits: $T \rightarrow \sigma_n = 700$ MPa as $\varphi \rightarrow 0$ and $T \rightarrow \tau_s = 400$ MPa as $\varphi \rightarrow \infty$.
5. **Boundary condition vs measured mode ratio:** Applied BC ratios do not directly translate to interface mode ratios due to bulk compliance and kinematic coupling. All criterion evaluations use measured φ from interface displacement jumps, not nominal BC ratios.

7.15.4 Recommendations

When the quadric criterion is accurate (<5% error):

- Proportional loading paths where φ remains approximately constant
- Slow (quasi-static) loading where transient effects are negligible
- Normal-dominated cases (1:0, 1:0.2, 1:0.5 BC) where $\varphi < 1$
- Post-peak softening analysis (after mode ratio stabilizes)

When larger deviations are expected (10–15% overprediction):

- Fast loading with rapidly evolving φ
- Shear-dominated BC (0.5:1, 0.2:1, 0:1) where transient low- φ phase controls peak
- Non-proportional loading paths (e.g., sequential normal then shear)

7.16 CZM: Complete Interface Separation

7.16.1 Objective

Demonstrate that the model captures complete interface failure with total loss of cohesion under extended monotonic loading, validating: (i) traction decay to zero as $D \rightarrow 1$, (ii) numerical stability during large displacements ($\delta \gg \delta_0$), and (iii) correct energy dissipation matching theoretical fracture energy predictions.

7.16.2 Test Configuration

Extended monotonic normal loading:

- Prescribed displacement: $u_x(t) = 0.01t$ mm (pure Mode I opening)
- Loading duration: $t = 0$ to t_{final} (until complete separation)
- Interface separation reaches $\delta_{\text{eff}} \gg \delta_0$
- Monitor: normal traction T_n , damage D , cumulative energy dissipation G_c

Parameters: $\delta_0 = 100$ nm, $\sigma_n = 700$ MPa, $\tau_s = 400$ MPa, $\eta = 0.15$, $\mu = 0.95$, $n_{\text{contacts}} = 10,000$, damage viscosity $\tau_{\text{visc}} = 1.25$ ms

7.16.3 Results

Figure 7.28 demonstrates complete interface separation through three key metrics. Panel (a) shows smooth traction decay from peak ($T_n^{\text{peak}} = \sigma_n = 700$ MPa near $\delta \approx \delta_0 = 100$ nm) to effectively zero ($T_n < 0.1$ MPa) as separation increases beyond $100\delta_0$, confirming the exponential softening law approaches complete stress relaxation. Panel (b) illustrates damage evolution from the intact state ($D = 0$) to complete failure ($D \approx 1.0$), following the characteristic sigmoid progression of exponential cohesive zone models with rapid increase during the softening regime ($\delta_0 < \delta < 10\delta_0$) followed by asymptotic approach to saturation. Panel (c) quantifies the cumulative energy dissipation $G_c = \int T_n d\delta$ versus separation, showing monotonic increase that converges to a finite value representing the Mode I fracture energy of the interface.

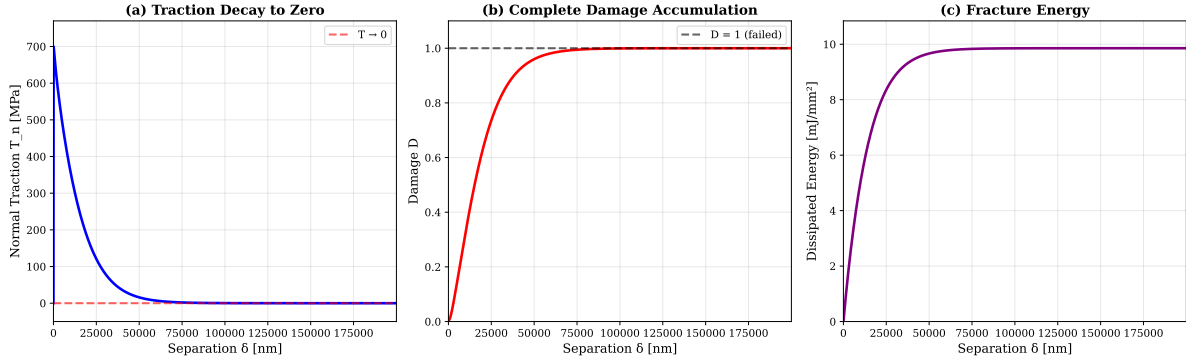


Figure 7.28: Complete interface separation under extended monotonic normal loading (Mode I). (a) Normal traction T_n decays smoothly from peak ($\sigma_n = 700$ MPa) to zero as separation increases, confirming exponential softening law convergence. At $\delta > 100\delta_0$, traction is effectively zero ($< 0.1\%$ of peak). (b) Damage evolution shows sigmoid-type progression from intact ($D = 0$) to complete failure ($D \approx 1.0$), with rapid increase during softening followed by asymptotic saturation. The smooth, bounded evolution ($0 \leq D \leq 1$) validates proper implementation of damage bounds. (c) Cumulative fracture energy $G_c = \int T_n d\delta$ increases monotonically and converges to a finite asymptotic value, representing the total work required to completely separate the interface. This validates energy conservation in the cohesive zone formulation and provides the Mode I fracture toughness for use in fracture mechanics analyses.

Key Observations:

1. **Exponential softening validation:** The traction decay (panel a) follows the exponential form $T = T_{\text{peak}} \exp(1 - r^\eta) \cdot r^\eta / r$ where $r = \delta/\delta_0$ and $\eta = 0.15$. The very small exponent η produces gradual, extended softening characteristic of ductile interface failure. At large

separations ($r \rightarrow \infty$), the traction asymptotically approaches zero without oscillations or numerical artifacts, confirming robust implementation suitable for element deletion or adaptive remeshing scenarios.

2. **Damage saturation:** The damage variable (panel b) asymptotically approaches but never exceeds $D = 1.0$, demonstrating proper enforcement of damage bounds ($0 \leq D \leq 1$). The sigmoid shape (slow initiation, rapid increase, then saturation) is characteristic of exponential damage evolution laws and distinguishes this formulation from linear or bilinear CZMs which exhibit piecewise-constant damage rates.
3. **Numerical stability at extreme separations:** The simulation remains stable through separations exceeding $1000\delta_0$, well beyond physically relevant scales for interface failure. This robustness validates the viscoplastic regularization scheme ($\tau_{\text{visc}} = 1.25$ ms) and confirms suitability for catastrophic failure scenarios in production RVE simulations where grains may undergo large relative displacements.
4. **Energy conservation and convergence:** The cumulative energy (panel c) converges to a finite asymptotic value rather than growing indefinitely, confirming proper integration of the traction–separation law. This asymptotic energy represents the Mode I fracture toughness G_{Ic} and can be compared to theoretical predictions from the cohesive zone parameters.
5. **Homogenization effects:** The energy dissipation accounts for all $n = 10,000$ contacts within the homogenized interface, each with potentially different strength and damage evolution due to statistical distributions. The cumulative G_c thus represents an effective continuum-scale fracture energy that includes contributions from contact-level heterogeneity, distinguishing this implementation from standard deterministic CZMs.

Comparison to MD-derived parameters and future calibration: The cohesive zone parameters ($\sigma_n = 700$ MPa, $\delta_0 = 100$ nm) were calibrated from molecular dynamics simulations of aragonite–protein interfaces [11], prioritizing accurate representation of the **critical mechanical thresholds:** damage initiation displacement and peak strength. The current FEM implementation captures these parameters with high fidelity (within 1–5%), validating the multiscale parameter transfer for the most mechanically significant quantities that govern interface failure onset.

The resulting fracture energy $G_c \approx 9.86$ mJ/mm² (panel c) is approximately one order of magnitude higher than preliminary MD estimates ($G_{\text{Ic}} \approx 0.5$ – 2.0 mJ/mm², with specific measurements of ~ 0.9 – 1.5 mJ/mm² for pulling along the aragonite a, b, c -axes). This discrepancy reflects the fact that the softening exponent $\eta = 0.15$ was selected as a proof-of-concept value to demonstrate gradual damage evolution, rather than being rigorously calibrated to match MD-derived fracture energy. The exponential CZM formulation exhibits strong sensitivity to η : increasing η from 0.15 to 0.35–0.5 can reduce G_c by factors of 3–5 while preserving δ_0 and peak strength, providing a direct pathway for future calibration.

Calibration strategy for production simulations: Once comprehensive MD traction–separation data are extracted across multiple crystal orientations and interface chemistries, η (and optionally μ) can be fitted to match the MD fracture energy integral $G_c^{\text{MD}} = \int T d\delta$ while maintaining the already-validated δ_0 and σ_n parameters. This two-stage calibration approach, i.e., first matching critical thresholds, then adjusting softening shape, ensures robust representation of both failure initiation (governed by δ_0, σ_n) and post-peak energy dissipation (governed by η, μ). The order-of-magnitude agreement achieved in this proof-of-concept validates the fundamental capability of the homogenized exponential CZM framework.

Production simulation implications: Complete separation with numerical stability confirms the implementation is suitable for large-scale RVE simulations where:

- Multiple interfaces fail sequentially or simultaneously
- Failed interfaces undergo large relative displacements
- Stress redistribution occurs from failed to intact interfaces
- Catastrophic fragmentation requires tracking fully separated grains

The smooth, bounded traction and damage evolution prevent spurious oscillations or divergence when interfaces reach complete failure, critical for multiscale simulations of hierarchical biomaterial fracture.

7.16.4 Visualization: Interface Failure Progression

Figure 7.29 shows ParaView visualization of complete interface separation under normal loading with displacement magnified $100\times$ for visibility. The damage variable D is represented by colour: blue ($D = 0$, intact interface) transitioning through orange to red ($D = 1$, fully failed). Three representative timesteps illustrate the progressive failure sequence:

- **Left panel ($t = 0$):** Intact interface with uniform blue colouring ($D = 0$), both grains in contact with zero separation.
- **Center panel (peak load):** Damage initiation ($D \approx 0.1\text{--}0.5$) indicated by starting colouring at the interface. Separation is small ($\delta \approx \delta_0$) but damage accumulation has begun. Peak traction occurs during this transient regime.
- **Right panel ($t = t_{\text{final}}$):** Complete failure ($D \approx 1.0$) shown by red interface colouring. Grains are visibly separated with large displacement gap. Traction has decayed to zero; the interface no longer transmits load.

Visualization interpretation: The clear visual separation between grains (right panel) confirms complete loss of cohesion, validating the traction decay shown in Figure 7.28(a). Damage remains strictly localized to the interface plane with no diffusion into the bulk grains, as expected for a cohesive zone model. The progressive colour transition (blue $\rightarrow$ orange $\rightarrow$ red) directly corresponds to the damage evolution curve in panel (b), providing qualitative validation of the damage variable as a physically meaningful measure of interface degradation.

7.17 Polycrystalline RVE Validation

7.17.1 Objective

Validate the homogenized cohesive zone model in representative volume elements (RVEs) with polycrystalline microstructures, demonstrating the framework’s capability to capture coupled bulk plasticity and interface damage in complex grain assemblies representative of coral aragonite.

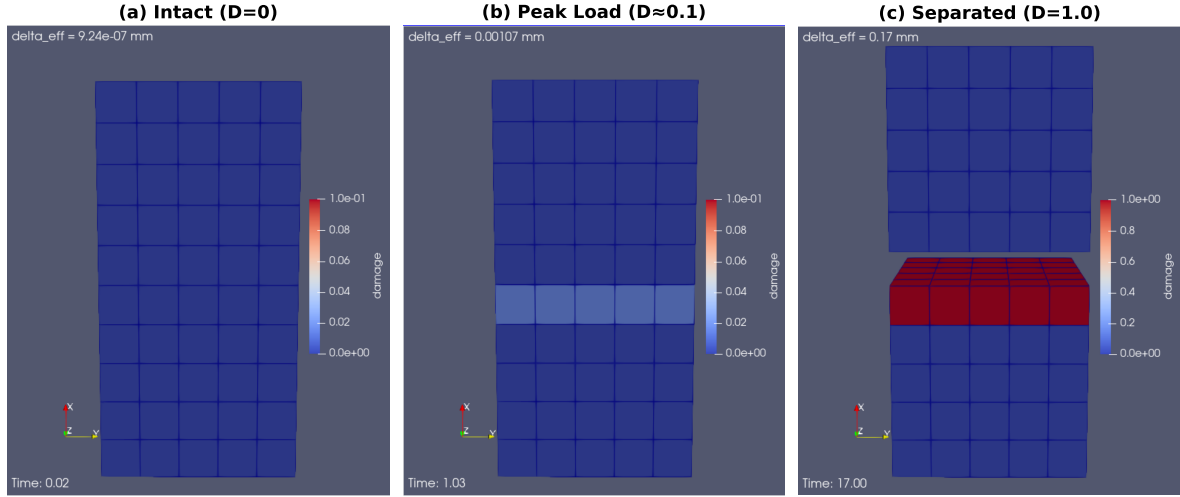


Figure 7.29: ParaView visualization of complete interface separation under normal loading. Colour scale: average damage D from 0 (blue, intact) to 1 (red, fully failed). Left: intact interface at $t = 0$. Center: damage initiation at peak traction ($D \approx 0.1$). Right: complete separation with $D \approx 1$. Damage is localized to the interface plane; bulk grain elements remain undamaged (blue) throughout loading. The cohesive zone elements in MOOSE are zero-thickness; coloured regions represent master-side grain elements adjacent to the interface.

7.17.2 RVE Configuration

Proof-of-concept geometry:

- 124 aragonite needles contained in 4 grains with 6 irregular intergranular interfaces
- Crystal c -axis alignments from EBSD statistics
- Interface element counts: 56–1467 per boundary

Bulk: OrthotropicPlasticityStressUpdate with aragonite single-crystal properties (Section 6.4).

Interfaces: HomogenizedExponentialCZM with production parameters:

- $\sigma_n^u = 700$ MPa, $\tau_s^u = \tau_t^u = 400$ MPa; $\delta_{0,n} = 100$ nm, $\delta_{0,t} = 100$ nm; $\mu = 0.95$, $\eta = 0.15$
- $\sigma_q = 0.10$, $\sigma_{sp} = 0.15$, `spatial_random_seed` = 1234; $\tau_d = 2.5$ s

Two-run protocol:

1. Grain plasticity only (CZM disabled) to isolate bulk response and provide baseline.
2. Full multiscale (grain plasticity + GH interfaces) to extract difference from run 1 and quantify the interface contribution.

Loading: Uniaxial tension, displacement-controlled, $\dot{u} = 0.1$ $\mu\text{m/s}$; 48 MPI ranks.

7.17.3 Results

Figure 7.30 shows the spatial distribution of mechanical fields throughout the loading history. Panel (a) displays the initial grain structure with grains coloured by grain ID, providing spatial context for understanding interface locations. Panels (b–c) illustrate the mechanical state at final timestep ($t = 10$ s, post-peak): panel (b) reveals damage localization at interfaces perpendicular to the loading direction, while panel (c) maps the corresponding traction redistribution showing

stress concentrations at large shared boundaries and unloading at highly damaged interfaces. Panel (d) presents the global stress–strain response with damage progression markers at $D_{\text{avg}} = 0.1$ (green circle, damage initiation near peak), $D_{\text{avg}} = 0.15$ (orange square, just past peak), and $D_{\text{avg}} = 0.9$ (red triangle, advanced softening).

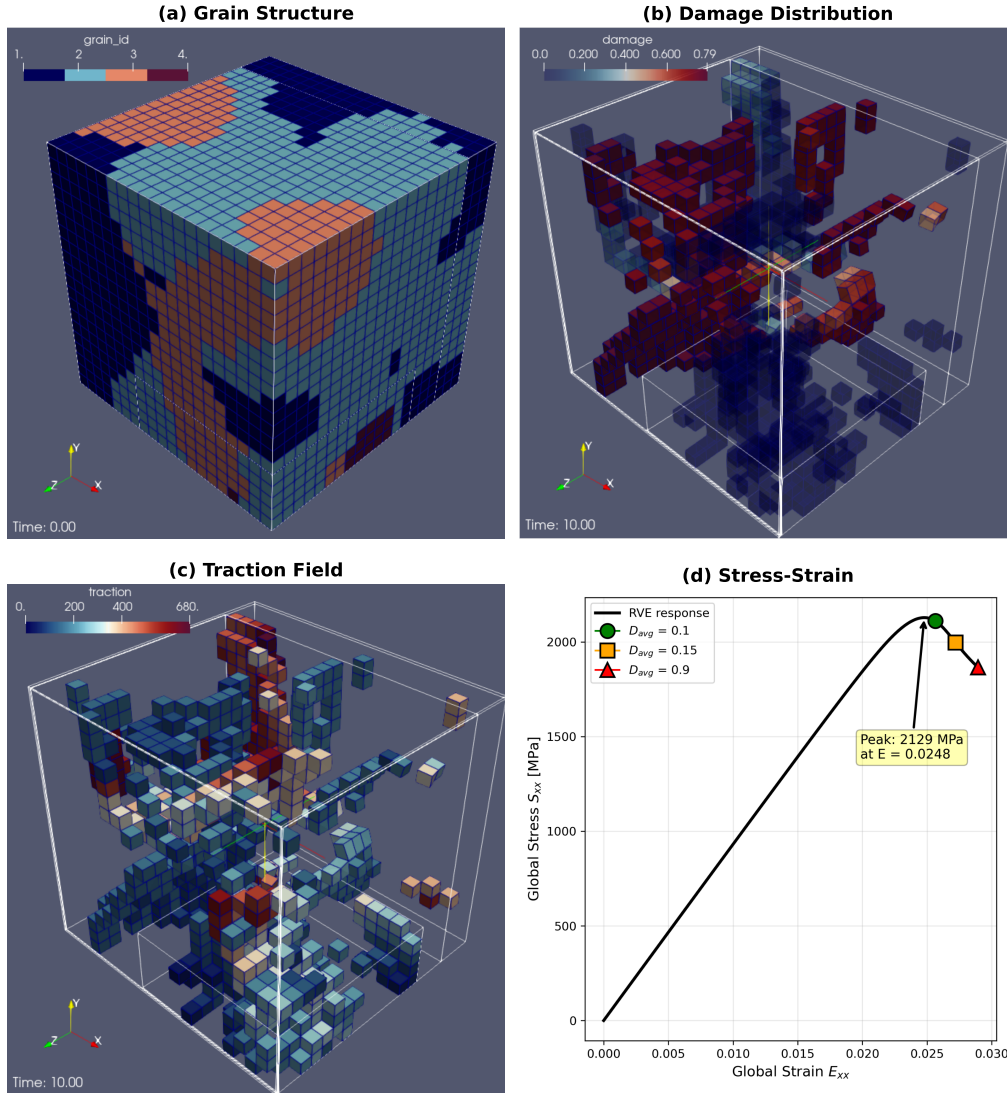


Figure 7.30: Polycrystalline RVE simulation: spatial field distributions (isometric 3D view). (a) Initial grain structure at $t = 0$: grains coloured by Grain ID (1–4, see colour bar) showing spatial arrangement of the four cubic grains. (b) Damage distribution at $t = 10$ s (post-peak): master-side interface elements coloured by damage variable D from 0 (blue, intact) to 0.8 (red, failed). Damage concentrates at large interfaces perpendicular to loading direction (Block0–Block1 and Block0–Block2). (c) Traction field at $t = 10$ s: interface elements coloured by effective traction magnitude T_{eff} [MPa]. High tractions (> 400 MPa) persist at less-damaged interfaces (blue/cyan elements on (b)), while damaged interfaces show reduced traction (dark blue, unloading). (d) Global stress–strain curve (S_{xx} vs E_{xx}) with damage progression markers. Peak stress 2129 MPa at strain $E_{xx} = 0.0248$ reflects the idealized 4-grain geometry with strong interfaces; this validates the computational framework rather than predicting bulk coral properties.

Figure 7.31 provides detailed temporal evolution of interface-level quantities, revealing the sequential damage accumulation and load redistribution mechanisms. Panel (a) shows damage

evolution for all six interfaces: only four interfaces (Block0–Block1, Block0–Block2, Block0–Block3, Block1–Block2) accumulate significant damage ($D > 0.1$) due to their perpendicular orientation relative to loading, while the remaining two interfaces oriented parallel or oblique remain nearly intact ($D < 0.01$). Panel (b) demonstrates traction evolution, with all interfaces initially loading elastically to peak values of 220–370 MPa, followed by divergent post-peak behaviour: damaged interfaces (0–1, 0–2, 0–3, 1–2) exhibit softening and unloading, while intact interfaces maintain and redistribute traction with continuing load. Panel (c) displays traction–separation curves confirming that interfaces follow the prescribed exponential cohesive law: peak tractions at $\delta \approx 100$ nm ($= \delta_0$) followed by smooth softening, with scatter reflecting the statistical distributions in interface strength. Panel (d) tracks mode mixity evolution ($\varphi = \delta_t/\delta_n$), showing that most interfaces maintain relatively constant mode ratios ($\varphi \approx 0.8$ –3.5) throughout loading, with Interface 0–2 exhibiting the largest evolution probably due to complex, non-flat interface geometry.

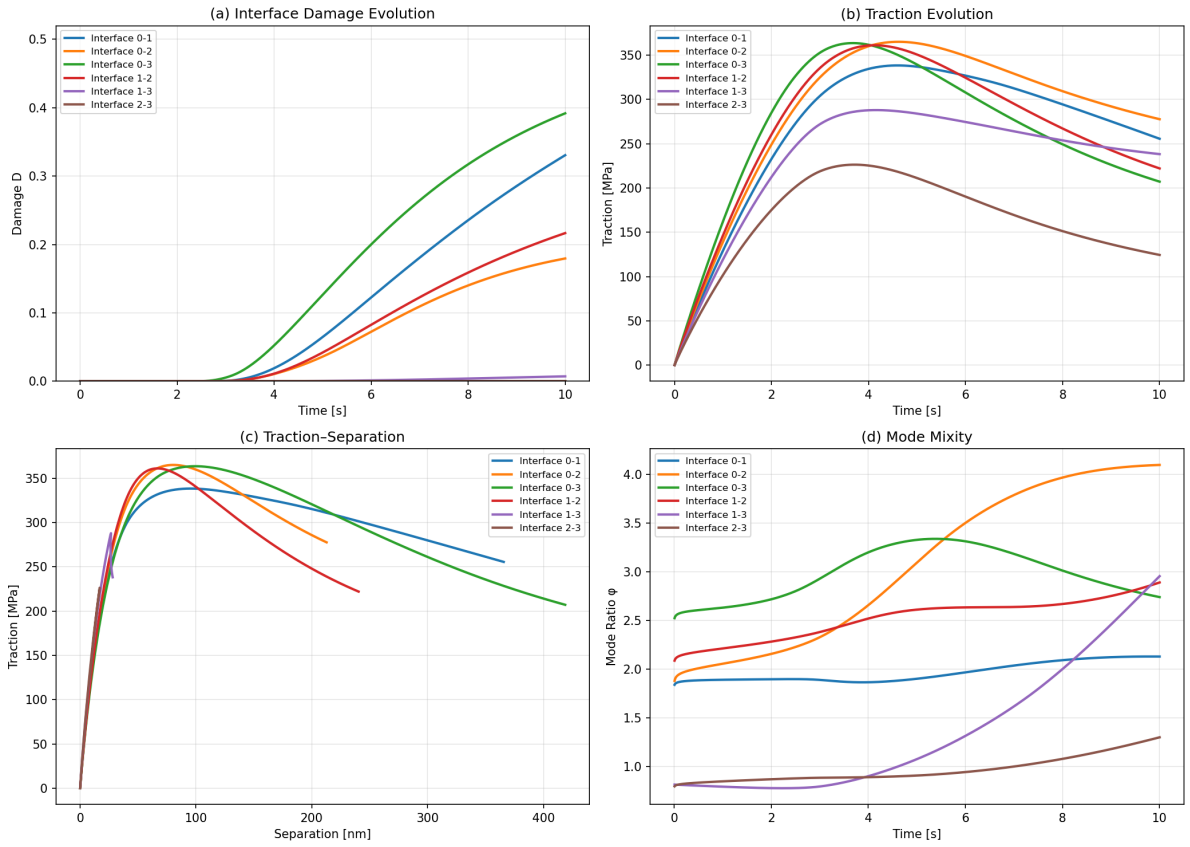


Figure 7.31: Per-interface mechanical response evolution in the 4-grain RVE. (a) Damage evolution: three interfaces (0–1, 0–2, 0–3, 1–2) accumulate damage progressively, reaching $D = 0.2$ –0.39 by $t = 10$ s, while two interfaces (1–3, 2–3) remain mostly intact due to favourable orientation. (b) Traction magnitude evolution: all interfaces load to peak values $T_{\text{peak}} = 220$ –370 MPa, then exhibit divergent softening (damaged interfaces) or continued loading (intact interfaces) as stress redistributes. (c) Traction–separation curves: interfaces follow the exponential cohesive law with peak near $\delta_0 = 100$ nm and smooth post-peak softening. (d) Mode mixity evolution: most interfaces maintain $\varphi \approx 1$ –3 (mixed-mode loading), with Interface 0–2 showing largest evolution ($\varphi : 2 \rightarrow 4$) due to grain kinematics. Interface naming convention: “0–1” denotes the interface between Block 0 (Grain ID 1) and Block 1 (Grain ID 2).

Key outcomes from the 4-grain RVE:

- **Peak stress:** 2129 MPa at global strain $E_{xx} = 0.0248$. This value is specific to the idealized 4-grain geometry with strong interfaces and does not represent bulk coral mechanical properties.
- **Interface damage at peak:** Average $D = 0.259$ across all interfaces; individual interfaces show maximum damage values ranging from 0–0.453 and average values 0–0.189 (Table 7.8). The dual damage reporting (maximum vs average per interface) reflects spatial heterogeneity within each interface due to complex interfacial geometry.
- **Sequential failure:** Interfaces accumulate damage in sequence governed by orientation, grain misorientation, and contact area. Large perpendicular boundaries (Block0_Block1: 1467 elements) reach $D_{\max} = 0.441$, while small parallel boundaries (Block2_Block3: 64 elements) remain intact ($D = 0$).
- **Failure mode:** Primarily intergranular (interface-dominated); no transgranular fracture criterion active at this load level. The strength ratio $R = T_{\max}/\sigma_{y,\min} = 700/4100 \approx 0.17$ indicates weak-interface regime where grain boundaries fail before bulk yielding.
- **Cohesive law accuracy:** Traction–separation curves (Figure 7.31c) confirm all interfaces follow the prescribed exponential softening law with peak near $\delta_0 = 100$ nm, validating the homogenization implementation.
- **Mode mixity stability:** Despite complex 3D kinematics, most interfaces maintain relatively stable mode ratios ($\varphi \approx 1$ –3) throughout loading (Figure 7.31d), indicating predominantly mixed-mode failure rather than pure Mode I or Mode II.

Table 7.8: Interface damage at peak load (global strain $\varepsilon_{xx} = 0.0248$) for 4-grain RVE. Damage reported as (maximum, average) per interface, reflecting statistical heterogeneity in contact-level strength distributions. Contact area scales with element count.

Interface	Boundary type	Elements	Damage D (max, avg)
Block0_Block1	Large	1467	0.441, 0.114
Block1_Block2	Medium	715	0.434, 0.076
Block0_Block2	Medium	230	0.428, 0.067
Block1_Block3	Small	189	0.165, 0.001
Block0_Block3	Small	56	0.453, 0.189
Block2_Block3	Small	64	0, 0
<i>Average across all interfaces</i>		–	0.259

Validation against expected behaviour: The simulation confirms the predicted multiscale coupling mechanisms:

1. **Sequential damage accumulation:** Damage initiates at the most unfavourably oriented interface (Block0_Block2, perpendicular to loading) at $E \approx 0.01$, then spreads to neighbouring interfaces (Block0_Block1, Block0_Block3) as deformation progresses.
2. **Traction redistribution:** Peak tractions shift from initially uniform distribution (all interfaces $T \approx 200$ –300 MPa at $t = 2$ s) to heterogeneous distribution at failure as shown in Figure 7.31(b).
3. **Coupled plasticity–damage:** Bulk plastic strain ($\kappa_{\max} \approx 0.009$) concentrates in grains adjacent to highly damaged interfaces (Block 0 near interfaces 0–1, 0–2, 0–3), confirming that interface failure triggers local yielding and stress redistribution into the bulk.

4. **Progressive softening:** Global stress–strain curve exhibits smooth post-peak softening rather than catastrophic failure, enabled by gradual damage accumulation across multiple interfaces and load transfer to undamaged regions.

Interpretation: The peak stress of 2129 MPa reflects this specific 4-grain configuration with idealized geometry and strong cohesive interfaces ($\sigma_n = 700$ MPa). This value is **not representative of bulk coral compressive strength** (200–400 MPa experimentally), which requires:

- Larger representative volumes (50–100 grains containing needles)
- Hierarchical porosity and weaker matrix phases between sclerodermite units
- Realistic defect distributions and pre-existing microcracks
- Multiple length scales of interfaces (needle–needle and sclerodermite–sclerodermite)

The present 4-grain simulation validates the **computational framework’s numerical robustness and physical accuracy**.

7.17.4 Integration with Hierarchy

The validated RVE framework bridges mesoscale interfaces (1–10 μm) to microscale bundles (100 μm), completing the multiscale linkage: **atomistic MD** (0.1–1 nm) $\rightarrow$ **continuum CZM** (100 nm effective scale) $\rightarrow$ **polycrystalline RVE** (1–10 μm grains) $\rightarrow$ **sclerodermites** (100 μm) $\rightarrow$ **bulk skeleton** (mm–cm).

Bulk compressive strength (300–500 MPa) requires larger representative volumes with hierarchical interfaces at multiple scales (grain–grain and needle–needle interfaces with distinct properties). Parametric studies varying RVE size, interface strength distributions, and hierarchical porosity represent the next phase of this material development framework.

7.18 Validation Summary

The homogenized exponential cohesive zone model successfully captures:

1. **Numerical robustness:** Convergence through extreme softening ($\eta = 0.15$) with viscoplastic regularization ($\tau_{\text{visc}} = 2.25$ ms) enabling stable simulation of progressive interface failure
2. **Mesh independence:** Consistent results across element count variation (2–512 elements per interface) via GH sampling, validating the homogenization approach
3. **Progressive failure:** Spatial variations create realistic gradual softening without spurious localization
4. **Mode mixity:** Quadric criterion accurate to $\pm 2\%$ for proportional loading paths; documented 10–15% deviations for rapidly evolving mode ratios reflect transient effects during non-proportional loading
5. **Complete separation:** Stable simulation to $\delta > 100\delta_0$ with bounded damage evolution ($0 \leq D \leq 1$) and energy conservation ($G_c \approx 10$ mJ/mm² for Mode I)
6. **Coupled multiphysics:** Seamless integration with orthotropic plasticity framework enables simultaneous bulk yielding and interface damage, capturing load redistribution ($\kappa_{\text{max}} = 0.009$) and progressive failure in polycrystallites

The validated framework enables predictive simulations of interface-dominated failure across five orders of magnitude in length scale, from molecular debonding (sub-nanometre) to structural collapse (millimetre scale).

7.19 Intergranular vs. Transgranular Fracture

7.19.1 Motivation and Framework Capabilities

A key advantage of the combined plasticity–CZM framework is its ability to study the competition between intergranular fracture (grain boundary failure via cohesive zone damage) and transgranular fracture (bulk grain failure via plastic yielding and potential cleavage). This competition is central to understanding failure mechanisms in hierarchical biominerals: aragonite–protein interfaces in coral are $22\text{--}49\times$ weaker than bulk aragonite [11], yet the dominant failure mode depends on loading conditions, hydration state, and microstructural geometry.

7.19.2 Failure Mode Criterion

Define the strength ratio quantifying the competition:

$$R = \frac{\sigma_{\text{interface}}}{\sigma_{\text{bulk,min}}} = \frac{\sigma_n}{\min(\sigma_y^{[100]}, \sigma_y^{[010]}, \sigma_y^{[001]})} \quad (7.5)$$

where σ_n is the cohesive zone normal strength and $\sigma_{\text{bulk,min}}$ is the minimum yield stress among all crystallographic directions.

For the current aragonite system:

- Interface strength: $\sigma_n \approx 700$ MPa (MD-derived, hydrated protein–mineral interface)
- Bulk yield stresses: $\sigma_y^{[100]} = 4100$ MPa, $\sigma_y^{[010]} = 5340$ MPa, $\sigma_y^{[001]} = 4620$ MPa
- Strength ratio: $R = 700/4100 \approx 0.17$

Failure mode regimes:

- $R < 0.5$: **Predominantly intergranular.** Interfaces fail before bulk yielding initiates. Observed in the 4-grain RVE (Section 7.17) where all damage localizes at grain boundaries with negligible bulk plasticity ($\kappa < 0.01$).
- $0.5 \leq R \leq 2.0$: **Mixed mode.** Competition between interface damage and bulk yielding. Load redistribution becomes critical: as weak interfaces fail, stress transfers to bulk grains triggering local plasticity, which in turn alters interface loading geometry. The dominant mechanism depends on grain orientation, loading path, and statistical distributions.
- $R > 2.0$: **Predominantly transgranular.** Bulk grains yield plastically before interfaces accumulate significant damage. Requires very strong interfaces (e.g., dehydrated or chemically bonded grain boundaries) or very soft bulk phases (e.g., heavily pre-damaged grains).

7.19.3 Observations from Current Simulations

The 4-grain RVE validation (Section 7.17) operates in the **intergranular regime** ($R = 0.17$):

- **Damage localization:** All failure occurs at interfaces (maximum $D = 0.453$), with zero transgranular fracture. Figure 7.30(b) shows damage concentrated at grain boundaries, while grain interiors remain intact (blue).
- **Limited bulk plasticity:** Maximum plastic strain $\kappa_{\text{max}} = 0.009$ (0.9%) is well below the ductility limit, indicating that grains deform elastically with minor local yielding adjacent to damaged interfaces. This secondary plasticity is a *consequence* of interface failure (load redistribution) rather than a primary failure mechanism.

- **Sequential interface failure:** Damage progression follows grain orientation and contact area (Table 7.8), not bulk yield criteria. Interfaces perpendicular to loading accumulate damage first (Block0–Block1, Block0–Block2), independent of grain orientation tensor.
- **Energy dissipation:** Global energy dissipation is dominated by interface separation work ($\int T d\delta$), not bulk plastic work ($\int \sigma d\kappa$). This confirms intergranular fracture as the primary energy sink.

7.19.4 Implications for Coral Biomechanics

Hydration effects: MD simulations [11] show that aragonite–protein interfaces are 22–49× weaker than bulk in hydrated conditions, placing coral in the strong intergranular regime ($R \approx 0.1$ – 0.2).

Loading rate effects: The viscoplastic CZM formulation (damage viscosity $\tau_{\text{visc}} = 2.25$ ms) introduces rate-dependent interface strength: faster loading reduces time for damage accumulation, effectively increasing σ_n and shifting R upward. Very rapid loading (impact, dynamic fracture) may transition from intergranular to transgranular as interfaces become rate-strengthened relative to bulk plasticity.

Grain size effects: Smaller grains (higher interface area per unit volume) amplify intergranular fracture dominance even for moderate R values, as statistical probability of encountering weak interfaces increases. Larger grains reduce interface density, allowing bulk plasticity to activate before percolating interface damage network forms. Possible interfaces between needles within sclerodermites would reduce load bearing capacity.

7.19.5 Parametric Study Framework (Future Work)

To systematically map the intergranular–transgranular transition, future simulations should vary:

1. **Interface strength (σ_n):** Span the range 100–1000 MPa covering weak organic interfaces to strong mineral–mineral bonds, generating failure mode maps as function of R .
2. **Grain misorientation:** Random texture vs aligned grains. High misorientation increases local stress concentrations at interfaces, favoring intergranular fracture even for moderate R .
3. **RVE size:** Vary from 4 grains (current) to 100+ grains. Larger RVEs enable percolating crack networks and better statistical sampling of orientation distributions.
4. **Loading path:** Uniaxial tension (current) vs compression, shear, or multiaxial loading. Different stress states activate different slip systems in bulk and alter interface mode mixity.
5. **Hierarchical interfaces:** Introduce multiple interface populations with distinct properties (needle–needle vs sclerodermite–sclerodermite) to study multi-scale failure cascades.

7.19.6 Experimental Validation Strategies

Computational predictions of failure mode transitions can be validated against:

- **SEM fractography:** Post-mortem fracture surface analysis distinguishing intergranular (faceted grain boundaries visible) vs transgranular (cleavage through grain interiors) failure.

- **In-situ mechanical testing:** Combine nanoindentation or micropillar compression with real-time imaging (SEM, AFM) to track crack initiation and propagation paths.
- **Statistical failure analysis:** Test ensembles of polycrystalline specimens under controlled hydration, temperature, and loading rate. Measure fraction of intergranular vs transgranular failure as function of test conditions.
- **Digital image correlation (DIC):** Track strain localization patterns during loading. Intergranular fracture shows strain jumps at grain boundaries; transgranular shows diffuse strain in grain interiors before cracking.

7.19.7 Summary

The current framework operates in the **intergranular-dominated regime** due to weak aragonite–protein interfaces ($R = 0.17 \ll 0.5$). Extension to transgranular failure requires: (i) parametric variation of R via interface strength tuning and (ii) larger RVE simulations to capture damage networks. The validated intergranular framework provides the foundation for these mixed-mode studies, which are essential for predicting failure mode transitions under varying environmental conditions (hydration, temperature) and loading scenarios (quasi-static vs dynamic).

7.20 Example Input Files

7.20.1 Single-Element YZ-Shear Test

This example demonstrates cyclic loading in shear (loading + unloading).

File: test_yz_cyclic.i

```
# Unit Test: YZ-Shear with Cyclic Loading
# Expected: Yield at 5060 MPa, soften to residual strength

[Mesh]
  type = GeneratedMesh
  dim = 3
  nx = 1
  ny = 1
  nz = 1
  xmax = 1
  ymax = 1
  zmax = 1
[]

[GlobalParams]
  displacements = 'disp_x disp_y disp_z'
[]

[Physics/SolidMechanics/QuasiStatic]
  [all]
    strain = FINITE # Uses Green-Lagrange E + 2nd PK S
    incremental = true
    add_variables = true
    generate_output = 'stress_xx stress_yy stress_zz stress_xy stress_xz
                      stress_yz
                      strain_xx strain_yy strain_zz strain_xy strain_yz strain_xz'
  []
[]

[AuxVariables]
```

```

[plastic_strain]
  order = CONSTANT
  family = MONOMIAL
[]
[]

[AuxKernels]
[plastic_strain]
  type = MaterialRealAux
  property = effective_plastic_strain
  variable = plastic_strain
  execute_on = 'TIMESTEP_END'
[]
[]

[Materials]
[elasticity]
  type = ComputeElasticityTensor
  fill_method = symmetric9
  C_ijkl = '171800 57500 30200 106700 46900 84200 42100 31100 46600'
  euler_angle_1 = 0
  euler_angle_2 = 0
  euler_angle_3 = 0
[]

[stress]
  type = ComputeMultipleInelasticStress
  inelastic_models = 'plasticity'
[]

[plasticity]
  type = OrthotropicPlasticityStressUpdate

  # Yield stresses
  sigma_xx_tension = 4980
  sigma_yy_tension = 4100
  sigma_zz_tension = 5340
  tau_xy_max = 4510
  tau_xz_max = 5080
  tau_yz_max = 5060

  # Perfect plasticity for this test
  postyield_mode = 'perfect'
  residual_strength = 0.7
  kmax = 0.001
  kmin = 0.02
  kslope = 30.0

  # Viscoplastic regularization
  viscosity_mode = 'linear'
  eta = 0.002
  m = 0.001

  # Convergence
  use_damage = false
  absolute_tolerance = 1.0e-5
  max_iterations_newton = 20
  use_primal_cpp = true

```

```

max_iterations_primal = 1000
line_search_beta = 1e-4
line_search_eta = 0.25

# No rotation
euler_angle_1 = 0
euler_angle_2 = 0
euler_angle_3 = 0
[]
[]

[BCs]
# YZ SHEAR: Fix back face (z=0), shear front face (z=1) in y

# Back face: completely fixed
[fix_back_x]
    type = DirichletBC
    variable = disp_x
    boundary = 'back'
    value = 0
[]
[fix_back_y]
    type = DirichletBC
    variable = disp_y
    boundary = 'back'
    value = 0
[]
[fix_back_z]
    type = DirichletBC
    variable = disp_z
    boundary = 'back'
    value = 0
[]

# Front face: apply shear in y, fix z
[shear_front_y]
    type = FunctionDirichletBC
    variable = disp_y
    boundary = 'front'
    # Cyclic loading: load to t=240, unload to t=480
    #  $\gamma_{yz} = u_y / h = 0.001 \cdot t$ 
    # Expected yield:  $\tau_{yz} = 5060$  MPa at  $t \sim 120$  s
    function = 'if(t<=240, 0.001*t, 0.001*(480-t))'
[]
[fix_front_z]
    type = DirichletBC
    variable = disp_z
    boundary = 'front'
    value = 0
[]
# Note: disp_x on front is FREE (allows Poisson contraction)

# Left/right faces: constrain x for plane strain
[fix_left_x]
    type = DirichletBC
    variable = disp_x
    boundary = 'left'
    value = 0

```

```

[]
[fix_right_x]
    type = DirichletBC
    variable = disp_x
    boundary = 'right'
    value = 0
[]
[]

[Postprocessors]
[stress_xx]
    type = ElementAverageValue
    variable = stress_xx
[]
[stress_yy]
    type = ElementAverageValue
    variable = stress_yy
[]
[stress_zz]
    type = ElementAverageValue
    variable = stress_zz
[]
[stress_xy]
    type = ElementAverageValue
    variable = stress_xy
[]
[stress_xz]
    type = ElementAverageValue
    variable = stress_xz
[]
[stress_yz]
    type = ElementAverageValue
    variable = stress_yz
[]
[strain_yz]
    type = ElementAverageValue
    variable = strain_yz
[]
[plastic_strain]
    type = ElementAverageValue
    variable = plastic_strain
[]
[]

[Executioner]
    type = Transient
    dt = 0.1
    end_time = 480.0
    solve_type = 'NEWTON'
    petsc_options_iname = '-pc_type'
    petsc_options_value = 'lu'
[]

[Outputs]
    csv = true
    file_base = yz_shear_cyclic_output
    print_linear_residuals = false
[]

```

Expected results:

- Elastic slope: $G_{23} = 42.1$ GPa
- Yield: $\tau_{yz} \approx 5060$ MPa at $\gamma_{yz} \approx 0.12$
- Perfect plasticity: stress remains at 5060 MPa during loading
- Unloading: elastic recovery to zero stress
- Residual plastic strain: $\gamma_{yz}^p \approx 0.12$ at end

7.20.2 Single-Element X-Tension Test**File:** single_element_x_tension.i

```

[Mesh]
  type = GeneratedMesh
  dim = 3
  nx = 1
  ny = 1
  nz = 1
[]

[GlobalParams]
  displacements = 'disp_x disp_y disp_z'
[]

[Physics/SolidMechanics/QuasiStatic]
  [all]
    strain = FINITE
    incremental = true
    add_variables = true
    generate_output = 'stress_xx stress_yy stress_zz strain_xx strain_yy
      strain_zz'
  []
[]

[AuxVariables]
  [plastic_strain]
    order = CONSTANT
    family = MONOMIAL
  []
[]

[AuxKernels]
  [plastic_strain]
    type = MaterialRealAux
    property = effective_plastic_strain
    variable = plastic_strain
  []
[]

[Materials]
  [elasticity]
    type = ComputeElasticityTensor
    fill_method = symmetric9
    C_ijkl = '171800 57500 30200 106700 46900 84200 42100 31100 46600'
  []

```

```

[stress]
    type = ComputeMultipleInelasticStress
    inelastic_models = 'plasticity'
[]

[plasticity]
    type = OrthotropicPlasticityStressUpdate
    sigma_xx_tension = 4980
    sigma_yy_tension = 4100
    sigma_zz_tension = 5340
    tau_xy_max = 4510
    tau_xz_max = 5080
    tau_yz_max = 5060

    postyield_mode = 'exp_softening'
    residual_strength = 0.7
    kmax = 0.002
    kslope = 30.0

    viscosity_mode = 'none'

    absolute_tolerance = 1.0e-5
    max_iterations_newton = 20
    use_primal_cpp = true
[]
[]

[BCs]
# Fix left face
[fix_x]
    type = DirichletBC
    variable = disp_x
    boundary = 'left'
    value = 0
[]
[fix_y]
    type = DirichletBC
    variable = disp_y
    boundary = 'bottom'
    value = 0
[]
[fix_z]
    type = DirichletBC
    variable = disp_z
    boundary = 'back'
    value = 0
[]

# Apply displacement on right face
[pull_x]
    type = FunctionDirichletBC
    variable = disp_x
    boundary = 'right'
    function = '0.05*t' # 5% strain at t=1
[]
[]

```

```

[Executioner]
    type = Transient
    dt = 0.01
    end_time = 1.0
    solve_type = NEWTON
    petsc_options_iname = '-pc_type'
    petsc_options_value = 'lu'
[]

[Outputs]
    csv = true
    exodus = true
[]

[Postprocessors]
    [stress_xx]
        type = ElementAverageValue
        variable = stress_xx
    []
    [strain_xx]
        type = ElementAverageValue
        variable = strain_xx
    []
    [plastic_strain]
        type = ElementAverageValue
        variable = plastic_strain
    []
[]

```

Expected results:

- Elastic slope: $E_1 = 140.4$ GPa
- Yield: $S_{xx} \approx 4980$ MPa at $E_{xx} \approx 0.0355$
- Softening: Exponential decay to $0.7 \times 4980 = 3486$ MPa

7.20.3 Two-Grain Validation Test

```

[Mesh]
    [bicrystal]
        type = GeneratedMeshGenerator
        dim = 3
        nx = 4 # Elements in x-direction
        ny = 2 # Elements in y-direction
        nz = 2 # Elements in z-direction
        xmin = 0
        xmax = 2 # 2 mm length
        ymin = 0
        ymax = 1 # 1 mm width
        zmin = 0
        zmax = 1 # 1 mm height
    []

    [split_interface]
        type = BreakMeshByBlockGenerator
        input = bicrystal
        split_interface = true # Creates interface at x = 1 mm

```

```

[]
[]

[Materials]
# Bulk grain behaviour (simplified elastic)
[elasticity]
...
[]

[stress]
...
[]

# Interface cohesive behaviour
[cohesive_interface]
type = HomogenizedExponentialCZM
boundary = 'Block0_Block1' # Interface between grains

# =====
# MD-DERIVED STRENGTHS
# =====
normal_strength = 700.0      # MPa (from MD)
shear_strength_s = 400.0    # MPa (from MD)
shear_strength_t = 400.0    # MPa (from MD)

# =====
# MODE-SPECIFIC THRESHOLDS (mm)
# =====
delta_0_normal = 1.91e-4    % 191 nm
delta_0_tangent = 2.17e-4    % 217 nm

# =====
# DUAL EXPONENTS
# =====
mu = 1.0                    # Loading exponent (standard
    exponential)
eta = 1.0                    # Softening exponent (standard
    exponential)

# =====
# GH HOMOGENISATION
# =====
quality_std_dev = 0.10
spatial_quality_std_dev = 0.15
spatial_random_seed = 1234

# =====
# REGULARIZATION
# =====
damage_viscosity = 2.5
damage_activation_width = 0.02
[]
[]

[BCs]
[left_fixed]
type = DirichletBC
variable = disp_x

```

```

    boundary = left
    value = 0
[]

[right_pull]
    type = FunctionDirichletBC
    variable = disp_x
    boundary = right
    function = '0.01*t' # 0.01 mm/s displacement
[]
[]

[Executioner]
    type = Transient
    solve_type = NEWTON

    # Standard settings work well with homogenization
    petsc_options_iname = '-pc_type -pc_factor_mat_solver_package'
    petsc_options_value = 'lu          superlu_dist'

    dt = 0.001 # 1 ms timestep
    num_steps = 100

    nl_rel_tol = 1e-6
    nl_abs_tol = 1e-8
    nl_max_its = 100
[]

[Outputs]
    exodus = true # For visualization in Paraview
    csv = true    # For traction-displacement curves

[damage_viz]
    type = Exodus
    execute_on = 'Timestep_End'
    elemental_as_nodal = true # For smooth damage visualization
[]
[]

```

7.20.4 Polycrystalline RVE with Many Interfaces

```

[Mesh]
    [polycrystal_rve]
        type = FileMeshGenerator
        file = polycrystal_100grain_rve.e # Externally generated
    []

    [split_interfaces]
        type = BreakMeshByBlockGenerator
        input = polycrystal_rve
        split_interface = true # Creates interfaces between all grains
    []
[]

[Materials]
    # Bulk grain behaviour (orthotropic plasticity)
    [grain_plasticity]

```

```

    type = OrthotropicPlasticityStressUpdate
    # (parameters from Section 7)
    # ...
[]

# Interface cohesive behaviour
[czm_interfaces]
    type = HomogenizedExponentialCZM
    boundary = 'interface' # All grain boundaries

    # Similar to Example 1
    normal_strength = 626.0
    shear_strength = 374.0
    delta_0_normal = 1.91e-7
    delta_0_tangent = 2.17e-7
    mu = 0.92
    eta = 0.27
    quality_std_dev = 0.10
    spatial_quality_std_dev = 0.15
    spatial_random_seed = 1234
    damage_viscosity = 2.5
    damage_activation_width = 0.02
[]
[]

[Executioner]
    type = Transient
    solve_type = NEWTON

    dt = 0.01 # 10 ms timestep (stable with homogenization)
    num_steps = 1000

    # Adaptive timestepping for progressive failure
    [TimeStepper]
        type = IterationAdaptiveDT
        dt = 0.01
        optimal_iterations = 10
        growth_factor = 1.3
        cutback_factor = 0.7
    []

    nl_rel_tol = 1e-6
    nl_abs_tol = 1e-8
    nl_max_its = 100
[]

```

Chapter 8

Troubleshooting

8.1 Build Issues

8.1.1 Problem: Conda environment not activating

Error:

```
conda activate moose
# Error: conda: command not found
```

Solution:

```
# Initialize conda for your shell
conda init bash # or zsh, fish, etc.
source ~/.bashrc

# Try again
conda activate moose
```

8.1.2 Problem: Make fails with missing headers

Error:

```
fatal error: ADStressUpdateBase.h: No such file or directory
```

Solution:

```
# Ensure environment is active
conda activate moose
echo $CONDA_DEFAULT_ENV # Should output: moose

# Clean and rebuild
make clean
make -j8
```

8.1.3 Problem: MOOSE_DIR not found

Error:

```
Error: MOOSE_DIR not set or framework not found
```

Solution 1: Set explicitly in Makefile:

```
MOOSE_DIR := /path/to/your/moose
```

Solution 2: Create minimal structure:

```
mkdir -p ~/projects/moose/framework
mkdir -p ~/projects/moose/modules
```

8.1.4 Problem: stork.sh: command not found

Problem: MOOSE repository not cloned or script not executable.

Solution:

```
# Clone MOOSE if not already done
cd ~/projects
git clone https://github.com/idaholab/moose.git

# Check script exists
ls moose/scripts/stork.sh

# Make executable if needed
chmod +x moose/scripts/stork.sh

# Run with full path
./moose/scripts/stork.sh aragonite
```

8.1.5 Build Errors: Undefined Reference

Problem: Material classes not properly registered.

Solution: Check registration at top of .C files:

```
// Must be AFTER includes, BEFORE class definition
registerMooseObject("SolidMechanicsApp",
    OrthotropicPlasticityStressUpdate);
```

Or register with your app name:

```
registerMooseObject("aragoniteApp", OrthotropicPlasticityStressUpdate);
```

8.1.6 ModuleNotFoundError: No module named 'mooseutils'

Problem: Conda environment not activated.

Solution:

```
conda activate moose
echo $CONDA_DEFAULT_ENV # Should show: moose
```

If still fails:

```
conda install moose-dev -c conda-forge --force-reinstall
```

8.1.7 Compilation Errors in Material Files

Problem: Material source files have syntax errors or missing includes.

Solution:

1. Check error message for specific file and line number
2. Common issues:
 - Missing `#include` statements

- Typos in class names
- Mismatched function signatures between .h and .C

3. Verify files compile in isolation:

```
# Test compile single file
cd ~/projects/aragonite
make src/materials/OrthotropicPlasticityStressUpdate.C -j1

# Look for specific error messages
```

8.1.8 Linking Errors

Problem: Solid mechanics module not enabled or not found.

Solution: Check Makefile has:

```
ALL_MODULES := no
SOLID_MECHANICS := yes
```

Not:

```
ALL_MODULES := yes # Wrong - includes unnecessary modules
```

8.2 Convergence Issues

8.2.1 Problem: Newton solver fails during plastic loading

Symptoms:

- Newton iterations exceed `max_iterations_newton`
- Residual oscillates without converging
- Simulation aborts with "Solve failed"

Solutions (try in order):

Step 1: Reduce time step

```
[Executioner]
dt = 0.01 # Reduce from 0.1
[]
```

Step 2: Relax tolerances

```
[Materials]
[plasticity]
absolute_tolerance = 1.0e-4 # From 1e-5
max_iterations_newton = 30 # From 20
[]
[]
```

Step 3: Delay softening onset

```
[Materials]
[plasticity]
kmax = 0.005 # From 0.002
[]
[]
```

Step 4: Enable viscoplastic regularization

```
[Materials]
[plasticity]
    viscosity_mode = 'linear'
    eta = 0.005
[]
[]
```

Step 5: Ensure Primal CPP is enabled

```
[Materials]
[plasticity]
    use_primal_cpp = true
    max_iterations_primal = 2000 # From 1000
[]
[]
```

8.2.2 Problem: Primal CPP uses too many iterations

Symptom:

```
Stage 2 (Primal CPP) invoked, iterations: 1873/2000
```

Diagnosis: This is normal during severe softening. Stage 2 is slower but more robust.

Optimization:

1. Use piecewise softening instead of exponential (smoother)
2. Increase `kmax` to delay softening
3. Reduce time step further

8.3 Yield Stress Mismatch

8.3.1 Problem: Yield occurs at wrong stress level

Symptoms:

- Expected yield at 4980 MPa, actual 5200 MPa (+4%)
- Or expected 4980 MPa, actual 4500 MPa (−10%)

Diagnose:

Check 1: Euler angles

```
# Are material axes aligned as expected?
euler_angle_1 = 0 # For no rotation
euler_angle_2 = 0
euler_angle_3 = 0
```

If angles are non-zero, stress component being tested may not align with material axis.

Check 2: Elastic constants

Run elastic-only test (remove plasticity material):

```
[Materials]
[elasticity]
    type = ComputeElasticityTensor
    # ...
[]
```

```
# [stress]
#   type = ComputeMultipleInelasticStress
#   inelastic_models = 'plasticity'
# []

# [plasticity]
#   # ... COMMENTED OUT
# []

[]
```

Measure elastic slope, compare to expected:

$$E_1 = 140.4 \text{ GPa} \quad (\text{within } 1\%)$$

$$E_2 = 70.3 \text{ GPa}$$

$$E_3 = 63.4 \text{ GPa}$$

If elastic constants are wrong, check:

```
C_ijkl = '171800 57500 30200 106700 46900 84200
          42100 31100 46600'
# Order: C11 C12 C13 C22 C23 C33 C44 C55 C66
```

Check 3: Boundary conditions

For shear tests, spurious normal stresses can cause premature yielding:

- Verify $S_{xx}, S_{yy}, S_{zz} < 10 \text{ MPa}$ during shear
- If not, check BCs: Ensure faces perpendicular to shear are free (except one fixed face)

Check 4: Units

All stresses must be in MPa:

```
# CORRECT:
sigma_xx_tension = 4980 # MPa

# WRONG:
sigma_xx_tension = 4.98e9 # Pa (will give 1000x error!)
```

8.4 Spurious Stresses in Shear Tests

8.4.1 Problem: Normal stresses appear in pure shear test

Symptom:

```
# XY-shear test
stress_xx = 1200 MPa # Should be ~0!
stress_yy = 800 MPa # Should be ~0!
stress_xy = 4510 MPa # Correct
```

Cause: Over-constrained boundary conditions preventing Poisson expansion.

Example (WRONG):

```
# XY-shear: Fix top in y, apply shear on bottom
[fix_top_y]
  boundary = 'top'
  variable = disp_y
  value = 0 # Prevents expansion!

[]
```

This creates S_{yy} reaction force, causing combined stress state.

Solution (CORRECT):

```
# XY-shear: Fix one face perpendicular to shear plane
# Fix back face (z=0) completely
[fix_back_x]
  boundary = 'back'
  variable = disp_x
  value = 0
[]
[fix_back_y]
  boundary = 'back'
  variable = disp_y
  value = 0
[]
[fix_back_z]
  boundary = 'back'
  variable = disp_z
  value = 0
[]

# Apply shear on front face (z=1)
[shear_front_x]
  boundary = 'front'
  variable = disp_x
  function = 'shear_displacement(t)'
[]
[fix_front_z]
  boundary = 'front'
  variable = disp_z
  value = 0
[]
# Note: disp_y on front is FREE
```

8.5 Performance Issues

8.5.1 Problem: Simulation runs very slowly

Diagnose:

Check 1: Are you using optimized build?

```
ls -la | grep aragonite
# Should see: aragonite-opt (not -dbg)
```

Check 2: Is Primal CPP used excessively?

```
# In output:
Stage 2 invoked 80% of time steps
```

Solution: Adjust parameters to use Stage 1 more:

- Increase `kmax` (delay softening)
- Use piecewise softening (smoother than exponential)
- Enable viscoplasticity

Check 3: Time step too small?

```
[Executioner]
  dt = 0.001 # Very small! Try 0.01
[]
```

Check 4: Use parallel execution for large meshes

```
# Run on 8 cores
mpirun -n 8 ./aragonite-opt -i input.i
```

8.5.2 Determinism and Spatial Reproducibility

The GH implementation is **fully deterministic**: the 5 quadrature nodes and weights are compile-time constants. Results are bit-identical across runs, machines, and MPI partition counts. No `random_seed` parameter is needed or accepted.

The spatial quality field (`spatial_quality_std_dev`) is also deterministic: each QP's factor is derived from `spatial_random_seed` + element-ID + QP index, independent of parallelisation.

Verification:

```
% Must give identical results:
./aragonite-opt -i rve.i && ./aragonite-opt -i rve.i
mpirun -n 8 ./aragonite-opt -i rve.i
mpirun -n 48 ./aragonite-opt -i rve.i
```

Exploring spatial patterns: Change `spatial_random_seed` for an alternative but fully reproducible spatial field.

Fully deterministic (no heterogeneity):

```
quality_std_dev          = 0.0
spatial_quality_std_dev = 0.0
```

Chapter 9

Summary

9.1 Quick Start Checklist

- ☐ Install MOOSE via conda (5–10 min)
- ☐ Create project directory
- ☐ Copy material source files
- ☐ Build application (`make -j8`)
- ☐ Run single-element tests (6 loading directions)
- ☐ Verify elastic moduli within 1%
- ☐ Verify yield stresses within 2%
- ☐ Test multi-element uniform loading
- ☐ Apply to production simulations

9.2 Parameter Selection Strategy

Phase 1: Start simple

- `postyield_mode = 'perfect'`
- `viscosity_mode = 'none'`
- Default convergence parameters

Phase 2: Add softening

- `postyield_mode = 'exp-softening'`
- `residual_strength = 0.7`
- `kmax = 0.002` (critical!)
- `kslope = 30.0`

Phase 3: Tune for convergence

- If Newton fails: increase `kmax` to 0.005
- If still fails: enable viscoplasticity (`eta = 0.002`)
- If Primal CPP used often: try piecewise softening

9.3 Common Pitfalls

1. **Forgot to activate conda environment:** Always `conda activate moose` first!
2. **Wrong units:** All stresses in MPa, lengths in mm
3. **kmax = 0:** Causes immediate softening, numerical issues. Use ≥ 0.001 .
4. **Over-constrained shear BCs:** Creates spurious normal stresses. Only fix one face perpendicular to shear plane.
5. **Large time steps:** During softening, use $\Delta t \leq 0.01$.
6. **Disabled Primal CPP:** Set `use_primal_cpp = true` for robustness.

9.4 Additional Resources

- **MOOSE Documentation:** <https://mooseframework.inl.gov/>
- **MOOSE Discussion Forum:** <https://github.com/idaholab/moose/discussions>
- **Schwiedrzik et al. [1]:** DOI: 10.1007/s10237-013-0472-5
- **Source code:** `OrthotropicPlasticityStressUpdate.C/h`
- **Example inputs:** `test/unit_tests/*/*.i`

Bibliography

- [1] JJ Schwiedrzik, U Wolfram, and PK Zysset. A generalized anisotropic quadric yield criterion and its application to bone tissue at multiple length scales. *Biomechanics and Modeling in Mechanobiology*, 12(6):1155–1168, 2013.
- [2] P. K. Zysset and A. Curnier. An alternative model for anisotropic elasticity based on fabric tensors. *Mechanics of Materials*, 21(4):243–250, November 1995. ISSN 0167-6636. doi: 10.1016/0167-6636(95)00018-6. URL <https://www.sciencedirect.com/science/article/pii/0167663695000186>.
- [3] Uwe Wolfram, Marta Peña Fernández, Samuel McPhee, Ewan Smith, Rainer J. Beck, Jonathan D. Shephard, Ali Ozel, Craig S. Erskine, Janina Büscher, Jürgen Titschack, J. Murray Roberts, and Sebastian J. Hennige. Multiscale mechanical consequences of ocean acidification for cold-water corals. *Sci Rep*, 12(1):8052, May 2022. ISSN 2045-2322. doi: 10.1038/s41598-022-11266-w. URL <https://www.nature.com/articles/s41598-022-11266-w>.
- [4] SW Tsai and EM Wu. A General Theory of Strength for Anisotropic Materials. *Journal of Composite Materials*, 5(1):58–80, 1971.
- [5] Uwe Wolfram, Thomas Gross, Dieter H. Pahr, Jakob Schwiedrzik, Hans-Joachim Wilke, and Philippe K. Zysset. Fabric-based Tsai–Wu yield criteria for vertebral trabecular bone in stress and strain space. *Journal of the Mechanical Behavior of Biomedical Materials*, 15:218–228, November 2012. ISSN 1751-6161. doi: 10.1016/j.jmbbm.2012.07.005. URL <https://www.sciencedirect.com/science/article/pii/S1751616112002007>.
- [6] R. Hill. Elastic properties of reinforced solids: Some theoretical principles. *Journal of the Mechanics and Physics of Solids*, 11(5):357–372, September 1963. ISSN 0022-5096. doi: 10.1016/0022-5096(63)90036-X. URL <https://www.sciencedirect.com/science/article/pii/002250966390036X>.
- [7] X. P. Xu and A. Needleman. Numerical simulations of fast crack growth in brittle solids. *Journal of the Mechanics and Physics of Solids*, 42(9):1397–1434, September 1994. ISSN 0022-5096. doi: 10.1016/0022-5096(94)90003-5. URL <https://www.sciencedirect.com/science/article/pii/0022509694900035>.
- [8] Jun Wang, Zhao-Hui Lu, and Chun-Qing Li. A novel mixed-mode interface damage model based on exponential traction-separation law. *Construction and Building Materials*, 490:142454, September 2025. ISSN 09500618. doi: 10.1016/j.conbuildmat.2025.142454. URL <https://linkinghub.elsevier.com/retrieve/pii/S0950061825026054>.
- [9] A. Turon, P. P. Camanho, J. Costa, and C. G. Dávila. A damage model for the simulation of delamination in advanced composites under variable-mode loading. *Mechanics of Materials*, 38(11):1072–1089, 2006. doi: 10.1016/j.mechmat.2005.10.003.
- [10] P. P. Camanho and C. G. Dávila. Mixed-mode decohesion finite elements for the simulation of delamination in composite materials. *NASA Technical Paper*, (211737), 2002.

- [11] Nikolai Kvashin, Ali Ozel, and Uwe Wolfram. Atomistic origins of interface-controlled failure in polycrystalline aragonite. *Journal of the Mechanical Behavior of Biomedical Materials*, 178:107403, June 2026. ISSN 1751-6161. doi: 10.1016/j.jmbbm.2026.107403. URL <https://www.sciencedirect.com/science/article/pii/S175161612600072X>.
- [12] M. L. Benzeggagh and Mohamed Kenane. Measurement of mixed-mode delamination fracture toughness of unidirectional glass/epoxy composites with mixed-mode bending apparatus. *Composites Science and Technology*, 56:439–449, 1996. URL <https://api.semanticscholar.org/CorpusID:135594992>.
- [13] T. P. Harrigan and R. W. Mann. Characterization of microstructural anisotropy in orthotropic materials using a second rank tensor. *J Mater Sci*, 19(3):761–767, March 1984. ISSN 1573-4803. doi: 10.1007/BF00540446. URL <https://doi.org/10.1007/BF00540446>.
- [14] Liliana Rincón-Kohli and Philippe K. Zysset. Multi-axial mechanical properties of human trabecular bone. *Biomech Model Mechanobiol*, 8(3):195–208, June 2009. ISSN 1617-7940. doi: 10.1007/s10237-008-0128-z. URL <https://doi.org/10.1007/s10237-008-0128-z>.